

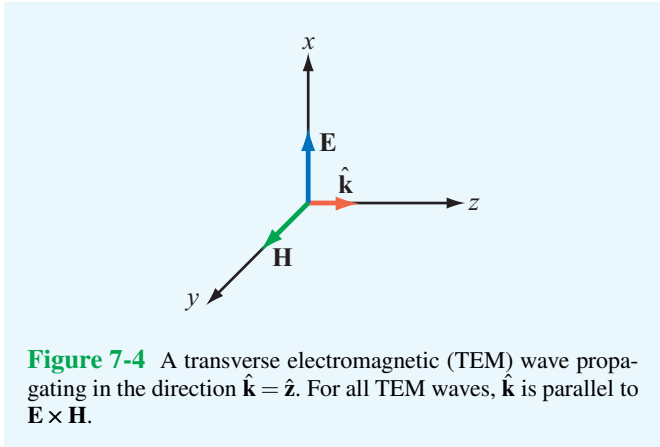


Figure 7-4 A transverse electromagnetic (TEM) wave propagating in the direction $\hat{\mathbf{k}} = \hat{\mathbf{z}}$. For all TEM waves, $\hat{\mathbf{k}}$ is parallel to $\mathbf{E} \times \mathbf{H}$.

and

$$\mathbf{H}(z, t) = \Re \left[\tilde{\mathbf{H}}(z) e^{j\omega t} \right] = \hat{\mathbf{y}} \frac{|E_{x0}^+|}{\eta} \cos(\omega t - kz + \phi^+) \quad (\text{A/m}). \quad (7.34b)$$

Because $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$ exhibit the same functional dependence on z and t , they are said to be **in phase**; when the amplitude of one of them reaches a maximum, the amplitude of the other does so too. The fact that $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ are in phase is characteristic of waves propagating in lossless media.

From the material on wave motion presented in Section 1-4, we deduce that the **phase velocity** of the wave is

$$u_p = \frac{\omega}{k} = \frac{\omega}{\omega \sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu \epsilon}} \quad (\text{m/s}), \quad (7.35)$$

and its wavelength is

$$\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m}). \quad (7.36)$$

In a vacuum, $\epsilon = \epsilon_0$ and $\mu = \mu_0$, and the phase velocity u_p and the intrinsic impedance η given by Eq. (7.31) are

$$u_p = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \quad (\text{m/s}), \quad (7.37)$$

$$\eta = \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377 \, (\Omega) \approx 120\pi \, (\Omega), \quad (7.38)$$

where c is the velocity of light and η_0 is called the **intrinsic impedance of free space**.

Example 7-1: EM Plane Wave in Air

This example is analogous to the “Sound Wave in Water” problem given by Example 1-1.

The electric field of a 1 MHz plane wave traveling in the $+z$ direction in air points along the x direction. If this field reaches a peak value of 1.2π (mV/m) at $t = 0$ and $z = 50$ m, obtain expressions for $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$. Then plot them as a function of z at $t = 0$.

Solution: At $f = 1$ MHz, the wavelength in air is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10^6} = 300 \text{ m},$$

and the corresponding wavenumber is $k = (2\pi/300)$ (rad/m). The general expression for an x -directed electric field traveling in the $+z$ direction is given by Eq. (7.34a) as

$$\begin{aligned} \mathbf{E}(z, t) &= \hat{\mathbf{x}} |E_{x0}^+| \cos(\omega t - kz + \phi^+) \\ &= \hat{\mathbf{x}} 1.2\pi \cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \phi^+\right) \quad (\text{mV/m}). \end{aligned}$$

The field $\mathbf{E}(z, t)$ is maximum when the argument of the cosine function equals zero or a multiple of 2π . At $t = 0$ and $z = 50$ m, this condition yields

$$-\frac{2\pi \times 50}{300} + \phi^+ = 0 \quad \text{or} \quad \phi^+ = \frac{\pi}{3}.$$

Hence,

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 1.2\pi \cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3}\right) \quad (\text{mV/m}),$$

and from Eq. (7.34b), we have

$$\begin{aligned} \mathbf{H}(z, t) &= \hat{\mathbf{y}} \frac{E(z, t)}{\eta_0} \\ &= \hat{\mathbf{y}} 10 \cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3}\right) \quad (\mu\text{A/m}), \end{aligned}$$

where we have used the approximation $\eta_0 \approx 120\pi$ (Ω).

At $t = 0$,

$$\mathbf{E}(z, 0) = \hat{\mathbf{x}} 1.2\pi \cos\left(\frac{2\pi z}{300} - \frac{\pi}{3}\right) \quad (\text{mV/m}),$$

$$\mathbf{H}(z, 0) = \hat{\mathbf{y}} 10 \cos\left(\frac{2\pi z}{300} - \frac{\pi}{3}\right) \quad (\mu\text{A/m}).$$

Plots of $\mathbf{E}(z, 0)$ and $\mathbf{H}(z, 0)$ as a function of z are shown in **Fig. 7-5**.

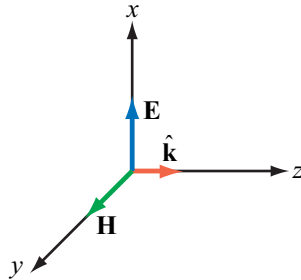


图 7-4 沿 $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ 方向传播的横向电磁 (TEM) 波。对于所有 TEM 波, $\hat{\mathbf{k}}$ 平行于 $\mathbf{E} \times \mathbf{H}$ 。

和

$$\mathbf{H}(z, t) = \Re \left[\tilde{\mathbf{H}}(z) e^{j\omega t} \right] = \hat{\mathbf{y}} \frac{|E_{x0}^+|}{\eta} \cos(\omega t - kz + \phi^+) \quad (\text{A/m}). \quad (7.34b)$$

因为 $\mathbf{E}(z, t)$ 和 $\mathbf{H}(z, t)$ 对 z 和 t 表现出相同的函数依赖性, 所以称它们同相; 当其中一个的幅度达到最大值时, 另一个的幅度也会达到最大值。 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 同相这一事实是波在无损介质中传播的特征。

根据第 1-4 节中介绍的有关波动的材料, 我们推断出波的相速度为

$$u_p = \frac{\omega}{k} = \frac{\omega}{\omega \sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu \epsilon}} \quad (\text{m/s}), \quad (7.35)$$

其波长为

$$\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m}). \quad (7.36)$$

在真空中, $\epsilon = \epsilon_0$ 和 $\mu = \mu_0$, 以及由方程给出的相速度 u_p 和固有阻抗 η 。(7.31) 是

$$u_p = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \quad (\text{m/s}), \quad (7.37)$$

$$\eta = \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377 \, (\Omega) \approx 120\pi \, (\Omega), \quad (7.38)$$

其中 c 是光速, η_0 称为自由空间的 **本征阻抗**。

例 7-1: 空气中的电磁平面波

此示例类似于示例 1-1 中给出的 “水中的声波” 问题。

空气中沿 $+z$ 方向传播的 1 MHz 平面波的电场沿 x 方向指向。如果该场在 $t = 0$ 和 $z = 50$ m 处达到峰值 1.2π (mV/m), 则获得 $\mathbf{E}(z, t)$ 和 $\mathbf{H}(z, t)$ 的表达式。然后将它们绘制为 $t = 0$ 处 z 的函数。

解: 在 $f = 1$ MHz 时, 空气中的波长为

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10^6} = 300 \text{ m},$$

对应的波数为 $k = (2\pi/300)$ (rad/m)。沿 $+z$ 方向传播的 x 方向电场的一般表达式由方程 1 给出。(7.34a) 为

$$\begin{aligned} \mathbf{E}(z, t) &= \hat{\mathbf{x}} |E_{x0}^+| \cos(\omega t - kz + \phi^+) \\ &= \hat{\mathbf{x}} 1.2\pi \cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \phi^+\right) \quad (\text{mV/m}). \end{aligned}$$

当余弦函数的参数等于零或 2π 的倍数时, 域 $\mathbf{E}(z, t)$ 最大。在 $t = 0$ 和 $z = 50$ m 处, 此条件产生

$$-\frac{2\pi \times 50}{300} + \phi^+ = 0 \quad \text{or} \quad \phi^+ = \frac{\pi}{3}.$$

因此,

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 1.2\pi \cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3}\right) \quad (\text{mV/m}),$$

并从方程式(7.34b), 我们有

$$\begin{aligned} \mathbf{H}(z, t) &= \hat{\mathbf{y}} \frac{E(z, t)}{\eta_0} \\ &= \hat{\mathbf{y}} 10 \cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3}\right) \quad (\mu\text{A/m}), \end{aligned}$$

我们使用了近似值 $\eta_0 \approx 120\pi$ (Ω)。

在 $t = 0$ 时,

$$\mathbf{E}(z, 0) = \hat{\mathbf{x}} 1.2\pi \cos\left(\frac{2\pi z}{300} - \frac{\pi}{3}\right) \quad (\text{mV/m}),$$

$$\mathbf{H}(z, 0) = \hat{\mathbf{y}} 10 \cos\left(\frac{2\pi z}{300} - \frac{\pi}{3}\right) \quad (\mu\text{A/m}).$$

$\mathbf{E}(z, 0)$ 和 $\mathbf{H}(z, 0)$ 作为 z 函数的图如图 7-5 所示。

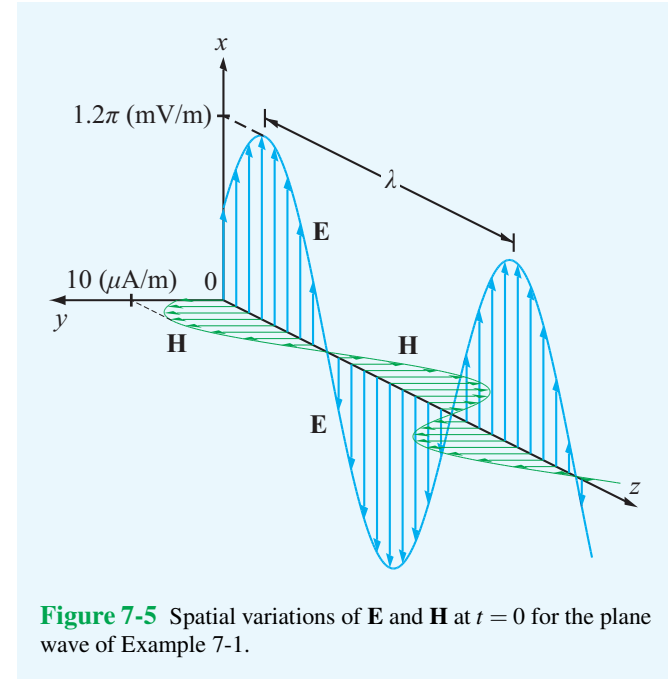


Figure 7-5 Spatial variations of $\mathbf{E}$ and $\mathbf{H}$ at $t = 0$ for the plane wave of Example 7-1.

7-2.2 General Relation between $\mathbf{E}$ and $\mathbf{H}$

It can be shown that, for any uniform plane wave traveling in an arbitrary direction denoted by the unit vector $\hat{\mathbf{k}}$, the electric and magnetic field phasors $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ are related as

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{k}} \times \tilde{\mathbf{E}}, \quad (7.39a)$$

$$\tilde{\mathbf{E}} = -\eta \hat{\mathbf{k}} \times \tilde{\mathbf{H}}. \quad (7.39b)$$

► The following right-hand rule applies: When we rotate the four fingers of the right hand from the direction of $\mathbf{E}$ toward that of $\mathbf{H}$, the thumb points in the direction of the wave travel, $\hat{\mathbf{k}}$.

The relations given by Eqs. (7.39a and b) are valid not only for lossless media but for lossy ones as well. As we see later in Section 7-4, the expression for η of a lossy medium is different from that given by Eq. (7.31). As long as the expression used for η is appropriate for the medium in which the wave is traveling, the relations given by Eqs. (7.39a and b) always hold.

Wave Propagating Along $+z$ with $\mathbf{E}$ Along $\hat{\mathbf{x}}$

Let us apply Eq. (7.39a) to the wave given by Eq. (7.32a). The direction of propagation $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ and $\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x^+(z)$. Hence,

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{k}} \times \tilde{\mathbf{E}} = \frac{1}{\eta} (\hat{\mathbf{z}} \times \hat{\mathbf{x}}) \tilde{E}_x^+(z) = \hat{\mathbf{y}} \frac{\tilde{E}_x^+(z)}{\eta}, \quad (7.40)$$

which is the same as the result given by Eq. (7.32b).

Wave Propagating Along $-z$ with $\mathbf{E}$ Along $\hat{\mathbf{x}}$

For a wave traveling in the $-z$ direction with electric field given by

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x^-(z) = \hat{\mathbf{x}} E_{x0}^- e^{jkz}, \quad (7.41)$$

application of Eq. (7.39a) gives

$$\tilde{\mathbf{H}} = \frac{1}{\eta} (-\hat{\mathbf{z}} \times \hat{\mathbf{x}}) \tilde{E}_x^-(z) = -\hat{\mathbf{y}} \frac{\tilde{E}_x^-(z)}{\eta} = -\hat{\mathbf{y}} \frac{E_{x0}^-}{\eta} e^{jkz}. \quad (7.42)$$

Hence, in this case, $\tilde{\mathbf{H}}$ points in the negative y direction.

Wave Propagating Along $+z$ with $\mathbf{E}$ Along $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$

In general, a uniform plane wave traveling in the $+z$ direction may have both x and y components, where $\tilde{\mathbf{E}}$ is given by

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x^+(z) + \hat{\mathbf{y}} \tilde{E}_y^+(z), \quad (7.43a)$$

and the associated magnetic field is

$$\tilde{\mathbf{H}} = \hat{\mathbf{x}} \tilde{H}_x^+(z) + \hat{\mathbf{y}} \tilde{H}_y^+(z). \quad (7.43b)$$

Application of Eq. (7.39a) gives

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{z}} \times \tilde{\mathbf{E}} = -\hat{\mathbf{x}} \frac{\tilde{E}_y^+(z)}{\eta} + \hat{\mathbf{y}} \frac{\tilde{E}_x^+(z)}{\eta}. \quad (7.44)$$

By equating Eq. (7.43b) to Eq. (7.44), we have

$$\tilde{H}_x^+(z) = -\frac{\tilde{E}_y^+(z)}{\eta}, \quad \tilde{H}_y^+(z) = \frac{\tilde{E}_x^+(z)}{\eta}. \quad (7.45)$$

These results are illustrated in **Fig. 7-6**. The wave may be considered the sum of two waves: one with electric and magnetic components (E_x^+, H_y^+) and another with components (E_y^+, H_x^+) . In general, a TEM wave may have an electric field in any direction in the plane orthogonal to the direction of wave travel, the associated magnetic field is also in the same plane, and its direction is dictated by Eq. (7.39a).

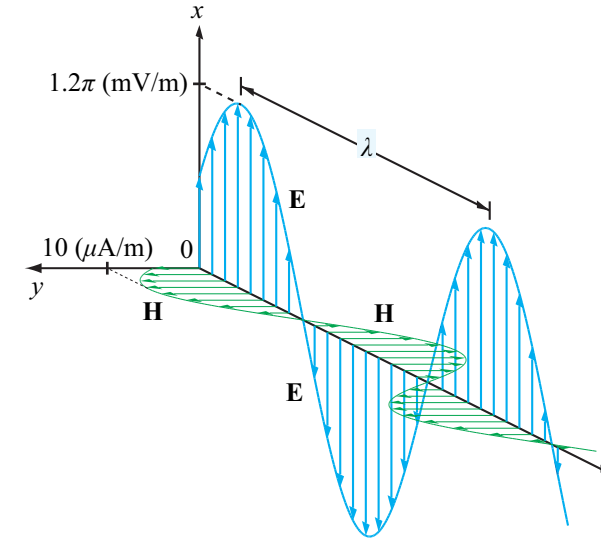


图 7-5 例 7-1 的平面波在 $t = 0$ 时 $\mathbf{E}$ 和 $\mathbf{H}$ 的空间变化。

7-2.2 E与H的一般关系

可以证明，对于以单位矢量 $\hat{\mathbf{k}}$ 表示的任意方向传播的任何均匀平面波，电场相量 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 的关系为

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{k}} \times \tilde{\mathbf{E}}, \quad (7.39a)$$

$$\tilde{\mathbf{E}} = -\eta \hat{\mathbf{k}} \times \tilde{\mathbf{H}}. \quad (7.39b)$$

► 以下右手定则适用：当我们将右手的四个手指从 $\mathbf{E}$ 方向向 $\mathbf{H}$ 方向旋转时，拇指指向波传播的方向 $\hat{\mathbf{k}}$ 。

等式给出的关系。(7.39a和b)不仅对于无损媒体有效，而且对于有损媒体也有效。正如我们稍后在第 7-4 节中看到的，有损介质的 η 的表达式与等式 1 给出的不同。(7.31)。只要用于 η 的表达式适用于波传播的介质，方程 1 给出的关系就可以得到。(7.39a 和 b) 始终成立。

波沿 $+z$ 传播， $\mathbf{E}$ 沿 $\hat{\mathbf{x}}$ 传播

让我们应用等式。(7.39a) 到方程给出的波。(7.32a)。传播方向 $\hat{\mathbf{k}} = \hat{\mathbf{z}}$ 和 $\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x^+(z)$ 。因此，

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{k}} \times \tilde{\mathbf{E}} = \frac{1}{\eta} (\hat{\mathbf{z}} \times \hat{\mathbf{x}}) \tilde{E}_x^+(z) = \hat{\mathbf{y}} \frac{\tilde{E}_x^+(z)}{\eta}, \quad (7.40)$$

这与等式给出的结果相同。(7.32b)。

波沿 $-z$ 传播， $\mathbf{E}$ 沿 $\hat{\mathbf{x}}$ 传播

对于沿 $-z$ 方向传播的波，其电场由下式给出

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x^-(z) = \hat{\mathbf{x}} E_{x0}^- e^{jkz}, \quad (7.41)$$

方程的应用(7.39a) 给出

$$\tilde{\mathbf{H}} = \frac{1}{\eta} (-\hat{\mathbf{z}} \times \hat{\mathbf{x}}) \tilde{E}_x^-(z) = -\hat{\mathbf{y}} \frac{\tilde{E}_x^-(z)}{\eta} = -\hat{\mathbf{y}} \frac{E_{x0}^-}{\eta} e^{jkz}. \quad (7.42)$$

因此，在这种情况下， $\tilde{\mathbf{H}}$ 指向负 y 方向。

波沿着 $+z$ 传播， $\mathbf{E}$ 沿着 $\hat{\mathbf{x}}$ 和 $\hat{\mathbf{y}}$

一般来说，沿 $+z$ 方向传播的均匀平面波可能同时具有 x 和 y 分量，其中 $\tilde{\mathbf{E}}$ 由下式给出

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x^+(z) + \hat{\mathbf{y}} \tilde{E}_y^+(z), \quad (7.43a)$$

相关的磁场是

$$\tilde{\mathbf{H}} = \hat{\mathbf{x}} \tilde{H}_x^+(z) + \hat{\mathbf{y}} \tilde{H}_y^+(z). \quad (7.43b)$$

方程的应用(7.39a) 给出

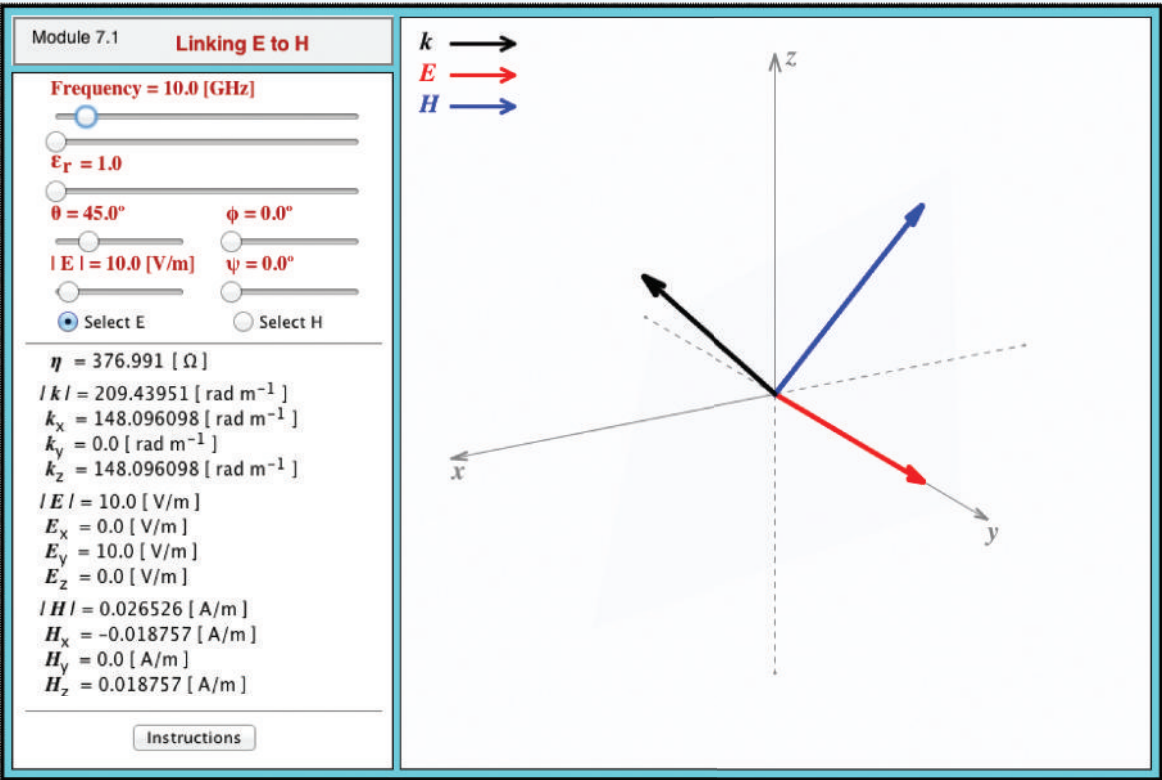
$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{z}} \times \tilde{\mathbf{E}} = -\hat{\mathbf{x}} \frac{\tilde{E}_y^+(z)}{\eta} + \hat{\mathbf{y}} \frac{\tilde{E}_x^+(z)}{\eta}. \quad (7.44)$$

通过等式。(7.43b) 至方程式 (7.44)，我们有

$$\tilde{H}_x^+(z) = -\frac{\tilde{E}_y^+(z)}{\eta}, \quad \tilde{H}_y^+(z) = \frac{\tilde{E}_x^+(z)}{\eta}. \quad (7.45)$$

这些结果如图 7-6 所示。该波可以被认为两个波的总和：一个具有电磁分量 (E_x^+, H_y^+) ，另一个具有分量 (E_y^+, H_x^+) 。一般来说，TEM 波可以在与波传播方向正交的平面内的任何方向上产生电场，相关的磁场也在同一平面内，其方向由方程 1 决定。(7.39a)。

Module 7.1 Linking E to H Select the directions and magnitudes of **E** and **H** and observe the resultant wave vector.



模块 7.1 将 E 链接到 H 选择 E 和 H 的方向和大小并观察合成的波矢量。

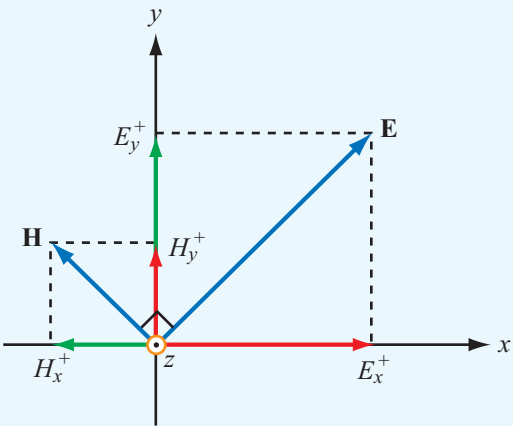
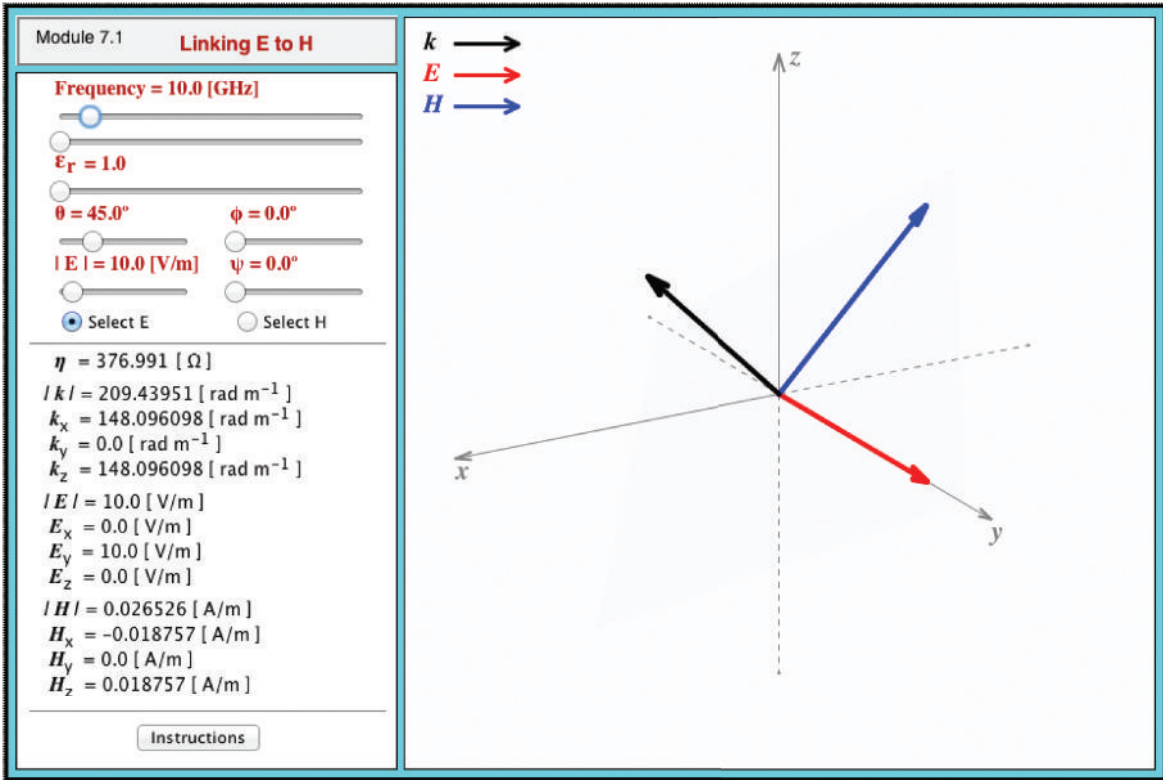


Figure 7-6 The wave (**E**,**H**) is equivalent to the sum of two waves: one with fields (E_x^+, H_y^+) and another with (E_y^+, H_x^+) with both traveling in the $+z$ direction.

Concept Question 7-1: What is a uniform plane wave? Describe its properties, both physically and mathematically. Under what conditions is it appropriate to treat a spherical wave as a plane wave?

Concept Question 7-2: Since $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ are governed by wave equations of the same form [Eqs. (7.15) and (7.16)], does it follow that $\tilde{\mathbf{E}} = \tilde{\mathbf{H}}$? Explain.

Concept Question 7-3: If a TEM wave is traveling in the $\hat{\mathbf{y}}$ direction, can its electric field have components along $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$? Explain.

Exercise 7-1: A 10-MHz uniform plane wave is traveling in a nonmagnetic medium with $\mu = \mu_0$ and $\epsilon_r = 9$. Find (a) the phase velocity, (b) the wavenumber, (c) the wavelength in the medium, and (d) the intrinsic impedance of the medium.

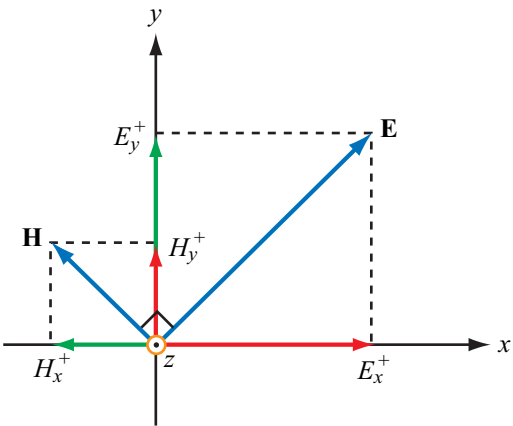


图 7-6 波 (**E**,**H**) 相当于两个波的和：一个波具有场 (E_x^+, H_y^+)，另一个波具有场 (E_y^+, H_x^+)，两者都沿 $+z$ 方向传播。

概念问题 7-1：什么是均匀平面波？从物理和数学角度描述它的属性。在什么条件下适合将球面波视为平面波？

概念问题 7-2：由于 $\tilde{\mathbf{E}}$ 和 $\tilde{\mathbf{H}}$ 受相同形式的波动方程控制 [Eqs. (7.15) 和 (7.16)]，它是否遵循 $\tilde{\mathbf{E}} = \tilde{\mathbf{H}}$ ？解释。

概念问题 7-3：如果 TEM 波沿 $\hat{\mathbf{y}}$ 方向传播，其电场是否可以有沿 $\hat{\mathbf{x}}$ 、 $\hat{\mathbf{y}}$ 和 $\hat{\mathbf{z}}$ 的分量？解释。

练习 7-1：10 MHz 均匀平面波在具有 $\mu = \mu_0$ 和 $\epsilon_r = 9$ 的非磁性介质中传播。求 (a) 相速度，(b) 波数，(c) 介质中的波长，以及 (d) 介质的固有阻抗。

Answer: (a) $u_p = 1 \times 10^8$ m/s, (b) $k = 0.2\pi$ rad/m, (c) $\lambda = 10$ m, (d) $\eta = 125.67 \Omega$. (See ⑤.)

Exercise 7-2: The electric field phasor of a uniform plane wave traveling in a lossless medium with an intrinsic impedance of 188.5Ω is given by $\tilde{\mathbf{E}} = \hat{\mathbf{z}} 10 e^{-j4\pi y}$ (mV/m). Determine (a) the associated magnetic field phasor and (b) the instantaneous expression for $\mathbf{E}(y, t)$ if the medium is nonmagnetic ($\mu = \mu_0$).

Answer: (a) $\tilde{\mathbf{H}} = \hat{\mathbf{x}} 53 e^{-j4\pi y}$ (μ A/m), (b) $\mathbf{E}(y, t) = \hat{\mathbf{z}} 10 \cos(6\pi \times 10^8 t - 4\pi y)$ (mV/m). (See ⑤.)

Exercise 7-3: If the magnetic field phasor of a plane wave traveling in a medium with intrinsic impedance $\eta = 100 \Omega$ is given by $\tilde{\mathbf{H}} = (\hat{\mathbf{y}} 10 + \hat{\mathbf{z}} 20) e^{-j4x}$ (mA/m), find the associated electric field phasor.

Answer: $\tilde{\mathbf{E}} = (-\hat{\mathbf{z}} + \hat{\mathbf{y}} 2) e^{-j4x}$ (V/m). (See ⑤.)

Exercise 7-4: Repeat Exercise 7-3 for a magnetic field given by $\tilde{\mathbf{H}} = \hat{\mathbf{y}}(10 e^{-j3x} - 20 e^{j3x})$ (mA/m).

Answer: $\tilde{\mathbf{E}} = -\hat{\mathbf{z}}(e^{-j3x} + 2e^{j3x})$ (V/m). (See ⑤.)

7-3 Wave Polarization

► The **polarization** of a uniform plane wave describes the locus traced by the tip of the $\mathbf{E}$ vector (in the plane orthogonal to the direction of propagation) at a given point in space as a function of time. ◀

In the most general case, the locus of the tip of $\mathbf{E}$ is an ellipse, and the wave is said to be **elliptically polarized**. Under certain conditions, the ellipse may degenerate into a circle or a straight line, in which case the **polarization state** is called **circular** or **linear**, respectively.

It was shown in Section 7-2 that the z components of the electric and magnetic fields of a z -propagating plane wave are both zero. Hence, in the most general case, the electric field phasor $\tilde{\mathbf{E}}(z)$ of a $+z$ -propagating plane wave may consist of an x component, $\hat{\mathbf{x}} \tilde{E}_x(z)$, and a y component, $\hat{\mathbf{y}} \tilde{E}_y(z)$, or

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x(z) + \hat{\mathbf{y}} \tilde{E}_y(z) \quad (7.46)$$

with

$$\tilde{E}_x(z) = E_{x0} e^{-jkz}, \quad (7.47a)$$

$$\tilde{E}_y(z) = E_{y0} e^{-jkz}, \quad (7.47b)$$

where E_{x0} and E_{y0} are the amplitudes of $\tilde{E}_x(z)$ and $\tilde{E}_y(z)$, respectively. For the sake of simplicity, the plus sign superscript has been suppressed; the negative sign in e^{-jkz} is sufficient to remind us that the wave is traveling in the positive z direction.

The two amplitudes E_{x0} and E_{y0} are, in general, complex quantities with each characterized by a magnitude and a phase angle. The phase of a wave is defined relative to a reference state, such as $z = 0$ and $t = 0$ or any other combination of z and t . As will become clear from the discussion that follows, the polarization of the wave described by Eqs. (7.46) and (7.47) depends on the phase of E_{y0} relative to that of E_{x0} but not on the absolute phases of E_{x0} and E_{y0} . Hence, for convenience, we assign E_{x0} a phase of zero and denote the phase of E_{y0} , relative to that of E_{x0} , as δ . Thus, δ is the **phase difference** between the y and x components of $\tilde{\mathbf{E}}$. Accordingly, we define E_{x0} and E_{y0} as

$$E_{x0} = a_x, \quad (7.48a)$$

$$E_{y0} = a_y e^{j\delta}, \quad (7.48b)$$

where $a_x = |E_{x0}| \geq 0$ and $a_y = |E_{y0}| \geq 0$ are the magnitudes of E_{x0} and E_{y0} , respectively. Thus, by definition, a_x and a_y may not assume negative values. Using Eqs. (7.48a and b) in Eqs. (7.47a and b), the total electric field phasor is

$$\tilde{\mathbf{E}}(z) = (\hat{\mathbf{x}} a_x + \hat{\mathbf{y}} a_y e^{j\delta}) e^{-jkz}, \quad (7.49)$$

and the corresponding instantaneous field is

$$\begin{aligned} \mathbf{E}(z, t) &= \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right] \\ &= \hat{\mathbf{x}} a_x \cos(\omega t - kz) + \hat{\mathbf{y}} a_y \cos(\omega t - kz + \delta). \end{aligned} \quad (7.50)$$

When characterizing an electric field at a given point in space, two of its attributes that are of particular interest are its magnitude and direction. The magnitude of $\mathbf{E}(z, t)$ is

$$\begin{aligned} |\mathbf{E}(z, t)| &= [E_x^2(z, t) + E_y^2(z, t)]^{1/2} \\ &= [a_x^2 \cos^2(\omega t - kz) + a_y^2 \cos^2(\omega t - kz + \delta)]^{1/2}. \end{aligned} \quad (7.51)$$

The electric field $\mathbf{E}(z, t)$ has components along the x and y directions. At a specific position z , the direction of $\mathbf{E}(z, t)$ is characterized by its **inclination angle** ψ , defined with respect to the x axis and given by

$$\psi(z, t) = \tan^{-1} \left[\frac{E_y(z, t)}{E_x(z, t)} \right]. \quad (7.52)$$

In the general case, both the intensity of $\mathbf{E}(z, t)$ and its direction are functions of z and t . Next, we examine some special cases.

答案: (a) $u_p = 1 \times 10^8$ m/s, (b) $k = 0.2\pi$ rad/m, (c) $\lambda = 10$ m, (d) $\eta = 125.67 \Omega$. (参见 ⑤.)

练习 7-2: 在固有阻抗为 188.5Ω 的无损介质中传播的均匀平面波的电场相量由 $\tilde{\mathbf{E}} = \hat{\mathbf{z}} 10 e^{-j4\pi y}$ (mV/m) 给出。如果介质是非磁性的 ($\mu = \mu_0$)，则确定 (a) 相关的磁场相量和 (b) $\mathbf{E}(y, t)$ 的瞬时表达式。

答案: (a) $\tilde{\mathbf{H}} = \hat{\mathbf{x}} 53 e^{-j4\pi y}$ (μ A/m), (b) $\mathbf{E}(y, t) = \hat{\mathbf{z}} 10 \cos(6\pi \times 10^8 t - 4\pi y)$ (mV/m)。 (参见 ⑤.)

练习 7-3: 如果在固有阻抗为 $\eta = 100 \Omega$ 的介质中传播的平面波的磁场相量由 $\tilde{\mathbf{H}} = (\hat{\mathbf{y}} 10 + \hat{\mathbf{z}} 20) e^{-j4x}$ (mA/m) 给出，求相关的电场相量。

Answer: $\tilde{\mathbf{E}} = (-\hat{\mathbf{z}} + \hat{\mathbf{y}} 2) e^{-j4x}$ (V/m). (See ⑤.)

练习 7-4: 对由 $\tilde{\mathbf{H}} = \hat{\mathbf{y}}(10 e^{-j3x} - 20 e^{j3x})$ (mA/m) 给出的磁场重复练习 7-3。

Answer: $\tilde{\mathbf{E}} = -\hat{\mathbf{z}}(e^{-j3x} + 2e^{j3x})$ (V/m). (See ⑤.)

7-3 波的偏振

► 均匀平面波的偏振描述了 $\mathbf{E}$ 矢量尖端（在与传播方向正交的平面中）在空间中给定点处随时间变化的轨迹。 ◀

在最一般的情况下， $\mathbf{E}$ 尖端的轨迹是椭圆，并且该波被称为椭圆偏振。在某些条件下，椭圆可能退化为圆形或直线，在这种情况下，偏振态分别称为圆形或线性。

第 7-2 节表明， z 方向传播的平面波的电场和磁场的 z 分量均为零。因此，在最一般的情况下， $+z$ 传播平面波的电场相量 $\tilde{\mathbf{E}}(z)$ 可能由 x 分量 $\hat{\mathbf{x}} \tilde{E}_x(z)$ 和 y 分量 $\hat{\mathbf{y}} \tilde{E}_y(z)$ 组成，或者

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x(z) + \hat{\mathbf{y}} \tilde{E}_y(z) \quad (7.46)$$

和

$$\tilde{E}_x(z) = E_{x0} e^{-jkz}, \quad (7.47a)$$

$$\tilde{E}_y(z) = E_{y0} e^{-jkz}, \quad (7.47b)$$

其中 E_{x0} 和 E_{y0} 分别是 $\tilde{E}_x(z)$ 和 $\tilde{E}_y(z)$ 的幅度。为了简单起见，加号上标已被取消； e^{-jkz} 中的负号足以提醒我们波正向 z 方向传播。

通常，两个幅度 E_{x0} 和 E_{y0} 是复量，每个幅度都由幅度和相位角表征。波的相位是相对于参考状态定义的，例如 $z = 0$ 和 $t = 0$ 或 z 和 t 的任何其他组合。从下面的讨论中可以清楚地看出，方程描述的波的偏振。(7.46) 和 (7.47) 取决于 E_{y0} 相对于 E_{x0} 的相位，但不取决于 E_{x0} 和 E_{y0} 的绝对相位。因此，为了方便起见，我们将 E_{x0} 的相位指定为零，并将 E_{y0} 相对于 E_{x0} 的相位表示为 δ 。因此， δ 是 $\tilde{\mathbf{E}}$ 的 y 和 x 分量之间的相位差。因此，我们将 E_{x0} 和 E_{y0} 定义为

$$E_{x0} = a_x, \quad (7.48a)$$

$$E_{y0} = a_y e^{j\delta}, \quad (7.48b)$$

其中 $a_x = |E_{x0}| \geq 0$ 和 $a_y = |E_{y0}| \geq 0$ 分别是 E_{x0} 和 E_{y0} 的大小。因此，根据定义， a_x 和 a_y 不能取负值。使用方程式。等式中的 (7.48a 和 b) (7.47a 和 b)，总电场相量为

$$\tilde{\mathbf{E}}(z) = (\hat{\mathbf{x}} a_x + \hat{\mathbf{y}} a_y e^{j\delta}) e^{-jkz}, \quad (7.49)$$

相应的瞬时场为

$$\begin{aligned} \mathbf{E}(z, t) &= \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right] \\ &= \hat{\mathbf{x}} a_x \cos(\omega t - kz) + \hat{\mathbf{y}} a_y \cos(\omega t - kz + \delta). \end{aligned} \quad (7.50)$$

当表征空间中给定点的电场时，特别令人感兴趣的两个属性是其大小和方向。 $\mathbf{E}(z, t)$ 的大小为

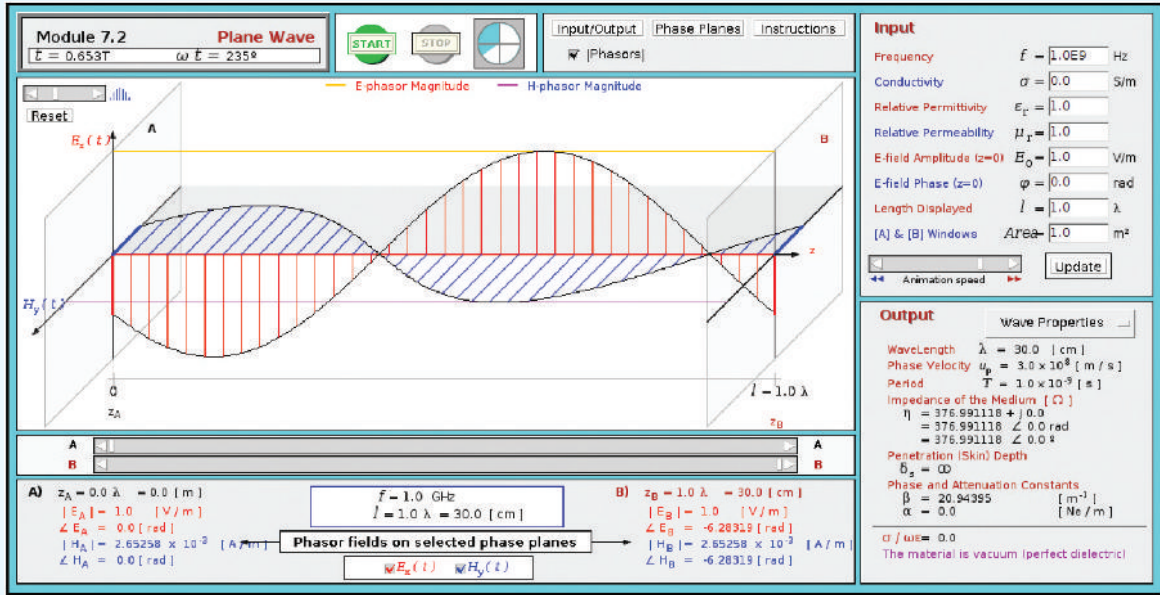
$$\begin{aligned} |\mathbf{E}(z, t)| &= [E_x^2(z, t) + E_y^2(z, t)]^{1/2} \\ &= [a_x^2 \cos^2(\omega t - kz) + a_y^2 \cos^2(\omega t - kz + \delta)]^{1/2}. \end{aligned} \quad (7.51)$$

电场 $\mathbf{E}(z, t)$ 具有沿 x 和 y 方向的分量。在特定位置 z ， $\mathbf{E}(z, t)$ 的方向由其倾斜角 ψ 表征，相对于 x 轴定义并由下式给出

$$\psi(z, t) = \tan^{-1} \left[\frac{E_y(z, t)}{E_x(z, t)} \right]. \quad (7.52)$$

一般情况下， $\mathbf{E}(z, t)$ 的强度及其方向都是 z 和 t 的函数。接下来，我们研究一些特殊情况。

Module 7.2 Plane Wave Observe a plane wave propagating along the z direction; note the temporal and spatial variations of $\mathbf{E}$ and $\mathbf{H}$. Examine how the wave properties change as a function of the values selected for the wave parameters—frequency and $\mathbf{E}$ field amplitude and phase—and the medium's constitutive parameters (ϵ, μ, σ).



7-3.1 Linear Polarization

► A wave is said to be linearly polarized if for a fixed z , the tip of $\mathbf{E}(z, t)$ traces a straight line segment as a function of time. This happens when $E_x(z, t)$ and $E_y(z, t)$ are **in phase** (i.e., $\delta = 0$) or **out of phase** ($\delta = \pi$). ◀

Under these conditions, Eq. (7.50) simplifies to

$$\mathbf{E}(0, t) = (\hat{x}a_x + \hat{y}a_y)\cos(\omega t - kz), \quad (\text{in phase}) \quad (7.53a)$$

$$\mathbf{E}(0, t) = (\hat{x}a_x - \hat{y}a_y)\cos(\omega t - kz). \quad (\text{out of phase}) \quad (7.53b)$$

Let us examine the out-of-phase case. The field's magnitude is

$$|\mathbf{E}(z, t)| = [a_x^2 + a_y^2]^{1/2} |\cos(\omega t - kz)|, \quad (7.54a)$$

and the inclination angle is

$$\psi = \tan^{-1} \left(\frac{-a_y}{a_x} \right). \quad (\text{out of phase}) \quad (7.54b)$$

We note that ψ is independent of both z and t . **Figure 7-7** displays the line segment traced by the tip of $\mathbf{E}$ at $z = 0$ over a half of a cycle. The trace would be the same at any other value of z as well. At $z = 0$ and $t = 0$, $|\mathbf{E}(0, 0)| = [a_x^2 + a_y^2]^{1/2}$.

The length of the vector representing $\mathbf{E}(0, t)$ decreases to zero at $\omega t = \pi/2$. The vector then reverses direction and increases in magnitude to $[a_x^2 + a_y^2]^{1/2}$ in the second quadrant of the x - y plane at $\omega t = \pi$. Since ψ is independent of both z and t , $\mathbf{E}(z, t)$ maintains a direction along the line making an angle ψ with the x axis while oscillating back and forth across the origin.

If $a_y = 0$, then $\psi = 0^\circ$ or 180° , and the wave is x -polarized; conversely, if $a_x = 0$, then $\psi = 90^\circ$ or -90° , and the wave is y -polarized.

7-3.2 Circular Polarization

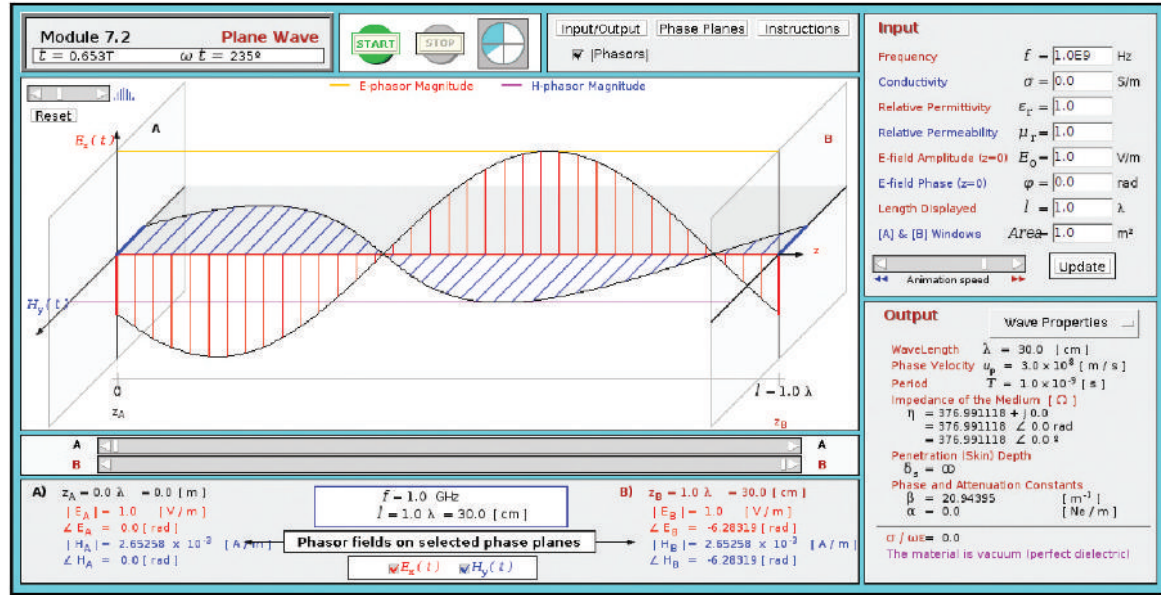
We now consider the special case when the magnitudes of the x and y components of $\tilde{\mathbf{E}}(z)$ are equal, and the phase difference $\delta = \pm\pi/2$. For reasons that become evident shortly, the wave polarization is called **left-hand circular** when $\delta = \pi/2$, and **right-hand circular** when $\delta = -\pi/2$.

Left-Hand Circular (LHC) Polarization

For $a_x = a_y = a$ and $\delta = \pi/2$, Eqs. (7.49) and (7.50) become

$$\tilde{\mathbf{E}}(z) = (\hat{x}a + \hat{y}ae^{j\pi/2})e^{-jkz} = a(\hat{x} + j\hat{y})e^{-jkz}, \quad (7.55a)$$

模块 7.2 平面波 观察沿 z 方向传播的平面波；注意 $\mathbf{E}$ 和 $\mathbf{H}$ 的时间和空间变化。检查波属性如何随着为波参数（频率和 $\mathbf{E}$ 场幅度和相位）以及介质的本构参数（ ϵ, μ, σ ）选择的值而变化。



7-3.1 线偏振

► 如果对于固定 z , $\mathbf{E}(z, t)$ 的尖端随时间描绘一条直线段，则称波是线性偏振的。当 $E_x(z, t)$ 和 $E_y(z, t)$ 同相（即 $\delta = 0$ ）或异相（ $\delta = \pi$ ）时，就会发生这种情况。 ◀

在这些条件下，方程 (7.50) 简化为

$$\mathbf{E}(0, t) = (\hat{x}a_x + \hat{y}a_y)\cos(\omega t - kz), \quad (\text{in phase}) \quad (7.53a)$$

$$\mathbf{E}(0, t) = (\hat{x}a_x - \hat{y}a_y)\cos(\omega t - kz). \quad (\text{out of phase}) \quad (7.53b)$$

让我们检查一下异相情况。场的大小为

$$|\mathbf{E}(z, t)| = [a_x^2 + a_y^2]^{1/2} |\cos(\omega t - kz)|, \quad (7.54a)$$

倾斜角为

$$\psi = \tan^{-1} \left(\frac{-a_y}{a_x} \right). \quad (\text{out of phase}) \quad (7.54b)$$

我们注意到 ψ 独立于 z 和 t 。图 7-7 显示了 $\mathbf{E}$ 尖端在 $z = 0$ 处经过半个周期所描绘的线段。对于任何其他 z 值，轨迹也将相同。在 $z = 0$ 和 $t = 0$ 处， $|\mathbf{E}(0, 0)| = [a_x^2 + a_y^2]^{1/2}$ 。

表示 $\mathbf{E}(0, t)$ 的向量长度在 $\omega t = \pi/2$ 处减小到零。然后，矢量反转方向并在 x - y 平面第二象限的 $\omega t = \pi$ 处增大幅度到 $[a_x^2 + a_y^2]^{1/2}$ 。由于 ψ 与 z 和 t 无关，因此 $\mathbf{E}(z, t)$ 保持沿与 x 轴形成角度 ψ 的线的方向，同时穿过原点来回振荡。

如果 $a_y = 0$ ，则 $\psi = 0^\circ$ 或 180° ，并且波是 x 偏振的；反之，如果 $a_x = 0$ ，则 $\psi = 90^\circ$ 或 -90° ，且波为 y 偏振。

7-3.2 圆偏振

我们现在考虑当 $\tilde{\mathbf{E}}(z)$ 的 x 和 y 分量的幅度相等且相位差 $\delta = \pm\pi/2$ 时的特殊情况。由于很快就会显而易见的原因，当 $\delta = \pi/2$ 时，波偏振称为左旋圆偏振，而当 $\delta = -\pi/2$ 时，波偏振称为右旋圆偏振。

左手圆 (LHC) 偏振

对于 $a_x = a_y = a$ 和 $\delta = \pi/2$ ，等式 (7.49) 和 (7.50) 变为

$$\tilde{\mathbf{E}}(z) = (\hat{x}a + \hat{y}ae^{j\pi/2})e^{-jkz} = a(\hat{x} + j\hat{y})e^{-jkz}, \quad (7.55a)$$

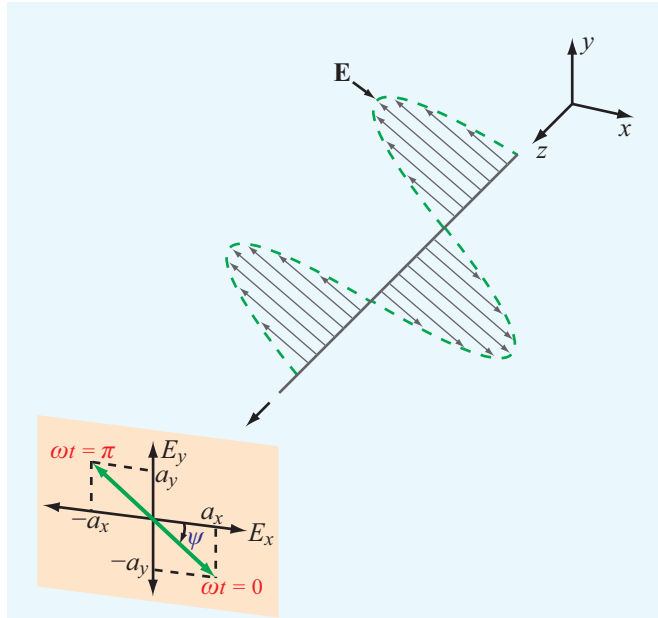


Figure 7-7 Linearly polarized wave traveling in the $+z$ direction (out of the page).

$$\begin{aligned} \mathbf{E}(z, t) &= \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right] \\ &= \hat{\mathbf{x}} a \cos(\omega t - kz) + \hat{\mathbf{y}} a \cos(\omega t - kz + \pi/2) \\ &= \hat{\mathbf{x}} a \cos(\omega t - kz) - \hat{\mathbf{y}} a \sin(\omega t - kz). \end{aligned} \quad (7.55b)$$

The corresponding field magnitude and inclination angle are

$$\begin{aligned} |\mathbf{E}(z, t)| &= [E_x^2(z, t) + E_y^2(z, t)]^{1/2} \\ &= [a^2 \cos^2(\omega t - kz) + a^2 \sin^2(\omega t - kz)]^{1/2} = a \end{aligned} \quad (7.56a)$$

and

$$\begin{aligned} \psi(z, t) &= \tan^{-1} \left[\frac{E_y(z, t)}{E_x(z, t)} \right] \\ &= \tan^{-1} \left[\frac{-a \sin(\omega t - kz)}{a \cos(\omega t - kz)} \right] = -(\omega t - kz). \end{aligned} \quad (7.56b)$$

We observe that the magnitude of $\mathbf{E}$ is independent of both z and t , whereas ψ depends on both variables. These functional dependencies are the converse of those for the linear polarization case.

At $z = 0$, Eq. (7.56b) gives $\psi = -\omega t$; the negative sign implies that the inclination angle decreases as time increases. As illustrated in **Fig. 7-8(a)**, the tip of $\mathbf{E}(t)$ traces a circle in

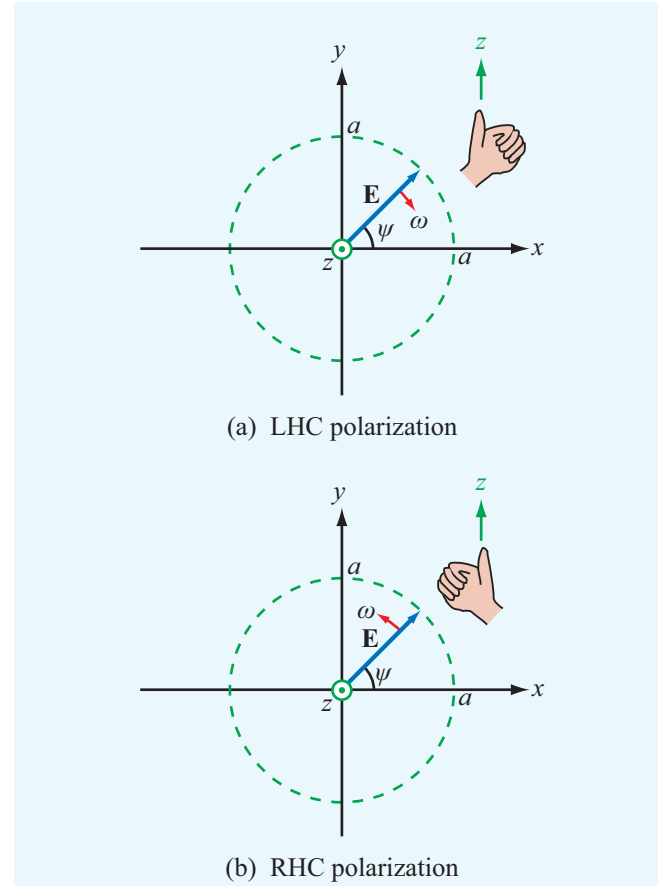


Figure 7-8 Circularly polarized plane waves propagating in the $+z$ direction (out of the page).

the x - y plane and rotates in a clockwise direction as a function of time (when viewing the wave approaching). Such a wave is called **left-hand circularly polarized** because when the thumb of the left hand points along the direction of propagation (the z direction in this case) the other four fingers point in the direction of rotation of $\mathbf{E}$.

Right-Hand Circular (RHC) Polarization

For $a_x = a_y = a$ and $\delta = -\pi/2$, we have

$$|\mathbf{E}(z, t)| = a, \quad \psi = (\omega t - kz). \quad (7.57)$$

The trace of $\mathbf{E}(0, t)$ as a function of t is shown in **Fig. 7-8(b)**. For RHC polarization, the fingers of the right hand point in the direction of rotation of $\mathbf{E}$ when the thumb is along the propagation direction. **Figure 7-9** depicts a right-hand circularly polarized wave radiated by a helical antenna.

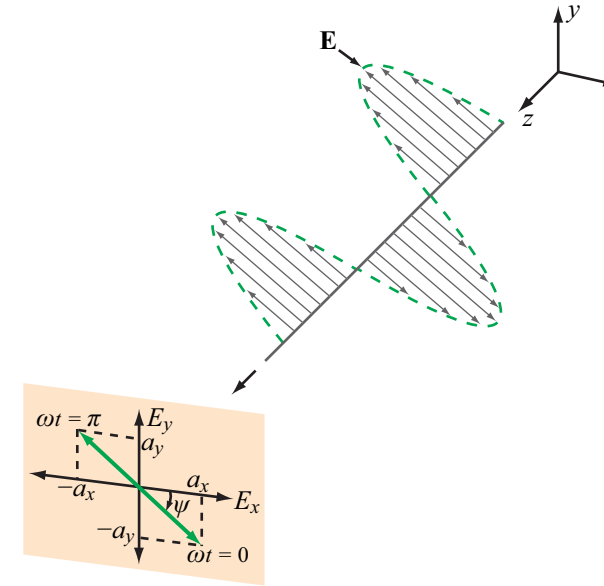


图 7-7 沿 $+z$ 方向传播的线偏振波（纸外）。

$$\begin{aligned} \mathbf{E}(z, t) &= \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right] \\ &= \hat{\mathbf{x}} a \cos(\omega t - kz) + \hat{\mathbf{y}} a \cos(\omega t - kz + \pi/2) \\ &= \hat{\mathbf{x}} a \cos(\omega t - kz) - \hat{\mathbf{y}} a \sin(\omega t - kz). \end{aligned} \quad (7.55b)$$

相应的场强和倾角为

$$\begin{aligned} |\mathbf{E}(z, t)| &= [E_x^2(z, t) + E_y^2(z, t)]^{1/2} \\ &= [a^2 \cos^2(\omega t - kz) + a^2 \sin^2(\omega t - kz)]^{1/2} = a \end{aligned} \quad (7.56a)$$

和

$$\begin{aligned} \psi(z, t) &= \tan^{-1} \left[\frac{E_y(z, t)}{E_x(z, t)} \right] \\ &= \tan^{-1} \left[\frac{-a \sin(\omega t - kz)}{a \cos(\omega t - kz)} \right] = -(\omega t - kz). \end{aligned} \quad (7.56b)$$

我们观察到 $\mathbf{E}$ 的大小与 z 和 t 无关，而 ψ 取决于这两个变量。这些函数依赖性跟线性偏振情况相反。

在 $z = 0$ 处，方程(7.56b)给出 $\psi = -\omega t$ ；负号表示倾角随着时间的增加而减小。如图7-8(a)所示， $\mathbf{E}(t)$ 的尖端在

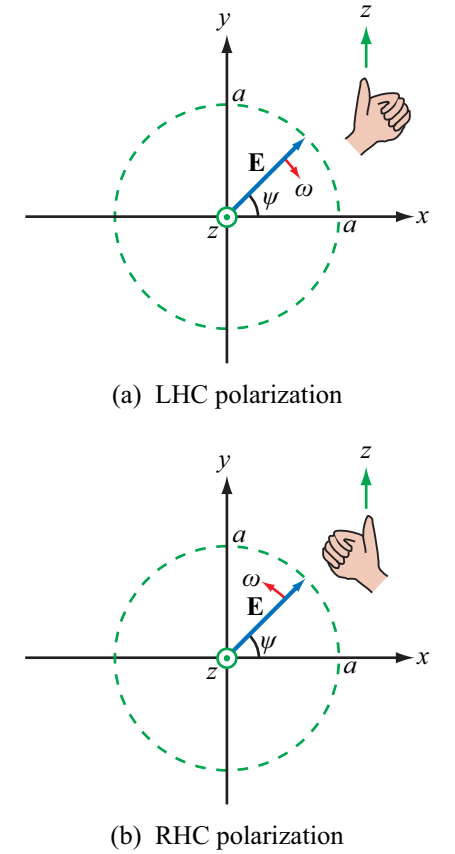


图 7-8 沿 $+z$ 方向（页面外）传播的圆偏振平面波。

x - y 平面并作为时间的函数沿顺时针方向旋转（当观察波接近时）。这种波称为左手圆偏振波，因为当左手拇指指向传播方向（本例中为 z 方向）时，其他四个手指指向 $\mathbf{E}$ 的旋转方向。

右旋圆 (RHC) 偏振

对于 $a_x = a_y = a$ 和 $\delta = -\pi/2$ ，我们有

$$|\mathbf{E}(z, t)| = a, \quad \psi = (\omega t - kz). \quad (7.57)$$

$\mathbf{E}(0, t)$ 作为 t 函数的轨迹如图 7-8(b) 所示。对于 RHC 偏振，当拇指沿传播方向时，右手手指指向 $\mathbf{E}$ 的旋转方向。图 7-9 描绘了螺旋天线辐射的右旋圆极化波。

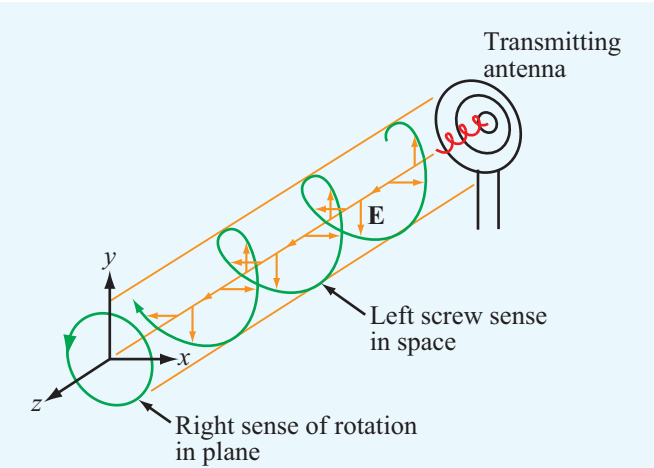


Figure 7-9 Right-hand circularly polarized wave radiated by a helical antenna.

► **Polarization handedness** is defined in terms of the rotation of $\mathbf{E}$ as a function of time in a fixed plane orthogonal to the direction of propagation, which is opposite the direction of rotation of $\mathbf{E}$ as a function of distance at a fixed point in time. ◀

Example 7-2: RHC-Polarized Wave

An RHC-polarized plane wave with electric field magnitude of 3 (mV/m) is traveling in the +y direction in a dielectric medium with $\epsilon = 4\epsilon_0$, $\mu = \mu_0$, and $\sigma = 0$. If the frequency is 100 MHz, obtain expressions for $\mathbf{E}(y, t)$ and $\mathbf{H}(y, t)$.

Solution: Since the wave is traveling in the +y direction, its field must have components along the x and z directions. The rotation of $\mathbf{E}(y, t)$ is depicted in **Fig. 7-10**, where $\hat{\mathbf{y}}$ is out of the page. By comparison with the RHC-polarized wave shown in **Fig. 7-8(b)**, we assign the z component of $\tilde{\mathbf{E}}(y)$ a phase angle of zero and the x component a phase shift of $\delta = -\pi/2$. Both components have magnitudes of $a = 3$ (mV/m). Hence,

$$\begin{aligned}\tilde{\mathbf{E}}(y) &= \hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{z}}\tilde{E}_z = \hat{\mathbf{x}}ae^{-j\pi/2}e^{-jky} + \hat{\mathbf{z}}ae^{-jky} \\ &= (-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky} \quad (\text{mV/m}),\end{aligned}$$

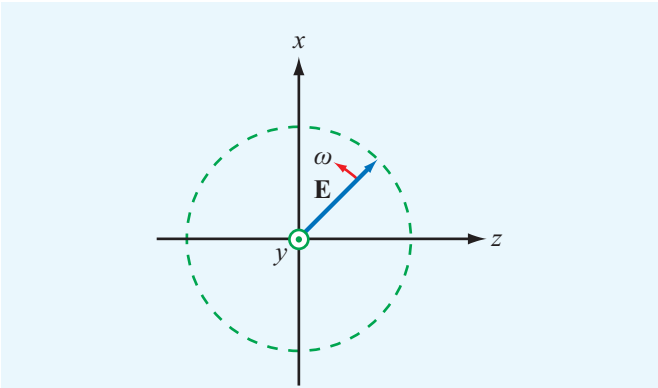


Figure 7-10 Right-hand circularly polarized wave of Example 7-2.

and application of Eq. (7.39a) gives

$$\begin{aligned}\tilde{\mathbf{H}}(y) &= \frac{1}{\eta} \hat{\mathbf{y}} \times \tilde{\mathbf{E}}(y) = \frac{1}{\eta} \hat{\mathbf{y}} \times (-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky} \\ &= \frac{3}{\eta} (\hat{\mathbf{z}}j + \hat{\mathbf{x}})e^{-jky} \quad (\text{mA/m}).\end{aligned}$$

With $\omega = 2\pi f = 2\pi \times 10^8$ (rad/s), the wavenumber k is

$$k = \frac{\omega\sqrt{\epsilon_r}}{c} = \frac{2\pi \times 10^8 \sqrt{4}}{3 \times 10^8} = \frac{4}{3}\pi \quad (\text{rad/m}),$$

and the intrinsic impedance η is

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \approx \frac{120\pi}{\sqrt{4}} = 60\pi \quad (\Omega).$$

The instantaneous fields $\mathbf{E}(y, t)$ and $\mathbf{H}(y, t)$ are

$$\begin{aligned}\mathbf{E}(y, t) &= \Re\left[\tilde{\mathbf{E}}(y) e^{j\omega t}\right] \\ &= \Re\left[(-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky} e^{j\omega t}\right] \\ &= 3[\hat{\mathbf{x}}\sin(\omega t - ky) + \hat{\mathbf{z}}\cos(\omega t - ky)] \quad (\text{mV/m})\end{aligned}$$

and

$$\begin{aligned}\mathbf{H}(y, t) &= \Re\left[\tilde{\mathbf{H}}(y) e^{j\omega t}\right] \\ &= \Re\left[\frac{3}{\eta}(\hat{\mathbf{z}}j + \hat{\mathbf{x}})e^{-jky} e^{j\omega t}\right] \\ &= \frac{1}{20\pi}[\hat{\mathbf{x}}\cos(\omega t - ky) - \hat{\mathbf{z}}\sin(\omega t - ky)] \quad (\text{mA/m}).\end{aligned}$$

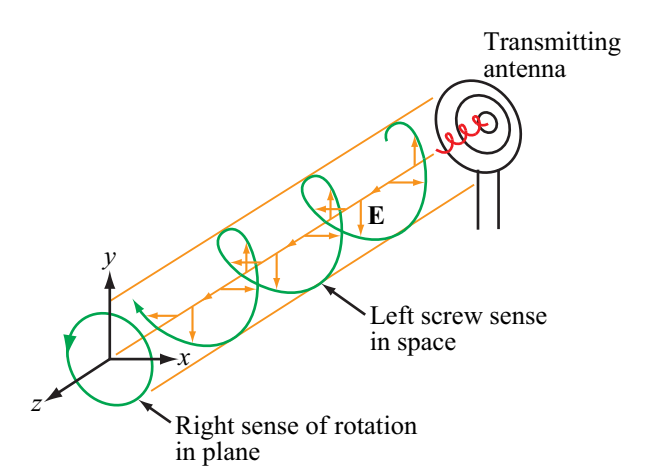


图 7-9 螺旋天线辐射的右旋圆极化波。

► 偏振旋向性是根据 E 在与传播方向正交的固定平面上随时间变化的旋转来定义的，这与 E 在固定时间点随距离变化的旋转方向相反。 ◀

例 7-2：RHC 偏振波

电场强度为 3 (mV/m) 的 RHC 极化平面波在 $\epsilon = 4\epsilon_0$ 、 $\mu = \mu_0$ 和 $\sigma = 0$ 的介电介质中沿 +y 方向传播。如果频率为 100 MHz，则获得 $\mathbf{E}(y, t)$ 和 $\mathbf{H}(y, t)$ 的表达式。

解：由于波沿 +y 方向传播，因此其场必须具有沿 x 和 z 方向的分量。E(y,t) 的旋转如图 7-10 所示，其中 y 超出了页面。通过与图 7-8(b) 所示的 RHC 偏振波进行比较，我们将 $\tilde{\mathbf{E}}(y)$ 的 z 分量指定为 0 相位角，将 x 分量指定为 $\delta = -\pi/2$ 的相移。两个分量的幅度均为 3 (mV/m)。因此，

$$\begin{aligned}\tilde{\mathbf{E}}(y) &= \hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{z}}\tilde{E}_z = \hat{\mathbf{x}}ae^{-j\pi/2}e^{-jky} + \hat{\mathbf{z}}ae^{-jky} \\ &= (-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky} \quad (\text{mV/m}),\end{aligned}$$

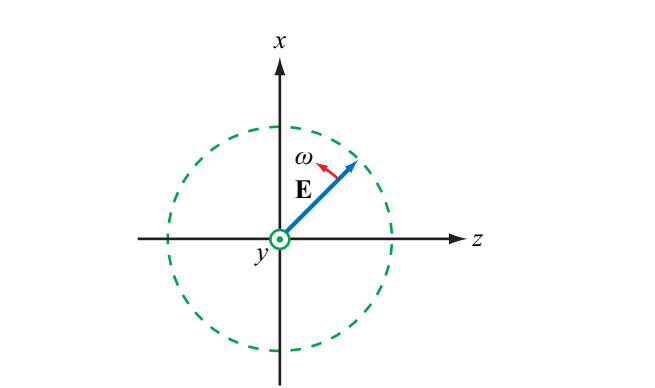


图 7-10 例 7-2 的右旋圆偏振波。

和方程的应用。(7.39a) 给出

$$\begin{aligned}\tilde{\mathbf{H}}(y) &= \frac{1}{\eta} \hat{\mathbf{y}} \times \tilde{\mathbf{E}}(y) = \frac{1}{\eta} \hat{\mathbf{y}} \times (-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky} \\ &= \frac{3}{\eta} (\hat{\mathbf{z}}j + \hat{\mathbf{x}})e^{-jky} \quad (\text{mA/m}).\end{aligned}$$

对于 $\omega = 2\pi f = 2\pi \times 10^8$ (rad/s)，波数 k 为

$$k = \frac{\omega\sqrt{\epsilon_r}}{c} = \frac{2\pi \times 10^8 \sqrt{4}}{3 \times 10^8} = \frac{4}{3}\pi \quad (\text{rad/m}),$$

固有阻抗 η 为

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \approx \frac{120\pi}{\sqrt{4}} = 60\pi \quad (\Omega).$$

瞬时场 $\mathbf{E}(y, t)$ 和 $\mathbf{H}(y, t)$ 为

$$\begin{aligned}\mathbf{E}(y, t) &= \Re\left[\tilde{\mathbf{E}}(y) e^{j\omega t}\right] \\ &= \Re\left[(-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky} e^{j\omega t}\right] \\ &= 3[\hat{\mathbf{x}}\sin(\omega t - ky) + \hat{\mathbf{z}}\cos(\omega t - ky)] \quad (\text{mV/m})\end{aligned}$$

和

$$\begin{aligned}\mathbf{H}(y, t) &= \Re\left[\tilde{\mathbf{H}}(y) e^{j\omega t}\right] \\ &= \Re\left[\frac{3}{\eta}(\hat{\mathbf{z}}j + \hat{\mathbf{x}})e^{-jky} e^{j\omega t}\right] \\ &= \frac{1}{20\pi}[\hat{\mathbf{x}}\cos(\omega t - ky) - \hat{\mathbf{z}}\sin(\omega t - ky)] \quad (\text{mA/m}).\end{aligned}$$

Technology Brief 13: RFID Systems

In 1973, two separate patents were issued in the United States for Radio Frequency Identification (RFID) concepts. The first, granted to Mario Cardullo, was for an **active RFID tag** with rewritable memory. An active tag has a power source (such as a battery) of its own, whereas a **passive RFID tag** does not. The second patent was granted to Charles Walton, who proposed the use of a passive tag for keyless entry (unlocking a door without a key). Shortly thereafter, a passive RFID tag was developed for tracking cattle (**Fig. TF13-1**), and the technology rapidly expanded into many commercial enterprises—from tracking vehicles and consumer products to supply chain management and automobile anti-theft systems.

RFID System Overview

In an RFID system, communication occurs between a **reader**—which actually is a **transceiver**—and a **tag** (**Fig. TF13-2**). When *interrogated* by the reader, a tag responds with information about its identity, as well as other relevant information depending on the specific application.

► The tag is, in essence, a **transponder** commanded by the reader. ◀

The functionality and associated capabilities of the RFID tag depend on two important attributes: (a) whether the tag is of the active or passive type and (b) the tag's operating frequency. Usually the RFID tag remains dormant (*asleep*) until activated by an electromagnetic signal radiated by the reader's antenna. The magnetic field of the EM signal induces a current in the coil contained in the tag's circuit (**Fig. TF13-3**). For a passive tag, the induced current has to be sufficient to generate the power necessary to activate the chip as well as to transmit the response to the reader.

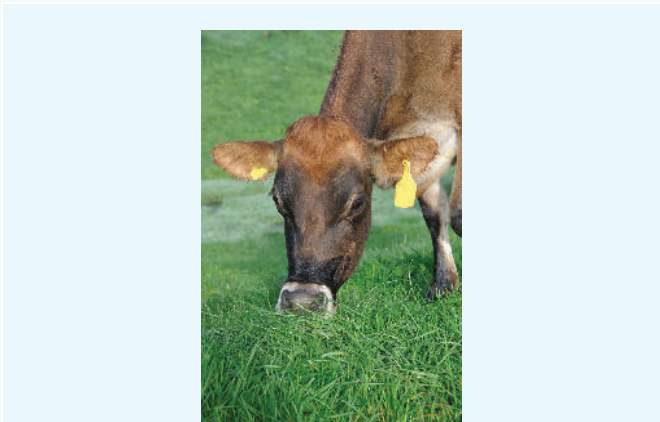


Figure TF13-1 Passive RFID tags were developed in the 1970s for tracking cows.

► Passive RFID systems are limited to short **read ranges** (between reader and tag) on the order of 30 cm to 3 m, depending on the system's frequency band (as noted in **Table TT13-1**). ◀

The obvious advantage of active RFID systems is that they can operate over greater distances and do not require reception of a signal from the reader's antenna to get activated. However, active tags are significantly more expensive to fabricate than their passive cousins.

RFID Frequency Bands

Table TT13-1 provides a comparison among the four frequency bands commonly used for RFID systems. Generally speaking, the higher-frequency tags can operate over longer read ranges and can carry higher data rates, but they are more expensive to fabricate.

技术简介 13: RFID 系统

1973 年，美国就射频识别 (RFID) 概念颁发了两项单独的专利。第一个授予 Mario Cardullo，用于具有可重写内存的有源 RFID 标签。有源标签有自己的电源（例如电池），而无源 RFID 标签则没有。第二项专利授予查尔斯·沃尔顿 (Charles Walton)，他提议使用无源标签进行无钥匙进入（无需钥匙即可解锁门）。此后不久，用于跟踪牛的无源 RFID 标签被开发出来（图 TF13-1），并且该技术迅速扩展到许多商业企业——从跟踪车辆和消费品到供应链管理和汽车防盗系统。

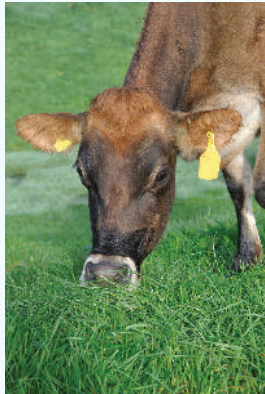


图 TF13-1 无源 RFID 标签于 20 世纪 70 年代开发，用于跟踪奶牛。

RFID系统概述

在 RFID 系统中，通信发生在读取器（实际上是收发器）和标签之间（图 TF13-2）。当阅读器询问时，标签会响应有关其身份的信息，以及取决于特定应用的其他相关信息。

► 标签本质上是一个由阅读器控制的应答器。 ◀

RFID 标签的功能和相关能力取决于两个重要属性：(a) 标签是有源类型还是无源类型，以及 (b) 标签的工作频率。通常，RFID 标签保持休眠（睡眠）状态，直到被读取器天线辐射的电磁信号激活。EM 信号的磁场会在标签电路中的线圈中感应出电流（图 TF13-3）。对于无源标签，感应电流必须足以产生激活芯片以及将响应传输到读取器所需的功率。

► 无源 RFID 系统仅限于 30 cm 至 3 m 左右的短读取范围（读取器和标签之间），具体取决于系统的频段（如表 TT13-1 中所述）。 ◀

有源 RFID 系统的明显优势是它们可以在更远的距离上运行，并且不需要接收来自阅读器天线的信号即可激活。然而，有源标签的制造成本明显高于无源标签。

RFID 频段

表 TT13-1 提供了 RFID 系统常用的四个频段之间的比较。一般来说，高频标签可以在更长的读取范围内运行，并且可以承载更高的数据速率，但它们的制造成本更高。

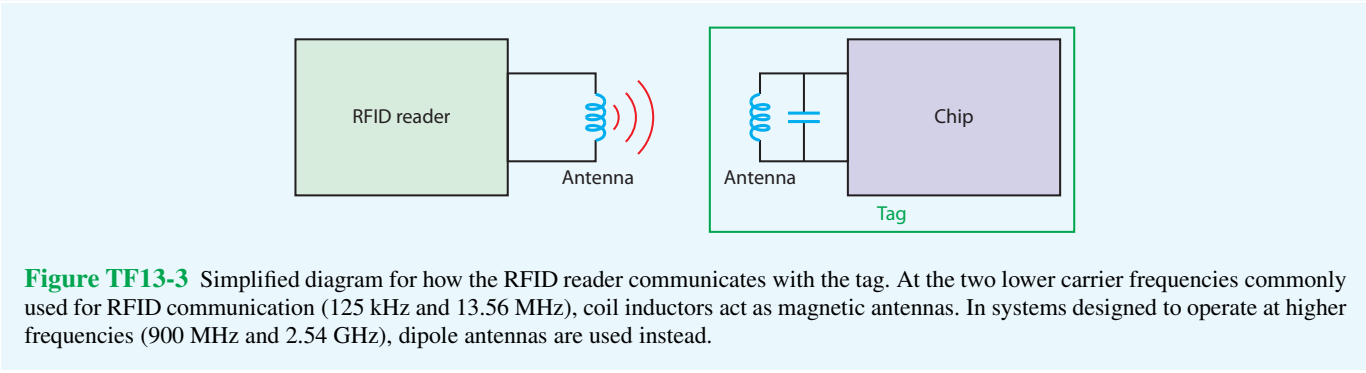
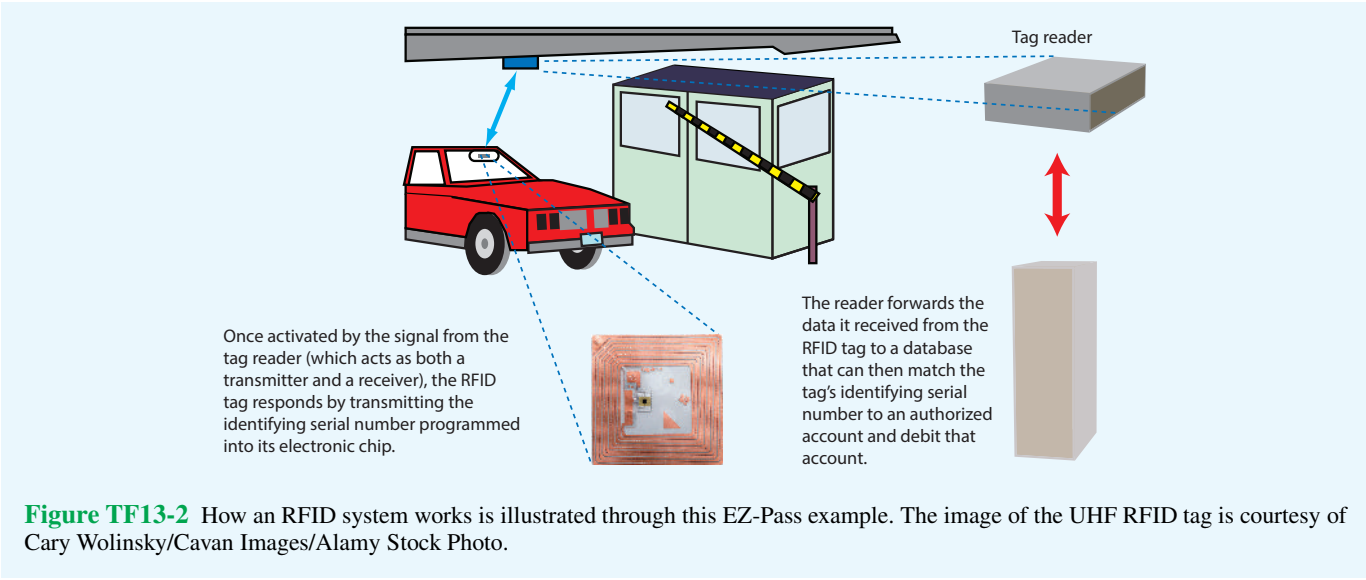


Table TT13-1 Comparison of RFID frequency bands.				
Band	LF	HF	UHF	Microwave
RFID frequency	125–134 kHz	13.56 MHz	865–956 MHz	2.45 GHz
Read range	≤ 0.5 m	≤ 1.5 m	≤ 5 m	≤ 10 m
Data rate	1 kbit/s	25 kbit/s	30 kbit/s	100 kbit/s
Typical applications	• Animal ID • Automobile key/antitheft • Access control	• Smart cards • Article surveillance • Airline baggage tracking • Library book tracking	• Supply chain management • Logistics	• Vehicle toll collection • Railroad car monitoring

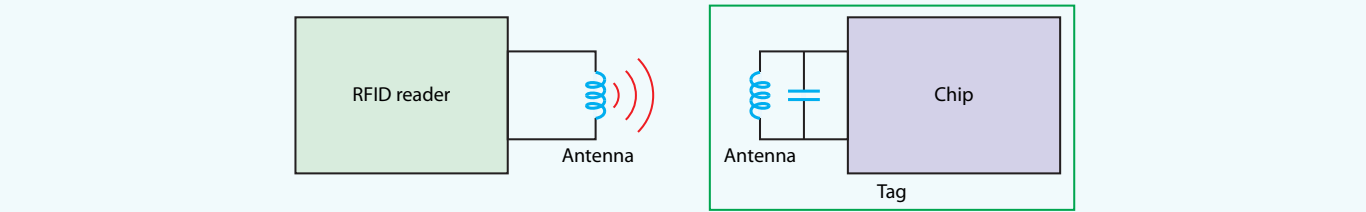
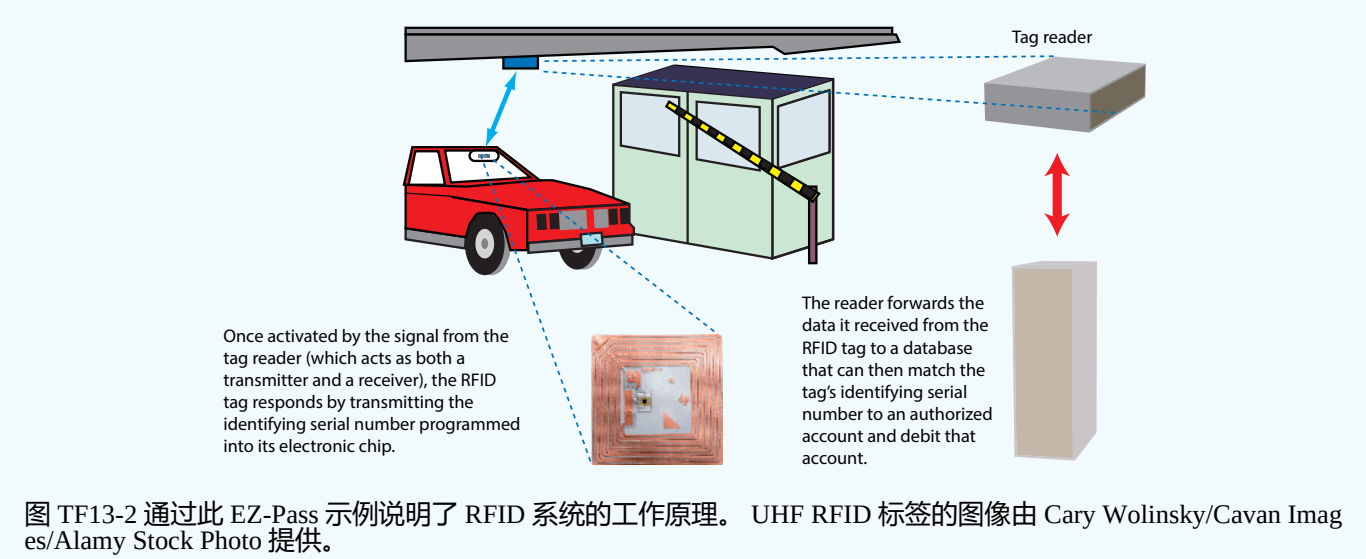


图 TF13-3 RFID 阅读器如何与标签通信的简化图。在 RFID 通信用两个较低载波频率（125 kHz 和 13.56 MHz）下，线圈电感器充当磁性天线。在设计为在较高频率（900 MHz 和 2.54 GHz）下运行的系统中，改为使用偶极天线。

表TT13-1 RFID频段比较。				
Band	LF	HF	UHF	Microwave
RFID frequency	125–134 kHz	13.56 MHz	865–956 MHz	2.45 GHz
Read range	≤ 0.5 m	≤ 1.5 m	≤ 5 m	≤ 10 m
Data rate	1 kbit/s	25 kbit/s	30 kbit/s	100 kbit/s
Typical applications	• Animal ID • Automobile key/antitheft • Access control	• Smart cards • Article surveillance • Airline baggage tracking • Library book tracking	• Supply chain management • Logistics	• Vehicle toll collection • Railroad car monitoring

7-3.3 Elliptical Polarization

Plane waves that are not linearly or circularly polarized are elliptically polarized. That is, the tip of $\mathbf{E}(z, t)$ traces an ellipse in the plane perpendicular to the direction of propagation. The shape of the ellipse and the field's handedness (left-hand or right-hand) are determined by the values of the ratio (a_y/a_x) and the phase difference δ .

The polarization ellipse shown in **Fig. 7-11** has its major axis with length a_ξ along the ξ direction and its minor axis with length a_η along the η direction. The **rotation angle** γ is defined as the angle between the major axis of the ellipse and a reference direction, which is chosen here to be the x axis, with γ being bounded within the range $-\pi/2 \leq \gamma \leq \pi/2$. The shape of the ellipse and its handedness are characterized by the **ellipticity angle** χ , which is defined as

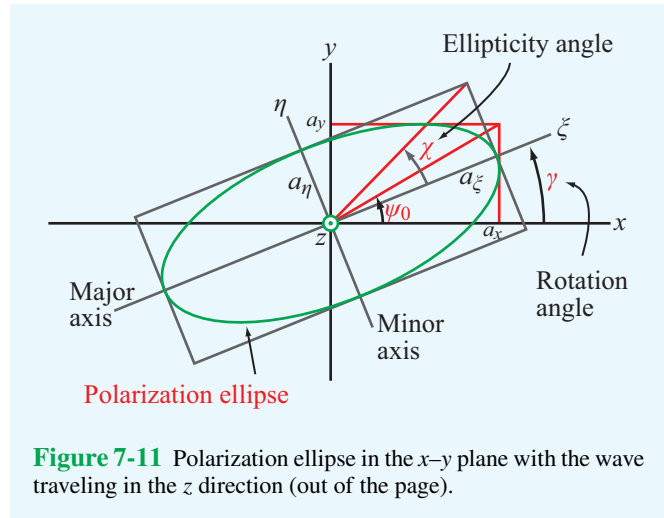
$$\tan \chi = \pm \frac{a_\eta}{a_\xi} = \pm \frac{1}{R}, \quad (7.58)$$

with the plus sign corresponding to left-handed rotation and the minus sign corresponding to right-handed rotation. The limits for χ are $-\pi/4 \leq \chi \leq \pi/4$. The quantity $R = a_\xi/a_\eta$ is called the **axial ratio** of the polarization ellipse, and it varies between 1 for circular polarization and ∞ for linear polarization. The polarization angles γ and χ are related to the wave parameters a_x , a_y , and δ by*

$$\tan 2\gamma = (\tan 2\psi_0) \cos \delta \quad (-\pi/2 \leq \gamma \leq \pi/2), \quad (7.59a)$$

$$\sin 2\chi = (\sin 2\psi_0) \sin \delta \quad (-\pi/4 \leq \chi \leq \pi/4), \quad (7.59b)$$

*From M. Born and E. Wolf, *Principles of Optics*, New York: Macmillan, 1965, p. 27.



where ψ_0 is an **auxiliary angle** defined by

$$\tan \psi_0 = \frac{a_y}{a_x} \quad \left(0 \leq \psi_0 \leq \frac{\pi}{2}\right). \quad (7.60)$$

Sketches of the polarization ellipse are shown in **Fig. 7-12** for various combinations of the angles (γ, χ) . The ellipse reduces to a circle for $\chi = \pm 45^\circ$ and to a line for $\chi = 0$.

► Positive values of χ , corresponding to $\sin \delta > 0$, are associated with left-handed rotation, and negative values of χ , corresponding to $\sin \delta < 0$, are associated with right-handed rotation. ◀

Since the magnitudes a_x and a_y are, by definition, nonnegative numbers, the ratio a_y/a_x may vary between zero for an x -polarized linear polarization and ∞ for a y -polarized linear polarization. Consequently, the angle ψ_0 is limited to the range $0 \leq \psi_0 \leq 90^\circ$. Application of Eq. (7.59a) leads to two possible solutions for the value of γ , both of which fall within the defined range from $-\pi/2$ to $\pi/2$. The correct choice is governed by the following rule:

$$\begin{aligned} \gamma &> 0 \text{ if } \cos \delta > 0, \\ \gamma &< 0 \text{ if } \cos \delta < 0. \end{aligned}$$

► In summary, the sign of the rotation angle γ is the same as the sign of $\cos \delta$, and the sign of the ellipticity angle χ is the same as the sign of $\sin \delta$. ◀

Example 7-3: Polarization State

Determine the polarization state of a plane wave with electric field

$$\begin{aligned} \mathbf{E}(z, t) = & \hat{\mathbf{x}} 3 \cos(\omega t - kz + 30^\circ) \\ & - \hat{\mathbf{y}} 4 \sin(\omega t - kz + 45^\circ) \quad (\text{mV/m}). \end{aligned}$$

Solution: We begin by converting the second term to a cosine reference:

$$\begin{aligned} \mathbf{E} = & \hat{\mathbf{x}} 3 \cos(\omega t - kz + 30^\circ) - \hat{\mathbf{y}} 4 \cos(\omega t - kz + 45^\circ - 90^\circ) \\ = & \hat{\mathbf{x}} 3 \cos(\omega t - kz + 30^\circ) - \hat{\mathbf{y}} 4 \cos(\omega t - kz - 45^\circ). \end{aligned}$$

The corresponding field phasor $\tilde{\mathbf{E}}(z)$ is

$$\begin{aligned} \tilde{\mathbf{E}}(z) = & \hat{\mathbf{x}} 3 e^{-jkz} e^{j30^\circ} - \hat{\mathbf{y}} 4 e^{-jkz} e^{-j45^\circ} \\ = & \hat{\mathbf{x}} 3 e^{-jkz} e^{j30^\circ} + \hat{\mathbf{y}} 4 e^{-jkz} e^{-j45^\circ} e^{j180^\circ} \\ = & \hat{\mathbf{x}} 3 e^{-jkz} e^{j30^\circ} + \hat{\mathbf{y}} 4 e^{-jkz} e^{j135^\circ}, \end{aligned}$$

7-3.3 椭圆偏振

非线偏振或圆偏振的平面波为椭圆偏振。即， $\mathbf{E}(z, t)$ 的尖端在垂直于传播方向的平面中描绘出椭圆形。椭圆的形状和场的旋向（左旋或右旋）由比率 (a_y/a_x) 和相位差 δ 的值决定。

图7-11所示的偏振椭圆的长轴沿 ξ 方向长度为 a_ξ ，短轴沿 η 方向长度为 a_η 。旋转角 γ 定义为椭圆长轴与参考方向（此处选择为 x 轴）之间的角度，其中 γ 被限制在范围 $-\pi/2 \leq \gamma \leq \pi/2$ 内。椭圆的形状及其旋向由椭圆度角 χ 来表征，其定义为

$$\tan \chi = \pm \frac{a_\eta}{a_\xi} = \pm \frac{1}{R}, \quad (7.58)$$

其中加号对应于左手旋转，减号对应于右手旋转。 χ 的限制为 $-\pi/4 \leq \chi \leq \pi/4$ 。 $R = a_\xi/a_\eta$ 称为偏振椭圆的轴比，它在圆偏振的1和线偏振的 ∞ 之间变化。偏振角 γ 和 χ 与波参数 a_x 、 a_y 和 δ 通过*相关

$$\tan 2\gamma = (\tan 2\psi_0) \cos \delta \quad (-\pi/2 \leq \gamma \leq \pi/2), \quad (7.59a)$$

$$\sin 2\chi = (\sin 2\psi_0) \sin \delta \quad (-\pi/4 \leq \chi \leq \pi/4), \quad (7.59b)$$

*来自 M. Born 和 E. Wolf, 《光学原理》，纽约：Macmillan, 1965 年，第 14 页。27.

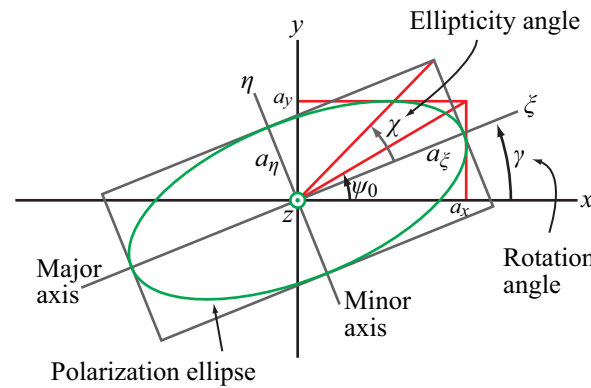


图 7-11 x - y 平面中的偏振椭圆，波沿 z 方向传播（页面外）。

其中 ψ_0 是由下式定义的辅助角

$$\tan \psi_0 = \frac{a_y}{a_x} \quad \left(0 \leq \psi_0 \leq \frac{\pi}{2}\right). \quad (7.60)$$

各种角度组合 (γ, χ) 的偏振椭圆草图如图 7-12 所示。对于 $\chi = \pm 45^\circ$ ，椭圆缩小为圆形；对于 $\chi = 0$ ，椭圆缩小为直线。

► χ 的正值对应于 $\sin \delta > 0$ ，与左手旋转相关联，而 χ 的负值对应于 $\sin \delta < 0$ ，与右手旋转相关联。◀

由于根据定义，幅度 a_x 和 a_y 是非负数，因此比率 a_y/a_x 可以在 x 偏振线性偏振的零和 y 偏振线性偏振的 ∞ 之间变化。因此，角度 ψ_0 被限制在 $0 \leq \psi_0 \leq 90^\circ$ 的范围内。方程的应用(7.59a)得出 γ 值的两个可能的解，它们都落在从 $-\pi/2$ 到 $\pi/2$ 的定义范围内。正确的选择遵循以下规则：

$$\begin{aligned} \gamma &> 0 \text{ if } \cos \delta > 0, \\ \gamma &< 0 \text{ if } \cos \delta < 0. \end{aligned}$$

► 综上所述，旋转角 γ 的符号与 $\cos \delta$ 的符号相同，椭圆角 χ 的符号与 $\sin \delta$ 的符号相同。◀

例 7-3：偏振态

确定电场中平面波的偏振态

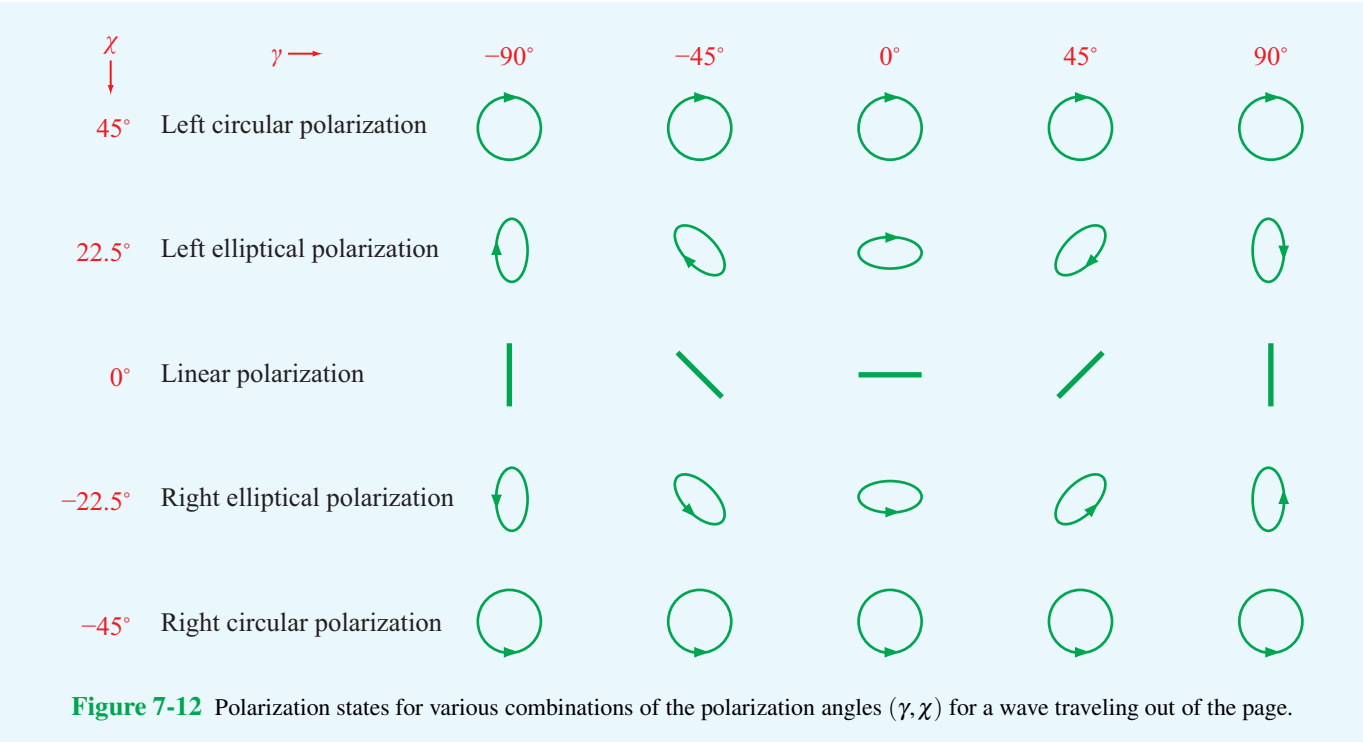
$$\begin{aligned} \mathbf{E}(z, t) = & \hat{\mathbf{x}} 3 \cos(\omega t - kz + 30^\circ) \\ & - \hat{\mathbf{y}} 4 \sin(\omega t - kz + 45^\circ) \quad (\text{mV/m}). \end{aligned}$$

解决方案：我们首先将第二项转换为余弦参考：

$$\begin{aligned} \mathbf{E} = & \hat{\mathbf{x}} 3 \cos(\omega t - kz + 30^\circ) - \hat{\mathbf{y}} 4 \cos(\omega t - kz + 45^\circ - 90^\circ) \\ = & \hat{\mathbf{x}} 3 \cos(\omega t - kz + 30^\circ) - \hat{\mathbf{y}} 4 \cos(\omega t - kz - 45^\circ). \end{aligned}$$

相应的场相量 $\tilde{\mathbf{E}}(z)$ 为

$$\begin{aligned} \tilde{\mathbf{E}}(z) = & \hat{\mathbf{x}} 3 e^{-jkz} e^{j30^\circ} - \hat{\mathbf{y}} 4 e^{-jkz} e^{-j45^\circ} \\ = & \hat{\mathbf{x}} 3 e^{-jkz} e^{j30^\circ} + \hat{\mathbf{y}} 4 e^{-jkz} e^{-j45^\circ} e^{j180^\circ} \\ = & \hat{\mathbf{x}} 3 e^{-jkz} e^{j30^\circ} + \hat{\mathbf{y}} 4 e^{-jkz} e^{j135^\circ}, \end{aligned}$$



where we have replaced the negative sign of the second term with e^{j180° in order to have positive amplitudes for both terms, thereby allowing us to use the definitions given earlier. According to the expression for $\tilde{\mathbf{E}}(z)$, the phase angles of the x and y components are $\delta_x = 30^\circ$ and $\delta_y = 135^\circ$, giving a phase difference of $\delta = \delta_y - \delta_x = 135^\circ - 30^\circ = 105^\circ$. The auxiliary angle ψ_0 is obtained from

$$\psi_0 = \tan^{-1} \left(\frac{a_y}{a_x} \right) = \tan^{-1} \left(\frac{4}{3} \right) = 53.1^\circ.$$

From Eq. (7.59a),

$$\tan 2\gamma = (\tan 2\psi_0) \cos \delta = \tan 106.2^\circ \cos 105^\circ = 0.89,$$

which gives two solutions for γ , namely $\gamma = 20.8^\circ$ and $\gamma = -69.2^\circ$. Since $\cos \delta < 0$, the correct value of γ is -69.2° . From Eq. (7.59b),

$$\begin{aligned} \sin 2\chi &= (\sin 2\psi_0) \sin \delta \\ &= \sin 106.2^\circ \sin 105^\circ = 0.93 \quad \text{or} \quad \chi = 34.0^\circ. \end{aligned}$$

The magnitude of χ indicates that the wave is elliptically polarized, and its positive polarity specifies its rotation as left handed.

Concept Question 7-4: An elliptically polarized wave is characterized by amplitudes a_x and a_y and by the phase difference δ . If a_x and a_y are both nonzero, what should δ be in order for the polarization state to reduce to linear polarization?

Concept Question 7-5: Which of the following two descriptions defines an RHC-polarized wave: A wave incident upon an observer is RHC-polarized if its electric field appears to the observer to rotate in a counterclockwise direction (a) as a function of time in a fixed plane perpendicular to the direction of wave travel or (b) as a function of travel distance at a fixed time t ?

Exercise 7-5: The electric field of a plane wave is given by

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 3 \cos(\omega t - kz) + \hat{\mathbf{y}} 4 \cos(\omega t - kz) \quad (\text{V/m}).$$

Determine (a) the polarization state, (b) the modulus of $\mathbf{E}$, and (c) the auxiliary angle.

Answer: (a) Linear, (b) $|\mathbf{E}| = 5 \cos(\omega t - kz)$ (V/m), (c) $\psi_0 = 53.1^\circ$. (See [EM](#).)

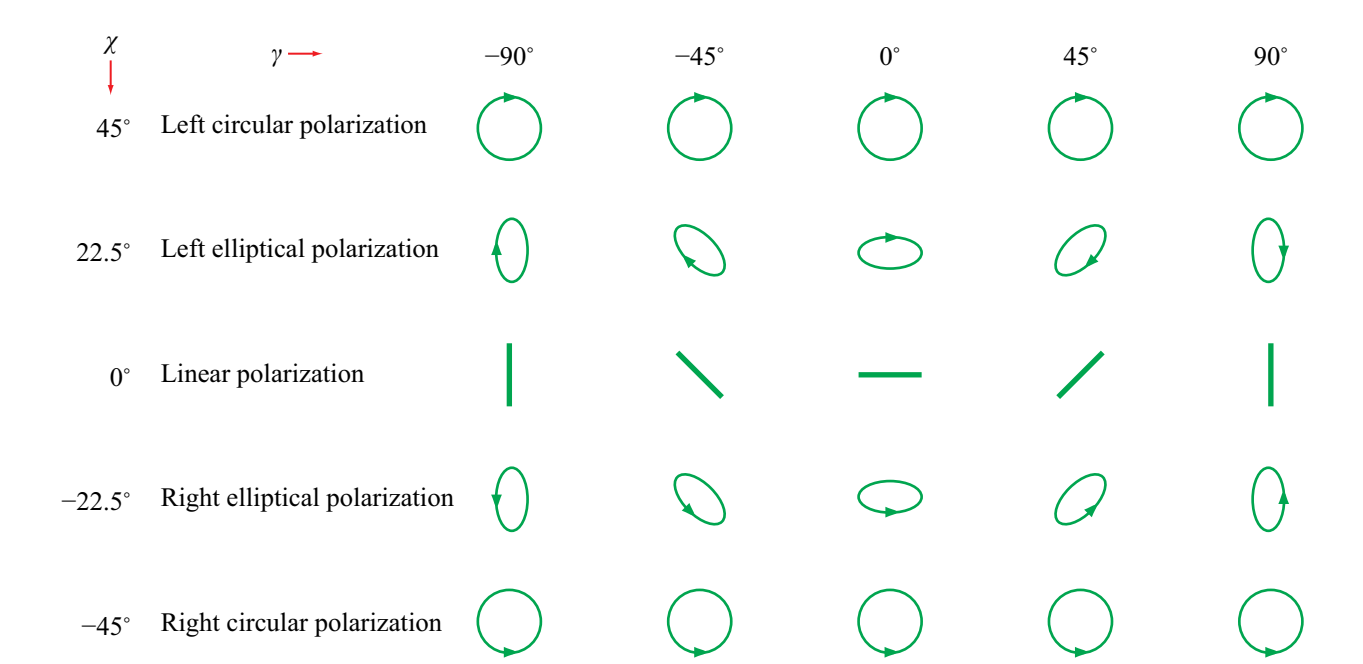


图 7-12 传播到页面之外的波的偏振角 (γ 、 χ) 的各种组合的偏振状态。

其中我们用 e^{j180° 替换了第二项的负号，以便两项都有正振幅，从而允许我们使用前面给出的定义。根据 $\tilde{\mathbf{E}}(z)$ 的表达式， x 和 y 分量的相位角为 $\delta_x = 30^\circ$ 和 $\delta_y = 135^\circ$ ，相位差为 $\delta = \delta_y - \delta_x = 135^\circ - 30^\circ = 105^\circ$ 。辅助角 ψ_0 由下式获得

$$\psi_0 = \tan^{-1} \left(\frac{a_y}{a_x} \right) = \tan^{-1} \left(\frac{4}{3} \right) = 53.1^\circ.$$

从方程。 (7.59a),

$$\tan 2\gamma = (\tan 2\psi_0) \cos \delta = \tan 106.2^\circ \cos 105^\circ = 0.89,$$

它给出了 γ 的两个解，即 $\gamma = 20.8^\circ$ 和 $\gamma = -69.2^\circ$ 。由于 $\cos \delta < 0$ ， γ 的正确值为 -69.2° 。从方程。 (7.59b),

$$\begin{aligned} \sin 2\chi &= (\sin 2\psi_0) \sin \delta \\ &= \sin 106.2^\circ \sin 105^\circ = 0.93 \quad \text{or} \quad \chi = 34.0^\circ. \end{aligned}$$

χ 的大小表明波是椭圆偏振的，其正极性指定其旋转为左手。

概念问题 7-4：椭圆偏振波的特征是振幅 a_x 和 a_y 以及相位差 δ 。如果 a_x 和 a_y 都不为零，那么 δ 应该是多少才能使偏振态降低为线偏振？

概念问题 7-5：以下哪两个描述定义了 RHC 偏振波：如果观察者认为入射到观察者的波的电场沿逆时针方向旋转，则该波是 RHC 偏振的 (a) 作为垂直于波传播方向的固定平面中时间的函数，或 (b) 作为固定时间 t 处传播距离的函数？

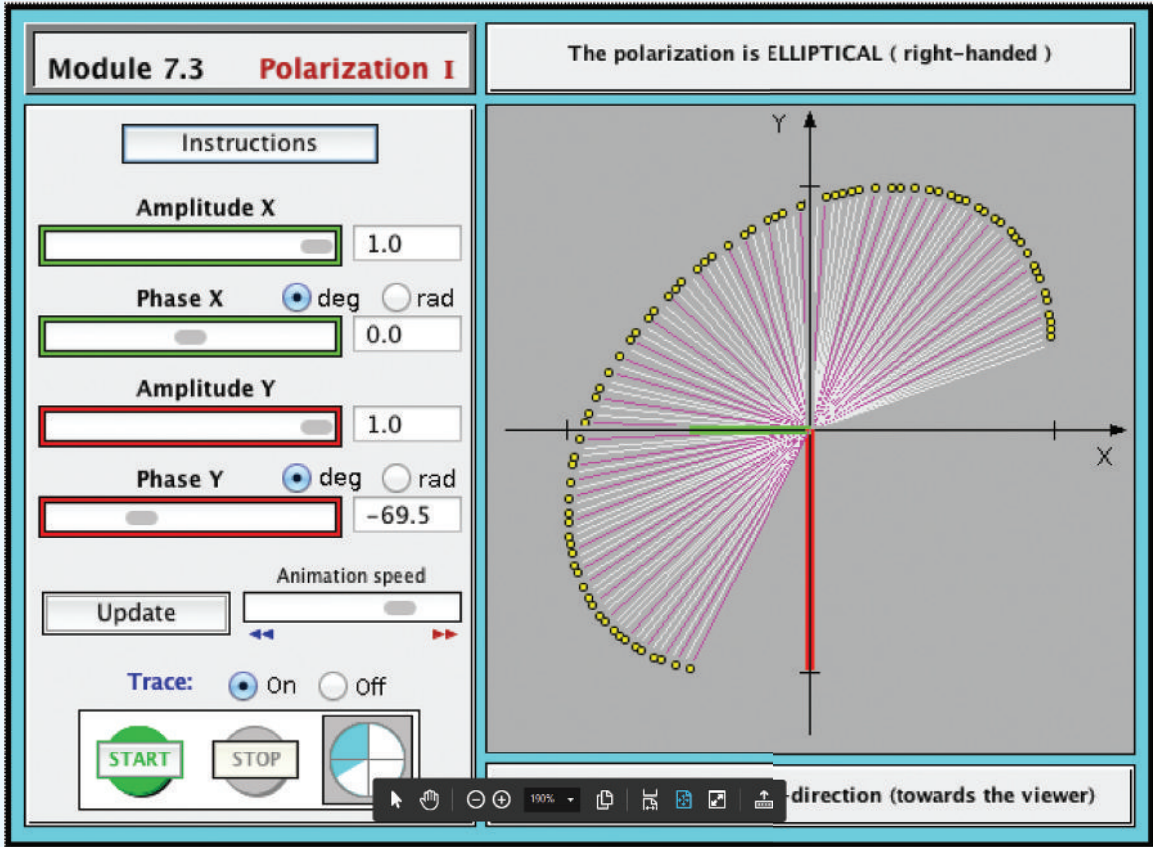
练习7-5：平面波的电场由下式给出

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 3 \cos(\omega t - kz) + \hat{\mathbf{y}} 4 \cos(\omega t - kz) \quad (\text{V/m}).$$

确定 (a) 偏振态、(b) E 模量和 (c) 辅助角。

答案：(a) 线性，(b) $|\mathbf{E}| = 5 \cos(\omega t - kz)$ (V/m)，(c) $\psi_0 = 53.1^\circ$ 。（参见^{EM}。）

Module 7.3 Polarization I Upon specifying the amplitudes and phases of the x and y components of $\mathbf{E}$, the user can observe the trace of $\mathbf{E}$ in the x - y plane.



Exercise 7-6: If the electric field phasor of a TEM wave is given by $\tilde{\mathbf{E}} = (\hat{\mathbf{y}} - \hat{\mathbf{z}}j)e^{-jkx}$, determine the polarization state.

Answer: RHC polarization. (See 7.6.)

7-4 Plane-Wave Propagation in Lossy Media

To examine wave propagation in a lossy (conducting) medium, we return to the wave equation given by Eq. (7.15),

$$\nabla^2 \tilde{\mathbf{E}} - \gamma^2 \tilde{\mathbf{E}} = 0 \quad (7.61)$$

with

$$\gamma^2 = -\omega^2 \mu \epsilon_c = -\omega^2 \mu (\epsilon' - j\epsilon''), \quad (7.62)$$

where $\epsilon' = \epsilon$ and $\epsilon'' = \sigma/\omega$. Since γ is complex, we express it as

$$\gamma = \alpha + j\beta, \quad (7.63)$$

where α is the medium's **attenuation constant** and β its **phase constant**. By replacing γ with $(\alpha + j\beta)$ in Eq. (7.62), we obtain

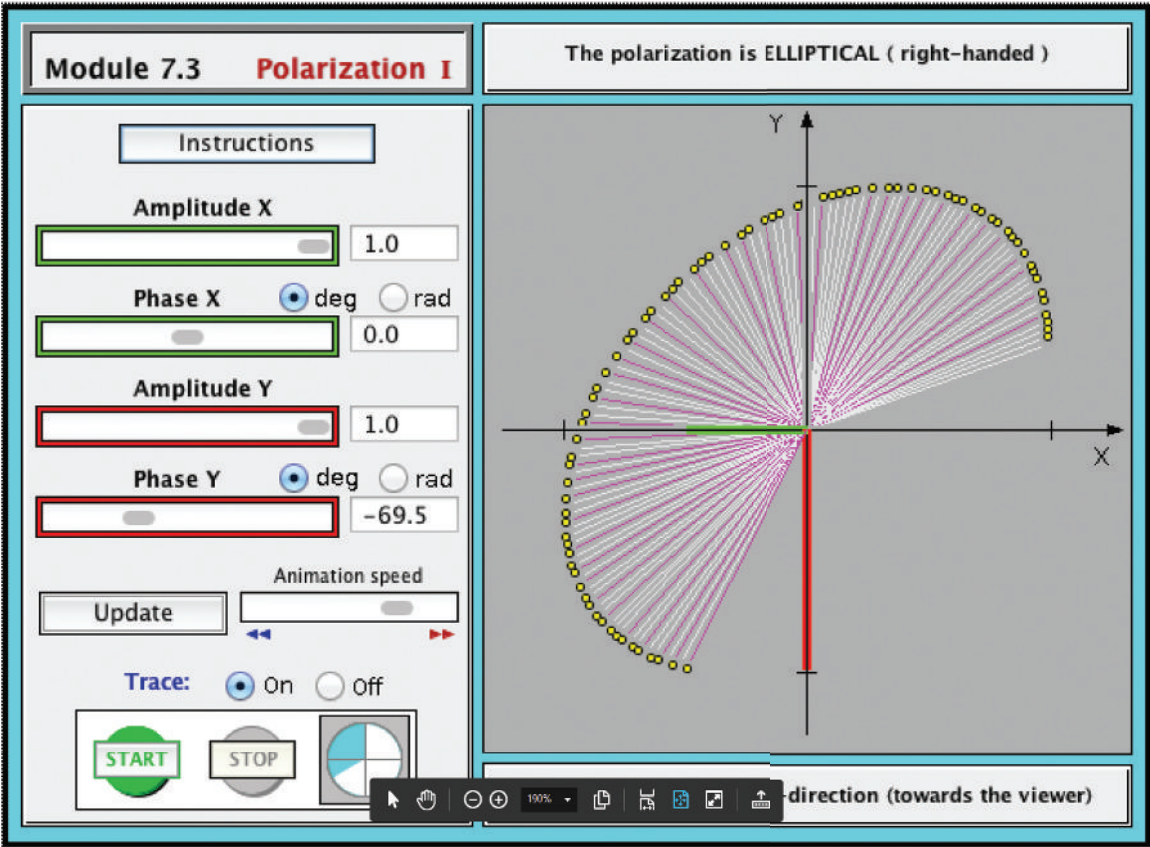
$$(\alpha + j\beta)^2 = (\alpha^2 - \beta^2) + j2\alpha\beta = -\omega^2 \mu \epsilon' + j\omega^2 \mu \epsilon''. \quad (7.64)$$

The rules of complex algebra require the real and imaginary parts on one side of an equation to equal, respectively, the real and imaginary parts on the other side. Hence,

$$\alpha^2 - \beta^2 = -\omega^2 \mu \epsilon', \quad (7.65a)$$

$$2\alpha\beta = \omega^2 \mu \epsilon''. \quad (7.65b)$$

模块 7.3 极化 I 在指定 $\mathbf{E}$ 的 x 和 y 分量的幅度和相位后，用户可以观察 $\mathbf{E}$ 在 x - y 平面上的迹线。



练习 7-6：如果 TEM 波的电场相量由 $\tilde{\mathbf{E}} = (\hat{\mathbf{y}} - \hat{\mathbf{z}}j)e^{-jkx}$ 给出，请确定偏振态。

答案：RHC 偏振。（参见 7.6。）

7-4 有损介质中的平面波传播

为了检查波在有损（传导）介质中的传播，我们回到方程式给出的波动方程。（7.15），

$$\nabla^2 \tilde{\mathbf{E}} - \gamma^2 \tilde{\mathbf{E}} = 0 \quad (7.61)$$

和

$$\gamma^2 = -\omega^2 \mu \epsilon_c = -\omega^2 \mu (\epsilon' - j\epsilon''), \quad (7.62)$$

其中 $\epsilon' = \epsilon$ 和 $\epsilon'' = \sigma/\omega$ 。由于 γ 很复杂，我们将其表示为

$$\gamma = \alpha + j\beta, \quad (7.63)$$

其中 α 是介质的衰减常数， β 是其相位常数。将等式中的 γ 替换为 $(\alpha + j\beta)$ 。（7.62），我们得到

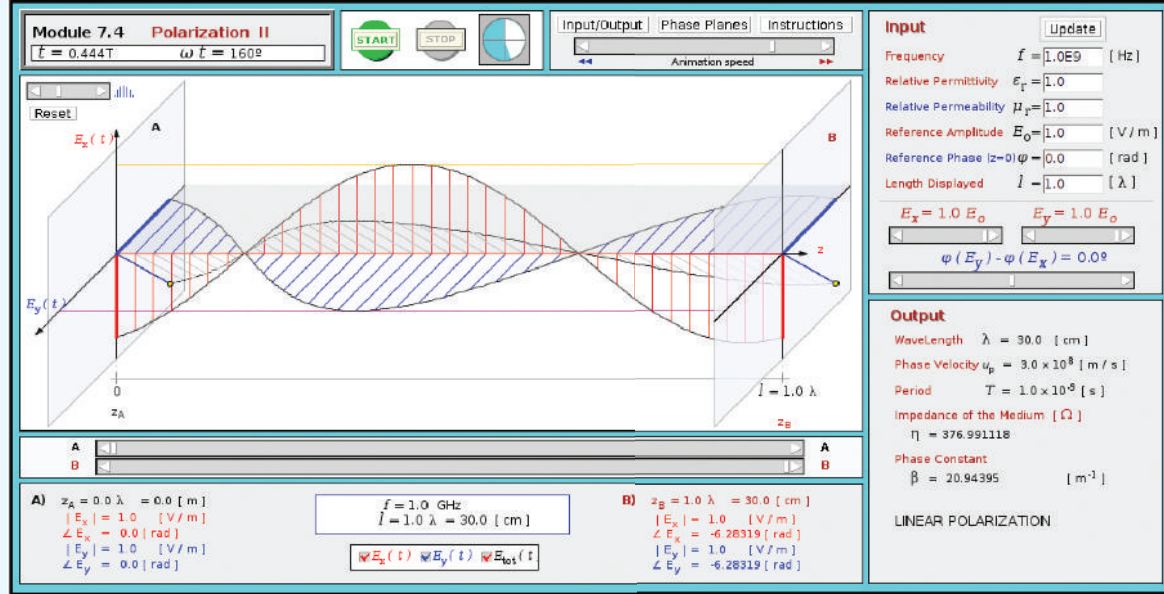
$$(\alpha + j\beta)^2 = (\alpha^2 - \beta^2) + j2\alpha\beta = -\omega^2 \mu \epsilon' + j\omega^2 \mu \epsilon''. \quad (7.64)$$

复代数规则要求方程一侧的实部和虚部分别等于另一侧的实部和虚部。因此，

$$\alpha^2 - \beta^2 = -\omega^2 \mu \epsilon', \quad (7.65a)$$

$$2\alpha\beta = \omega^2 \mu \epsilon''. \quad (7.65b)$$

Module 7.4 Polarization II Upon specifying the amplitudes and phases of the x and y components of $\mathbf{E}$, the user can observe the 3-D profile of the $\mathbf{E}$ vector over a specified length span.



Solving these two equations for α and β gives

$$\alpha = \omega \left\{ \frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right\}^{1/2} \quad (7.66a)$$

(Np/m),

$$\beta = \omega \left\{ \frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right\}^{1/2} \quad (7.66b)$$

(rad/m).

For a uniform plane wave with electric field $\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x(z)$ traveling along the z direction, the wave equation given by Eq. (7.61) reduces to

$$\frac{d^2 \tilde{E}_x(z)}{dz^2} - \gamma^2 \tilde{E}_x(z) = 0. \quad (7.67)$$

The general solution of the wave equation given by Eq. (7.67) comprises two waves: one traveling in the $+z$ direction and another traveling in the $-z$ direction. Assuming only the former is present, the solution of the wave equation leads to

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x(z) = \hat{\mathbf{x}} E_{x0} e^{-\gamma z} = \hat{\mathbf{x}} E_{x0} e^{-\alpha z} e^{-j\beta z}. \quad (7.68)$$

The associated magnetic field $\tilde{\mathbf{H}}$ can be determined by applying Eq. (7.2b): $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$, or using Eq. (7.39a): $\tilde{\mathbf{H}} = (\hat{\mathbf{k}} \times \tilde{\mathbf{E}})/\eta_c$, where η_c is the **intrinsic impedance of the lossy medium**. Both approaches give

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \tilde{H}_y(z) = \hat{\mathbf{y}} \frac{\tilde{E}_x(z)}{\eta_c} = \hat{\mathbf{y}} \frac{E_{x0}}{\eta_c} e^{-\alpha z} e^{-j\beta z}, \quad (7.69)$$

where

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon'}} \left(1 - j \frac{\epsilon''}{\epsilon'} \right)^{-1/2} \quad (\Omega). \quad (7.70)$$

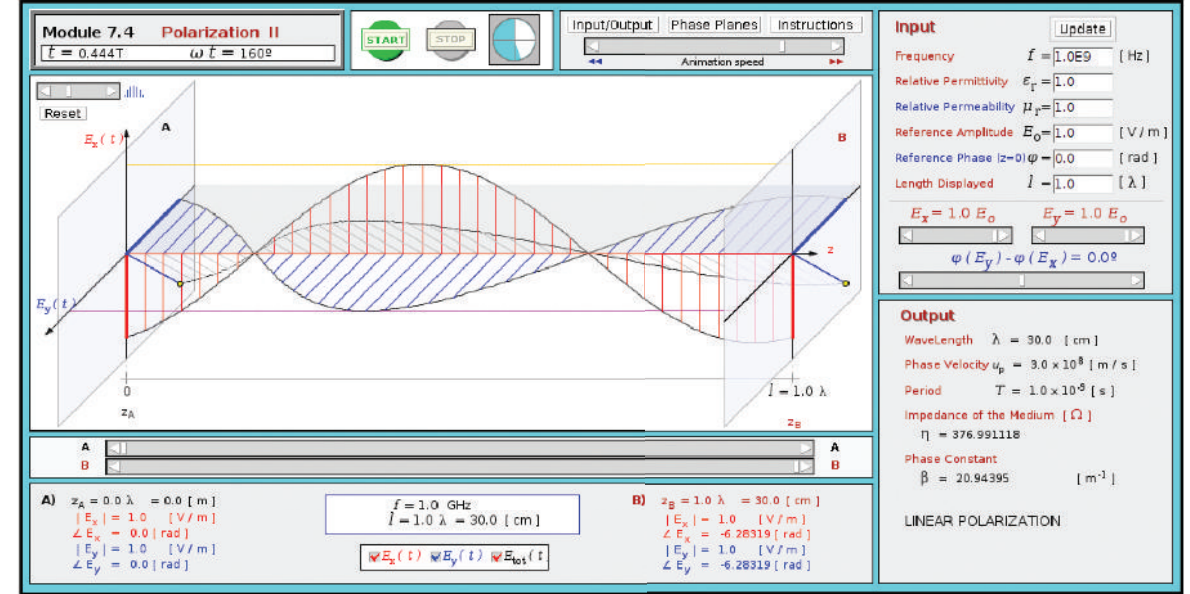
We noted earlier that in a lossless medium, $\mathbf{E}(z,t)$ is in phase with $\mathbf{H}(z,t)$. This property no longer holds true in a lossy medium because η_c is complex. This fact is demonstrated later in Example 7-4.

From Eq. (7.68), the magnitude of $\tilde{E}_x(z)$ is given by

$$|\tilde{E}_x(z)| = |E_{x0} e^{-\alpha z} e^{-j\beta z}| = |E_{x0}| e^{-\alpha z}, \quad (7.71)$$

which decreases exponentially with z at a rate dictated by the attenuation constant α . Since $\tilde{H}_y = \tilde{E}_x/\eta_c$, the magnitude of $\tilde{H}_y$ also decreases as $e^{-\alpha z}$. As the field attenuates, part of the energy carried by the electromagnetic wave is converted into

模块 7.4 极化 II 在指定 $\mathbf{E}$ 的 x 和 y 分量的幅度和相位后，用户可以在指定的长度范围内观察 $\mathbf{E}$ 矢量的 3-D 轮廓。



求解这两个方程的 α 和 β 给出

$$\alpha = \omega \left\{ \frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right\}^{1/2} \quad (7.66a)$$

(Np/m),

$$\beta = \omega \left\{ \frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right\}^{1/2} \quad (7.66b)$$

(rad/m).

对于沿 z 方向传播的电场 $\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x(z)$ 的均匀平面波，波动方程由式 (1) 给出。(7.61) 减少为

$$\frac{d^2 \tilde{E}_x(z)}{dz^2} - \gamma^2 \tilde{E}_x(z) = 0. \quad (7.67)$$

方程给出的波动方程的通解。(7.67) 式包含两个波：一个沿 $+z$ 方向传播，另一个沿 $-z$ 方向传播。假设仅存在前者，则波动方程的解导致

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x(z) = \hat{\mathbf{x}} E_{x0} e^{-\gamma z} = \hat{\mathbf{x}} E_{x0} e^{-\alpha z} e^{-j\beta z}. \quad (7.68)$$

相关的磁场 $\tilde{\mathbf{H}}$ 可以通过应用等式来确定。(7.2b): $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$, 或使用等式: (7.39a): $\tilde{\mathbf{H}} = (\hat{\mathbf{k}} \times \tilde{\mathbf{E}})/\eta_c$, 其中 η_c 是有损介质的固有阻抗。两种方法都给出

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \tilde{H}_y(z) = \hat{\mathbf{y}} \frac{\tilde{E}_x(z)}{\eta_c} = \hat{\mathbf{y}} \frac{E_{x0}}{\eta_c} e^{-\alpha z} e^{-j\beta z}, \quad (7.69)$$

在哪里

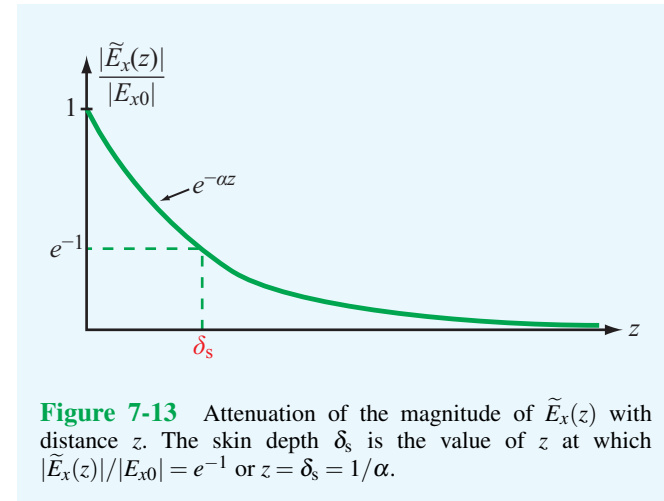
$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon'}} \left(1 - j \frac{\epsilon''}{\epsilon'} \right)^{-1/2} \quad (\Omega). \quad (7.70)$$

我们之前注意到，在无损耗介质中， $\mathbf{E}(z,t)$ 与 $\mathbf{H}(z,t)$ 同相。此属性在有损介质中不再成立，因为 η_c 很复杂。这个事实在后面的例 7-4 中得到了证明。

从方程。(7.68), $\tilde{E}_x(z)$ 的大小由下式给出

$$|\tilde{E}_x(z)| = |E_{x0} e^{-\alpha z} e^{-j\beta z}| = |E_{x0}| e^{-\alpha z}, \quad (7.71)$$

它随 z 以衰减常数 α 指定的速率呈指数下降。由于 $\tilde{H}_y = \tilde{E}_x/\eta_c$, $\tilde{H}_y$ 的幅度也随着 $e^{-\alpha z}$ 而减小。随着磁场衰减，电磁波携带的部分能量转化为



heat due to conduction in the medium. As the wave travels through a distance $z = \delta_s$ with

$$\delta_s = \frac{1}{\alpha} \quad (\text{m}), \quad (7.72)$$

the wave magnitude decreases by a factor of $e^{-1} \approx 0.37$ (Fig. 7-13). At depth $z = 3\delta_s$, the field magnitude is less than 5% of its initial value, and at $z = 5\delta_s$, it is less than 1%.

► This distance δ_s , called the **skin depth** of the medium, characterizes how deep an electromagnetic wave can penetrate into a conducting medium. ◀

In a perfect dielectric, $\sigma = 0$ and $\epsilon'' = 0$. Use of Eq. (7.66a) yields $\alpha = 0$; therefore, $\delta_s = \infty$. Thus, in free space, a plane wave can propagate indefinitely with no loss in magnitude. On the other extreme, in a perfect conductor, $\sigma = \infty$ and use of Eq. (7.66a) leads to $\alpha = \infty$. Hence $\delta_s = 0$. If the outer conductor of a coaxial cable is designed to be several skin depths thick, it prevents energy inside the cable from leaking outward and shields against penetration of electromagnetic energy from external sources into the cable.

The expressions given by Eqs. (7.66a), (7.66b), and (7.70) for α , β , and η_c are valid for any linear, isotropic, and homogeneous medium. For a perfect dielectric ($\sigma = 0$), these expressions reduce to those for the lossless case (Section 7-2), wherein $\alpha = 0$, $\beta = k = \omega\sqrt{\mu\epsilon}$, and $\eta_c = \eta$. For a lossy medium, the ratio $\epsilon''/\epsilon' = \sigma/\omega\epsilon$, which appears in all these expressions, plays an important role in classifying how lossy the medium is. When $\epsilon''/\epsilon' \ll 1$, the medium is considered

a **low-loss dielectric**, and when $\epsilon''/\epsilon' \gg 1$, it is considered a **good conductor**. In practice, the medium may be regarded as a low-loss dielectric if $\epsilon''/\epsilon' < 10^{-2}$, as a good conductor if $\epsilon''/\epsilon' > 10^2$, and as a **quasi-conductor** if $10^{-2} \leq \epsilon''/\epsilon' \leq 10^2$. For low-loss dielectrics and good conductors, the expressions given by Eq. (7.66) can be significantly simplified, as shown next.

7-4.1 Low-Loss Dielectric

From Eq. (7.62), the general expression for γ is

$$\gamma = j\omega\sqrt{\mu\epsilon'} \left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{1/2}. \quad (7.73)$$

For $|x| \ll 1$, the function $(1-x)^{1/2}$ can be approximated by the first two terms of its binomial series; that is, $(1-x)^{1/2} \approx 1 - x/2$. By applying this approximation to Eq. (7.73) for a low-loss dielectric with $x = j\epsilon''/\epsilon'$ and $\epsilon''/\epsilon' \ll 1$, we obtain

$$\gamma \approx j\omega\sqrt{\mu\epsilon'} \left(1 - j\frac{\epsilon''}{2\epsilon'}\right). \quad (7.74)$$

The real and imaginary parts of Eq. (7.74) are

$$\alpha \approx \frac{\omega\epsilon''}{2} \sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} \quad (\text{Np/m}), \quad (7.75a)$$

$$\beta \approx \omega\sqrt{\mu\epsilon'} = \omega\sqrt{\mu\epsilon} \quad (\text{rad/m}). \quad (7.75b)$$

(low-loss medium)

We note that the expression for β is the same as that for the wavenumber k of a lossless medium. Applying the binomial approximation $(1-x)^{-1/2} \approx (1+x/2)$ to Eq. (7.70) leads to

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j\frac{\epsilon''}{2\epsilon'}\right) = \sqrt{\frac{\mu}{\epsilon}} \left(1 + j\frac{\sigma}{2\omega\epsilon}\right). \quad (7.76a)$$

In practice, because $\epsilon''/\epsilon' = \sigma/\omega\epsilon < 10^{-2}$, the second term in Eq. (7.76a) often is ignored. Thus,

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}}, \quad (7.76b)$$

which is the same as Eq. (7.31) for the lossless case.

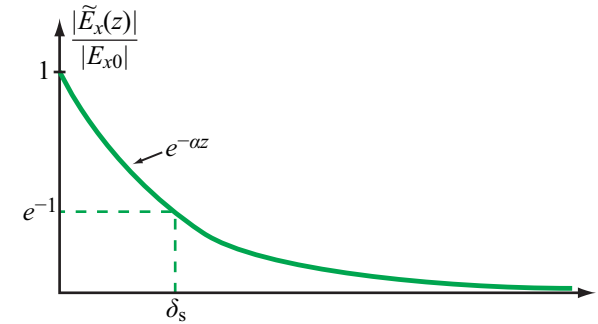


图7-13 $\tilde{E}_x(z)$ 的幅度随距离 z 的衰减。趋肤深度 δ_s 是 $|\tilde{E}_x(z)|/|E_{x0}| = e^{-1}$ 或 $z = \delta_s = 1/\alpha$ 处的 z 值。

由于介质传导而产生热量。当波传播距离 $z = \delta_s$ 时

$$\delta_s = \frac{1}{\alpha} \quad (\text{m}), \quad (7.72)$$

波幅减小了 $e^{-1} \approx 0.37$ 倍 (图 7-13)。在深度 $z = 3\delta_s$ 处，场大小小于其初始值的 5%，而在 $z = 5\delta_s$ 处，场大小小于其初始值的 1%。

► 这个距离 δ_s 称为介质的趋肤深度，表征电磁波可以穿透导电介质的深度。◀

在理想的电介质中， $\sigma = 0$ 和 $\epsilon'' = 0$ 。(7.66a) 产生 $\alpha = 0$ ；因此， $\delta_s = \infty$ 。因此，在自由空间中，平面波可以无限传播而不会损失幅度。在另一个极端，在理想导体中， $\sigma = \infty$ 和方程的使用。(7.66a) 导致 $\alpha = \infty$ 。因此 $\delta_s = 0$ 。如果同轴电缆的外导体被设计为几个趋肤深度厚，它可以防止电缆内部的能量向外泄漏，并屏蔽来自外部源的电磁能量渗透到电缆中。

等式给出的表达式。 α 、 β 和 η_c 的 (7.66a)、(7.66b) 和 (7.70) 对于任何线性、各向同性和均匀介质都有效。对于理想电介质 ($\sigma = 0$)，这些表达式简化为无损情况的表达式 (第 7-2 节)，其中 $\alpha = 0$ 、 $\beta = k = \omega\sqrt{\mu\epsilon}$ 和 $\eta_c = \eta$ 。对于有损介质，出现在所有这些表达式中的比率 $\epsilon''/\epsilon' = \sigma/\omega\epsilon$ 在对介质有损程度进行分类时起着重要作用。当 $\epsilon''/\epsilon' \ll 1$ 时，考虑介质

一种低损耗电介质，当 $\epsilon''/\epsilon' \gg 1$ 时，它被认为是良导体。实际上，如果 $\epsilon''/\epsilon' < 10^{-2}$ ，介质可以被视为低损耗电介质，如果 $\epsilon''/\epsilon' > 10^2$ ，则介质可以被视为良导体，如果 $10^{-2} \leq \epsilon''/\epsilon' \leq 10^2$ ，则可以被视为准导体。对于低损耗电介质和良导体，方程给出的表达式：(7.66) 可以被大大简化，如下所示。

7-4.1 低损耗电介质

从方程。(7.62)， γ 的一般表达式为

$$\gamma = j\omega\sqrt{\mu\epsilon'} \left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{1/2}. \quad (7.73)$$

对于 $|x| \ll 1$ ，函数 $(1-x)^{1/2}$ 可以通过其二项式级数的前两项来近似；即 $(1-x)^{1/2} \approx 1 - x/2$ 。通过将此近似应用于方程。(7.73) 对于 $x = j\epsilon''/\epsilon'$ 和 $\epsilon''/\epsilon' \ll 1$ 的低损耗电介质，我们得到

$$\gamma \approx j\omega\sqrt{\mu\epsilon'} \left(1 - j\frac{\epsilon''}{2\epsilon'}\right). \quad (7.74)$$

方程的实部和虚部。(7.74) 是

$$\alpha \approx \frac{\omega\epsilon''}{2} \sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} \quad (\text{Np/m}), \quad (7.75a)$$

$$\beta \approx \omega\sqrt{\mu\epsilon'} = \omega\sqrt{\mu\epsilon} \quad (\text{rad/m}). \quad (7.75b)$$

(low-loss medium)

我们注意到， β 的表达式与无损介质的波数 k 的表达式相同。将二项式近似 $(1-x)^{-1/2} \approx (1+x/2)$ 应用于等式：(7.70) 导致

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j\frac{\epsilon''}{2\epsilon'}\right) = \sqrt{\frac{\mu}{\epsilon}} \left(1 + j\frac{\sigma}{2\omega\epsilon}\right). \quad (7.76a)$$

实际上，因为 $\epsilon''/\epsilon' = \sigma/\omega\epsilon < 10^{-2}$ ，方程中的第二项。(7.76a) 经常被忽视。因此，

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}}, \quad (7.76b)$$

这与等式相同。(7.31) 对于无损情况。

Technology Brief 14:
Liquid Crystal Display (LCD)

LCDs are used in digital clocks, cellular phones, desktop and laptop computers, some televisions, and other electronic systems. They offer a decided advantage over former display technologies, such as cathode ray tubes, because they are much lighter and thinner and consume a lot less power to operate. LCD technology relies on special electrical and optical properties of a class of materials known as liquid crystals, which were first discovered in the 1880s by botanist **Friedrich Reinitzer**.

Physical Principle

► **Liquid crystals** are neither a pure solid nor a pure liquid; rather, they are a hybrid of both. ◀

One particular variety of interest is the **twisted nematic** liquid crystal whose rod-shaped molecules have a natural tendency to assume a **twisted spiral structure** when the material is sandwiched between finely **grooved glass substrates** with orthogonal orientations (Fig. TF14-1). Note that the molecules in contact with the grooved surfaces align themselves in parallel along the grooves, from a *y* orientation at the entrance substrate into an *x* orientation at the exit substrate. The molecular spiral causes the crystal to behave like a **wave polarizer**: unpolarized light incident upon the entrance substrate follows the orientation of the spiral, emerging through the exit substrate with its polarization (direction of electric field) parallel to the groove's direction, which in Fig. TF14-1 is along the *x* direction. Thus, of the *x* and *y* components of the incident light, only the *y* component is allowed to pass through the *y*-polarized filter, but as a consequence of the spiral action facilitated by the liquid crystal's molecules, the light that emerges from the LCD structure is *x*-polarized.

LCD Structure

A **single-pixel LCD** structure is shown in Fig. TF14-2 for the OFF and ON states, with OFF corresponding to a bright-looking pixel and ON to a dark-looking pixel.

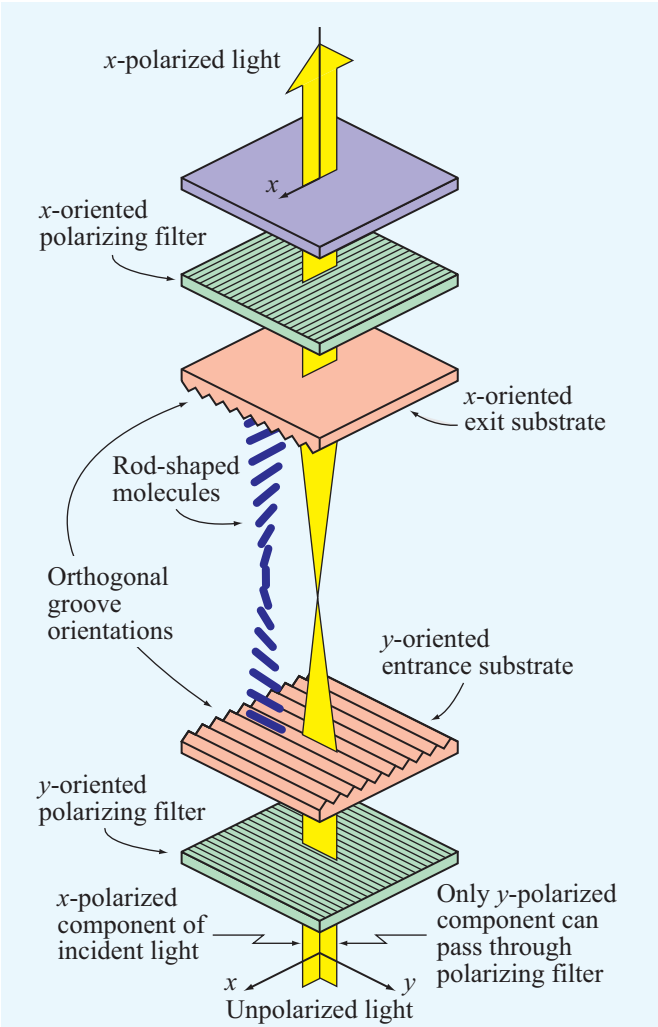


Figure TF14-1 The rod-shaped molecules of a liquid crystal sandwiched between grooved substrates with orthogonal orientations causes the electric field of the light passing through it to rotate by 90°.

► The sandwiched liquid-crystal layer (typically on the order of **5 microns in thickness** or 1/20 of the width of a human hair) is straddled by a pair of optical filters with orthogonal polarizations. ◀

When no voltage is applied across the crystal layer (Fig. TF14-2(a)), incoming unpolarized light gets polarized as it passes through the entrance polarizer, rotates by 90° as it follows the molecular spiral, and finally

技术简介 14: 液晶显示器 (LCD)

LCD 用于数字时钟、移动电话、台式电脑和笔记本电脑、某些电视以及其他电子系统。与以前的显示技术（例如阴极射线管）相比，它们具有明显的优势，因为它们更轻、更薄，并且运行时消耗的功率也少得多。LCD 技术依赖于液晶材料的特殊电学和光学特性，这种材料由植物学家 Friedrich Reinitzer 在 1880 年代首次发现。

物理原理

► 液晶既不是纯固体也不是纯液体；相反，它们是两者的混合物。◀

一种特别令人感兴趣的品种是扭曲向列液晶，当该材料夹在具有正交取向的精细凹槽玻璃基板之间时，其杆状分子自然倾向于呈现扭曲的螺旋结构（图TF14-1）。请注意，与凹槽表面接触分子沿着凹槽平行排列，从入口基板处的 *y* 方向到出口基板处的 *x* 方向。分子螺旋使晶体表现得像一个波偏振器：入射到入口基板上的非偏振光遵循螺旋的方向，通过出口基板出现，其偏振（电场方向）平行于凹槽的方向，在图 TF14-1 中沿着 *x* 方向。因此，在入射光的 *x* 和 *y* 分量中，只有 *y* 分量可以通过 *y* 偏振滤光片，但由于液晶分子促进的螺旋作用，从 LCD 结构中发出的光是 *x* 偏振的。

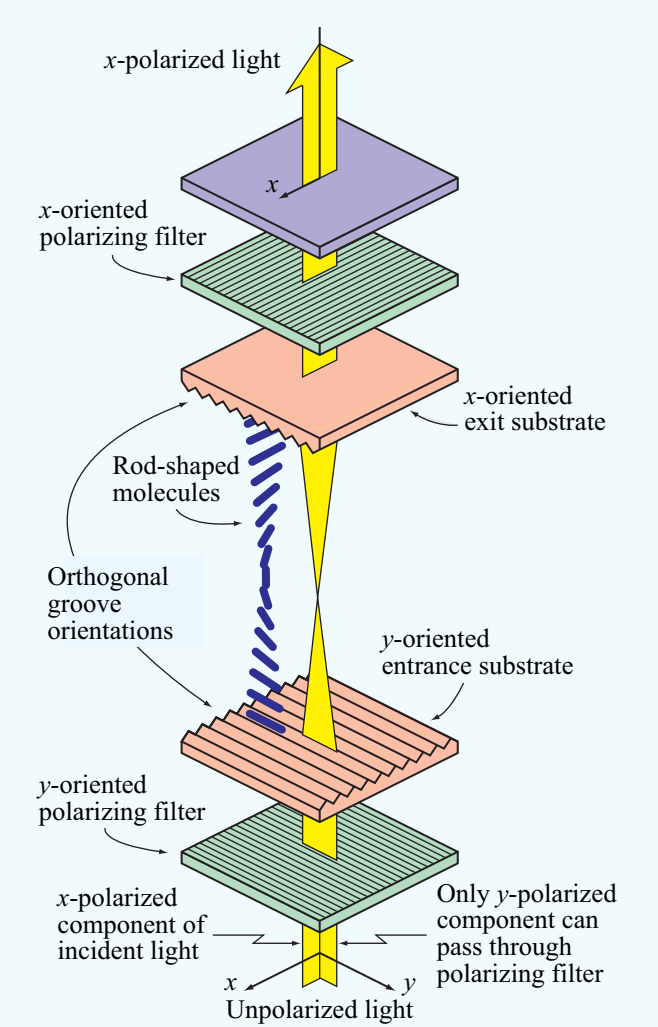


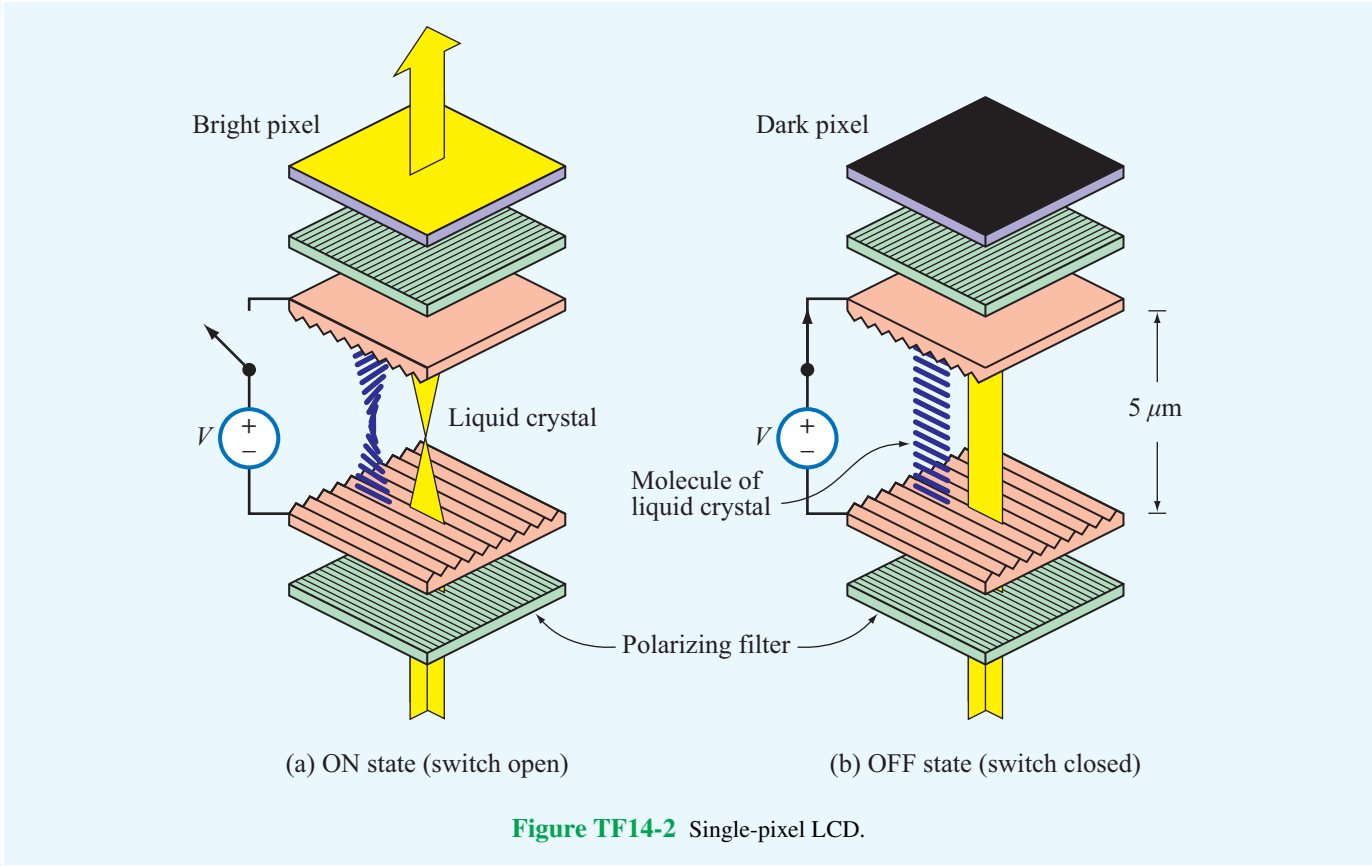
图 TF14-1 液晶的棒状分子夹在具有正交方向的凹槽基板之间，导致穿过它的光的电场旋转 90°。

► 夹在中间的液晶层（厚度通常约为 5 微米或人类头发宽度的 1/20）由一对正交偏振的光学滤光片横跨。◀

液晶显示屏结构

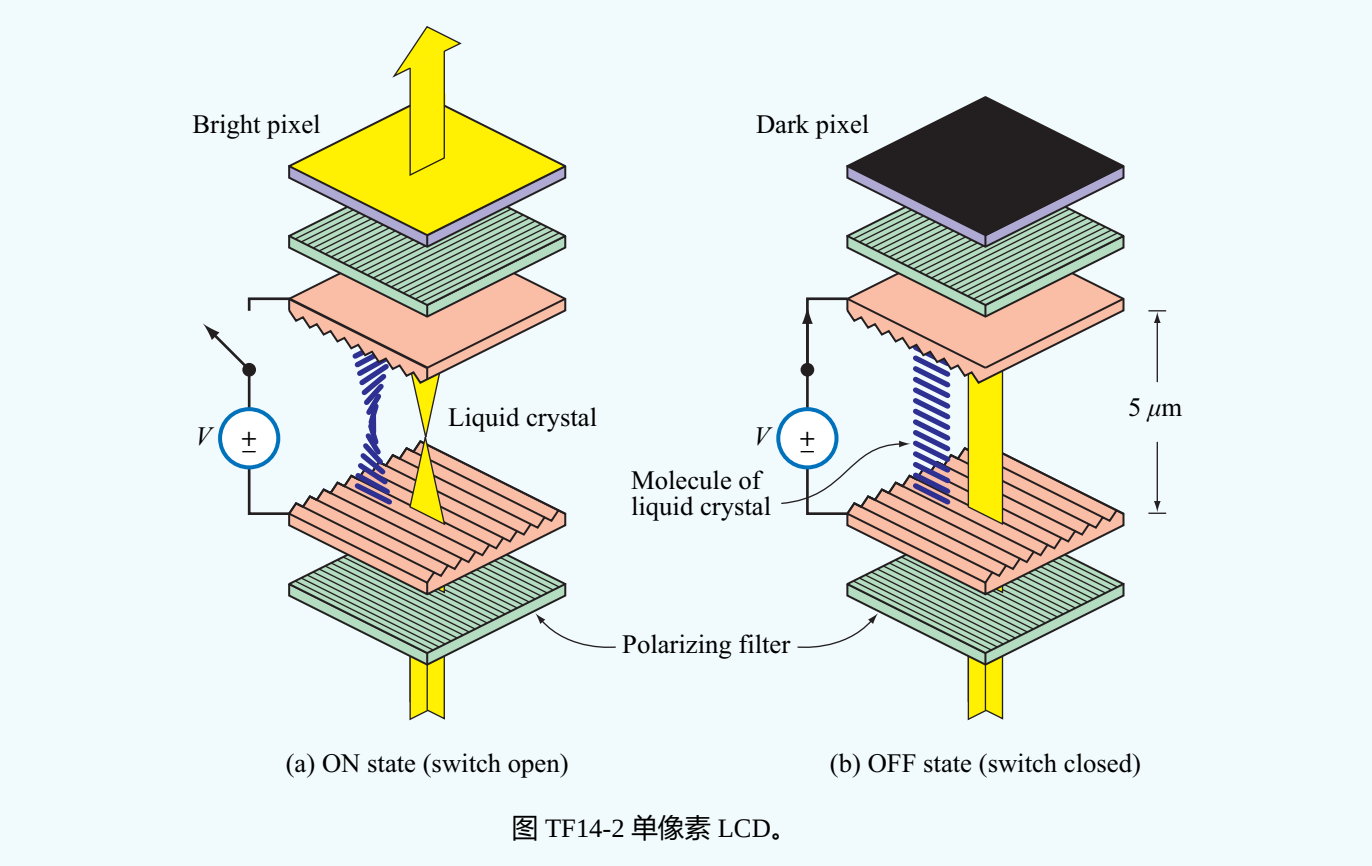
图 TF14-2 显示了单像素 LCD 的 OFF 和 ON 状态结构，其中 OFF 对应于亮像素，ON 对应于暗像素。

当晶体层上没有施加电压时（图 TF14-2(a)），入射的非偏振光在穿过入口偏振器时会发生偏振，并沿着分子螺旋旋转 90°，最后



emerges from the exit polarizer, giving the exited surface a **bright appearance**. A useful feature of nematic liquid crystals is that their spiral untwists (Fig. TF14-2(b)) under the influence of an electric field (induced by a voltage difference across the layer). The degree of untwisting depends on the strength of the electric field. With no spiral to rotate the wave polarization as the light travels through the crystal, the light polarization becomes orthogonal to that of the exit polarizer, allowing no light to pass through it. Hence, the pixel exhibits a **dark appearance**.

► By extending the concept to a **two-dimensional array** of pixels and devising a scheme to control the voltage across each pixel individually (usually by using a thin-film transistor), a complete image can be displayed as illustrated in Fig. TF14-3. For color displays (Fig. TF14-4), each pixel is made up of three subpixels with complementary color filters (red, green, and blue). ◀



从出射偏振器中射出，使出射表面具有明亮的外观。向列液晶的一个有用特征是它们的螺旋在电场（由层间电压差引起）的影响下解开（图TF14-2 (b)）。解绞的程度取决于电场的强度。当光穿过晶体时，由于没有螺旋来旋转波偏振，因此光偏振变得与出射偏振器的偏振正交，从而不允许光通过它。因此，像素呈现出暗的外观。

► 通过将概念扩展到二维像素阵列并设计一种方案来单独控制每个像素上的电压（通常通过使用薄膜晶体管），可以显示完整的图像，如图 TF14-3 所示。对于彩色显示器（图 TF14-4），每个像素由三个具有互补滤色镜（红、绿和蓝）的子像素组成。◀

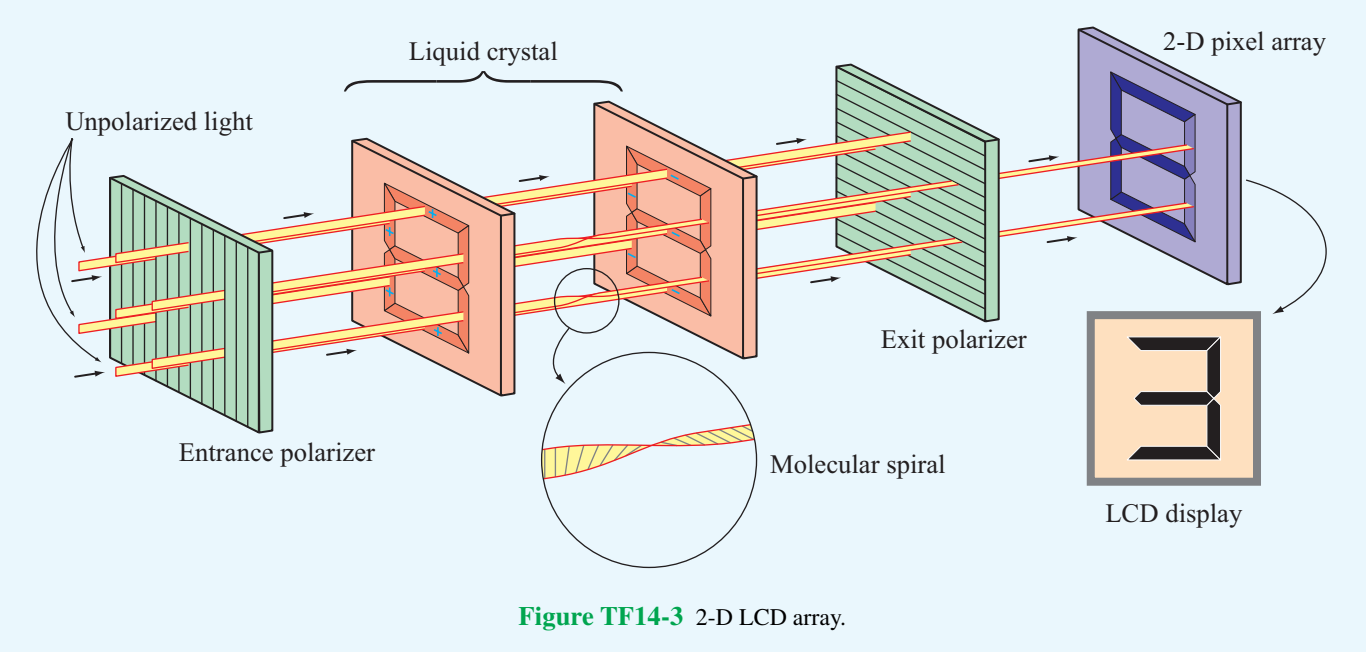
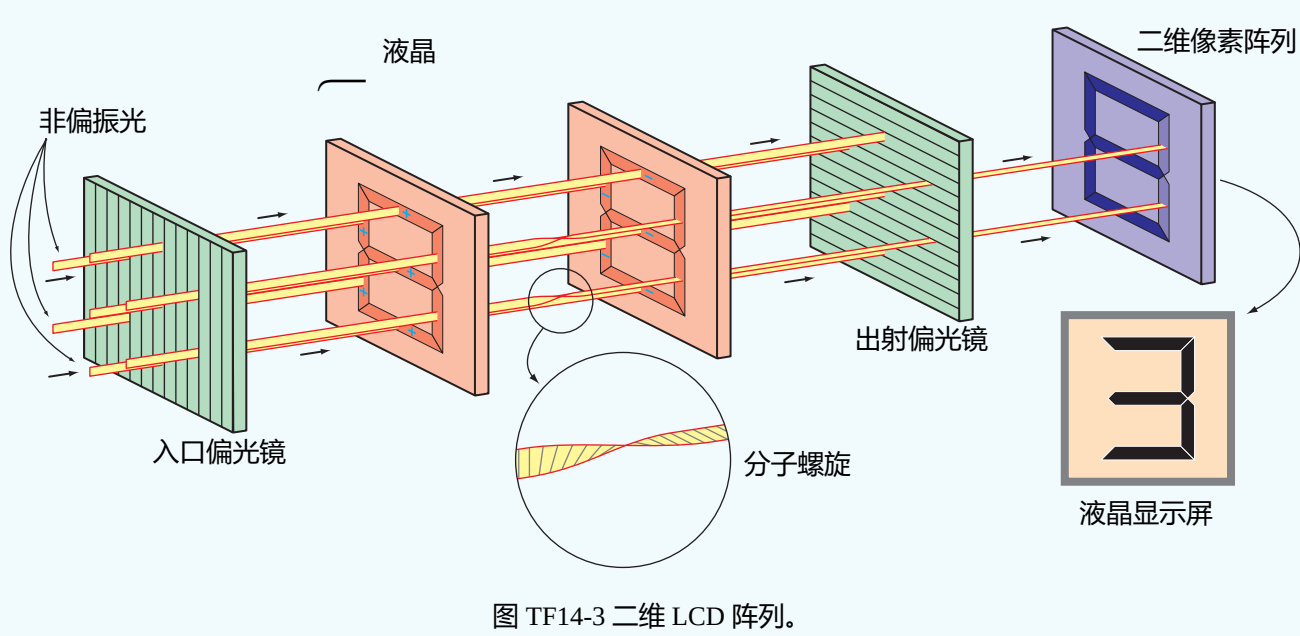


Figure TF14-4 LCD display.



图TF14-4液晶显示屏。

7-4.2 Good Conductor

When $\epsilon''/\epsilon' > 100$, Eqs. (7.66a), (7.66b), and (7.70) can be approximated as

$$\alpha \approx \omega \sqrt{\frac{\mu \epsilon''}{2}} = \omega \sqrt{\frac{\mu \sigma}{2\omega}} = \sqrt{\pi f \mu \sigma} \quad (\text{Np/m}), \quad (7.77\text{a})$$

$$\beta \approx \alpha \approx \sqrt{\pi f \mu \sigma} \quad (\text{rad/m}), \quad (7.77\text{b})$$

$$\eta_{\text{c}} \approx \sqrt{j \frac{\mu}{\epsilon''}} = (1+j) \sqrt{\frac{\pi f \mu}{\sigma}} = (1+j) \frac{\alpha}{\sigma} \quad (\Omega). \quad (7.77\text{c})$$

(good conductors)

In Eq. (7.77c), we used the relation given by Eq. (1.53): $\sqrt{j} = (1+j)/\sqrt{2}$. For a perfect conductor with $\sigma = \infty$, these expressions yield $\alpha = \beta = \infty$ and $\eta_{\text{c}} = 0$. A perfect conductor is equivalent to a short circuit in a transmission line equivalent. Expressions for the propagation parameters in various types of media are summarized in **Table 7-1**.

Example 7-4: Plane Wave in Seawater

A uniform plane wave is traveling in seawater. Assume that the x – y plane resides just below the sea surface and the wave travels in the $+z$ direction into the water. The constitutive parameters of seawater are $\epsilon_{\text{r}} = 80$, $\mu_{\text{r}} = 1$, and $\sigma = 4$ S/m. If the magnetic field at $z = 0$ is

$$\mathbf{H}(0, t) = \hat{\mathbf{y}} 100 \cos(2\pi \times 10^3 t + 15^\circ) \quad (\text{mA/m}),$$

- (a) obtain expressions for $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$, and
 (b) determine the depth at which the magnitude of $\mathbf{E}$ is 1% of its value at $z = 0$.

Solution: (a) Since $\mathbf{H}$ is along $\hat{\mathbf{y}}$ and the propagation direction is $\hat{\mathbf{z}}$, $\mathbf{E}$ must be along $\hat{\mathbf{x}}$. Hence, the general expressions for the phasor fields are

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} E_{x0} e^{-\alpha z} e^{-j\beta z}, \quad (7.78\text{a})$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{E_{x0}}{\eta_{\text{c}}} e^{-\alpha z} e^{-j\beta z}. \quad (7.78\text{b})$$

To determine α , β , and η_{c} for seawater, we begin by evaluating the ratio ϵ''/ϵ' . From the argument of the cosine function of $\mathbf{H}(0, t)$, we deduce that $\omega = 2\pi \times 10^3$ (rad/s). Therefore, $f = 1$ kHz. Hence,

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon} = \frac{\sigma}{\omega \epsilon_{\text{r}} \epsilon_0} = \frac{4}{2\pi \times 10^3 \times 80 \times (10^{-9}/36\pi)} = 9 \times 10^5.$$

This qualifies seawater as a good conductor at 1 kHz and allows us to use the good-conductor expressions given in **Table 7-1**:

$$\alpha = \sqrt{\pi f \mu \sigma}$$

$$= \sqrt{\pi \times 10^3 \times 4\pi \times 10^{-7} \times 4}$$

$$= 0.126 \quad (\text{Np/m}), \quad (7.79\text{a})$$

$$\beta = \alpha = 0.126 \quad (\text{rad/m}), \quad (7.79\text{b})$$

$$\eta_{\text{c}} = (1+j) \frac{\alpha}{\sigma}$$

$$= (\sqrt{2} e^{j\pi/4}) \frac{0.126}{4} = 0.044 e^{j\pi/4} \quad (\Omega). \quad (7.79\text{c})$$

As no explicit information has been given about the electric field amplitude E_{x0} , we should assume it to be complex; that is, $E_{x0} = |E_{x0}|e^{j\phi_0}$. The wave’s instantaneous electric and magnetic fields are given by

$$\mathbf{E}(z, t) = \Re \left[\hat{\mathbf{x}} |E_{x0}| e^{j\phi_0} e^{-\alpha z} e^{-j\beta z} e^{j\omega t} \right]$$

$$= \hat{\mathbf{x}} |E_{x0}| e^{-0.126z} \cos(2\pi \times 10^3 t - 0.126z + \phi_0) \quad (\text{V/m}), \quad (7.80\text{a})$$

$$\mathbf{H}(z, t) = \Re \left[\hat{\mathbf{y}} \frac{|E_{x0}| e^{j\phi_0}}{0.044 e^{j\pi/4}} e^{-\alpha z} e^{-j\beta z} e^{j\omega t} \right]$$

$$= \hat{\mathbf{y}} 22.5 |E_{x0}| e^{-0.126z} \cos(2\pi \times 10^3 t - 0.126z + \phi_0 - 45^\circ) \quad (\text{A/m}). \quad (7.80\text{b})$$

At $z = 0$,

$$\mathbf{H}(0, t) = \hat{\mathbf{y}} 22.5 |E_{x0}| \cos(2\pi \times 10^3 t + \phi_0 - 45^\circ) \quad (\text{A/m}). \quad (7.81)$$

By comparing Eq. (7.81) with the expression given in the problem statement,

$$\mathbf{H}(0, t) = \hat{\mathbf{y}} 100 \cos(2\pi \times 10^3 t + 15^\circ) \quad (\text{mA/m}),$$

we deduce that

$$22.5 |E_{x0}| = 100 \times 10^{-3}$$

or

$$|E_{x0}| = 4.44 \quad (\text{mV/m})$$

and

$$\phi_0 - 45^\circ = 15^\circ \quad \text{or} \quad \phi_0 = 60^\circ.$$

7-4.2 良导体

当 $\epsilon''/\epsilon' > 100$ 时，等式(7.66a)、(7.66b)和(7.70)可近似为

$$\alpha \approx \omega \sqrt{\frac{\mu \epsilon''}{2}} = \omega \sqrt{\frac{\mu \sigma}{2\omega}} = \sqrt{\pi f \mu \sigma} \quad (\text{Np/m}), \quad (7.77\text{a})$$

$$\beta \approx \alpha \approx \sqrt{\pi f \mu \sigma} \quad (\text{rad/m}), \quad (7.77\text{b})$$

$$\eta_{\text{c}} \approx \sqrt{j \frac{\mu}{\epsilon''}} = (1+j) \sqrt{\frac{\pi f \mu}{\sigma}} = (1+j) \frac{\alpha}{\sigma} \quad (\Omega). \quad (7.77\text{c})$$

(good conductors)

在等式中。(7.77c)，我们使用方程给出的关系。(1.53): $\sqrt{j} = (1+j)/\sqrt{2}$ 。对于具有 $\sigma = \infty$ 的理想导体，这些表达式产生 $\alpha = \beta = \infty$ 和 $\eta_{\text{c}} = 0$ 。理想导体相当于传输线等效中的短路。

表7-1总结了各种类型介质中传播参数的表达式。

例7-4：海水中的平面波

均匀的平面波在海水中传播。假设 x – y 平面位于海面正下方，并且波沿 $+z$ 方向传播到水中。海水的本构参数为 $\epsilon_{\text{r}} = 80$ 、 $\mu_{\text{r}} = 1$ 和 $\sigma = 4$ S/m。如果 $z = 0$ 处的磁场为

$$\mathbf{H}(0, t) = \hat{\mathbf{y}} 100 \cos(2\pi \times 10^3 t + 15^\circ) \quad (\text{mA/m}),$$

- (a) 获得 $\mathbf{E}(z, t)$ 和 $\mathbf{H}(z, t)$ 的表达式，并且
 (b) 确定 $\mathbf{E}$ 的幅度为其 $z = 0$ 处值的 1% 时的深度。

解：(a) 由于 $\mathbf{H}$ 沿 $\hat{\mathbf{y}}$ 且传播方向为 $\hat{\mathbf{z}}$ ，因此 $\mathbf{E}$ 必定沿 $\hat{\mathbf{x}}$ 。因此，相量场的一般表达式为

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} E_{x0} e^{-\alpha z} e^{-j\beta z}, \quad (7.78\text{a})$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{E_{x0}}{\eta_{\text{c}}} e^{-\alpha z} e^{-j\beta z}. \quad (7.78\text{b})$$

为了确定海水的 α 、 β 和 η_{c} ，我们首先评估比率 ϵ''/ϵ' 。根据 $\mathbf{H}(0,t)$ 的余弦函数的参数，我们推导出 $\omega = 2\pi \times 10^3$ (rad/s)。因此， $f = 1$ kHz。因此，

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon} = \frac{\sigma}{\omega \epsilon_{\text{r}} \epsilon_0} = \frac{4}{2\pi \times 10^3 \times 80 \times (10^{-9}/36\pi)} = 9 \times 10^5.$$

这使得海水在 1 kHz 下成为良导体，并允许我们使用表 7-1 中给出的良导体表达式：

$$\alpha = \sqrt{\pi f \mu \sigma}$$

$$= \sqrt{\pi \times 10^3 \times 4\pi \times 10^{-7} \times 4}$$

$$= 0.126 \quad (\text{Np/m}), \quad (7.79\text{a})$$

$$\beta = \alpha = 0.126 \quad (\text{rad/m}), \quad (7.79\text{b})$$

$$\eta_{\text{c}} = (1+j) \frac{\alpha}{\sigma}$$

$$= (\sqrt{2} e^{j\pi/4}) \frac{0.126}{4} = 0.044 e^{j\pi/4} \quad (\Omega). \quad (7.79\text{c})$$

由于没有给出关于电场幅度 E_{x0} 的明确信息，我们应该假设它是复数；即 $E_{x0} = |E_{x0}|e^{j\phi_0}$ 。波的瞬时电场和磁场由下式给出

$$\mathbf{E}(z, t) = \Re \left[\hat{\mathbf{x}} |E_{x0}| e^{j\phi_0} e^{-\alpha z} e^{-j\beta z} e^{j\omega t} \right]$$

$$= \hat{\mathbf{x}} |E_{x0}| e^{-0.126z} \cos(2\pi \times 10^3 t - 0.126z + \phi_0) \quad (\text{V/m}), \quad (7.80\text{a})$$

$$\mathbf{H}(z, t) = \Re \left[\hat{\mathbf{y}} \frac{|E_{x0}| e^{j\phi_0}}{0.044 e^{j\pi/4}} e^{-\alpha z} e^{-j\beta z} e^{j\omega t} \right]$$

$$= \hat{\mathbf{y}} 22.5 |E_{x0}| e^{-0.126z} \cos(2\pi \times 10^3 t - 0.126z + \phi_0 - 45^\circ) \quad (\text{A/m}). \quad (7.80\text{b})$$

At $z = 0$,

$$\mathbf{H}(0, t) = \hat{\mathbf{y}} 22.5 |E_{x0}| \cos(2\pi \times 10^3 t + \phi_0 - 45^\circ) \quad (\text{A/m}). \quad (7.81)$$

通过比较等式。(7.81) 用问题陈述中给出的表达式，

$$\mathbf{H}(0, t) = \hat{\mathbf{y}} 100 \cos(2\pi \times 10^3 t + 15^\circ) \quad (\text{mA/m}),$$

我们推断

$$22.5 |E_{x0}| = 100 \times 10^{-3}$$

或者

$$|E_{x0}| = 4.44 \quad (\text{mV/m})$$

和

$$\phi_0 - 45^\circ = 15^\circ \quad \text{or} \quad \phi_0 = 60^\circ.$$

Table 7-1 Expressions for α , β , η_c , u_p , and λ for various types of media.

	Any Medium	Lossless Medium ($\sigma = 0$)	Low-loss Medium ($\epsilon''/\epsilon' \ll 1$)	Good Conductor ($\epsilon''/\epsilon' \gg 1$)	Units
$\alpha =$	$\omega \left[\frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right]^{1/2}$	0	$\approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$	$\approx \sqrt{\pi f \mu \sigma}$	(Np/m)
$\beta =$	$\omega \left[\frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right]^{1/2}$	$\omega \sqrt{\mu \epsilon}$	$\approx \omega \sqrt{\mu \epsilon}$	$\approx \sqrt{\pi f \mu \sigma}$	(rad/m)
$\eta_c =$	$\sqrt{\frac{\mu}{\epsilon'}} \left(1 - j \frac{\epsilon''}{\epsilon'} \right)^{-1/2}$	$\sqrt{\frac{\mu}{\epsilon}}$	$\approx \sqrt{\frac{\mu}{\epsilon}}$	$\approx (1 + j) \frac{\alpha}{\sigma}$	(Ω)
$u_p =$	ω / β	$1 / \sqrt{\mu \epsilon}$	$\approx 1 / \sqrt{\mu \epsilon}$	$\approx \sqrt{4 \pi f / \mu \sigma}$	(m/s)
$\lambda =$	$2 \pi / \beta = u_p / f$	u_p / f	$\approx u_p / f$	$\approx u_p / f$	(m)
Notes: $\epsilon' = \epsilon$; $\epsilon'' = \sigma / \omega$; in free space, $\epsilon = \epsilon_0$, $\mu = \mu_0$; in practice, a material is considered a low-loss medium if $\epsilon'' / \epsilon' = \sigma / \omega \epsilon < 0.01$ and a good conducting medium if $\epsilon'' / \epsilon' > 100$.					

Hence, the final expressions for $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$ are

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 4.44 e^{-0.126 z} \cos(2 \pi \times 10^3 t - 0.126 z + 60^\circ) \quad (\text{mV/m}), \quad (7.82a)$$

$$\mathbf{H}(z, t) = \hat{\mathbf{y}} 100 e^{-0.126 z} \cos(2 \pi \times 10^3 t - 0.126 z + 15^\circ) \quad (\text{mA/m}). \quad (7.82b)$$

► Because $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$ in a lossy medium no longer have the same constant-phase angle, they no longer oscillate in sync with one another as a function of z and t . ◀

(b) The depth at which the amplitude of $\mathbf{E}$ has decreased to 1% of its initial value at $z = 0$ is obtained from

$$0.01 = e^{-0.126 z}$$

or

$$z = \frac{\ln(0.01)}{-0.126} = 36.55 \text{ m} \approx 37 \text{ m}.$$

Exercise 7-7: The constitutive parameters of copper are $\mu = \mu_0 = 4 \pi \times 10^{-7}$ (H/m), $\epsilon = \epsilon_0 \approx (1/36 \pi) \times 10^{-9}$ (F/m), and $\sigma = 5.8 \times 10^7$ (S/m). Assuming that these parameters are frequency independent, over what frequency range of the electromagnetic spectrum (see **Fig. 1-16**) is copper a good conductor?

Answer: $f < 1.04 \times 10^{16}$ Hz, which includes the radio, infrared, visible, and part of the ultraviolet regions of the EM spectrum. (See **EM**.)

Exercise 7-8: Over what frequency range may dry soil with $\epsilon_r = 3$, $\mu_r = 1$, and $\sigma = 10^{-4}$ (S/m) be regarded as a low-loss dielectric?

Answer: $f > 60$ MHz. (See **EM**.)

Exercise 7-9: For a wave traveling in a medium with a skin depth δ_s , what is the amplitude of $\mathbf{E}$ at a distance of $3 \delta_s$ compared with its initial value?

Answer: $e^{-3} \approx 0.05$ or 5%. (See **EM**.)

表 7-1 各种类型媒体的 α 、 β 、 η_c 、 u_p 和 λ 的表达式。

	Any Medium	Lossless Medium ($\sigma = 0$)	Low-loss Medium ($\epsilon''/\epsilon' \ll 1$)	Good Conductor ($\epsilon''/\epsilon' \gg 1$)	Units
$\alpha =$	$\omega \left[\frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right]^{1/2}$	0	$\approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}}$	$\approx \sqrt{\pi f \mu \sigma}$	(Np/m)
$\beta =$	$\omega \left[\frac{\mu \epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right]^{1/2}$	$\omega \sqrt{\mu \epsilon}$	$\approx \omega \sqrt{\mu \epsilon}$	$\approx \sqrt{\pi f \mu \sigma}$	(rad/m)
$\eta_c =$	$\sqrt{\frac{\mu}{\epsilon'}} \left(1 - j \frac{\epsilon''}{\epsilon'} \right)^{-1/2}$	$\sqrt{\frac{\mu}{\epsilon}}$	$\approx \sqrt{\frac{\mu}{\epsilon}}$	$\approx (1 + j) \frac{\alpha}{\sigma}$	(Ω)
$u_p =$	ω / β	$1 / \sqrt{\mu \epsilon}$	$\approx 1 / \sqrt{\mu \epsilon}$	$\approx \sqrt{4 \pi f / \mu \sigma}$	(m/s)
$\lambda =$	$2 \pi / \beta = u_p / f$	u_p / f	$\approx u_p / f$	$\approx u_p / f$	(m)

Notes: $\epsilon' = \epsilon$; $\epsilon'' = \sigma / \omega$; in free space, $\epsilon = \epsilon_0$, $\mu = \mu_0$; in practice, a material is considered a low-loss medium if $\epsilon'' / \epsilon' = \sigma / \omega \epsilon < 0.01$ and a good conducting medium if $\epsilon'' / \epsilon' > 100$.

因此， $\mathbf{E}(z, t)$ 和 $\mathbf{H}(z, t)$ 的最终表达式为

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 4.44 e^{-0.126 z} \cos(2 \pi \times 10^3 t - 0.126 z + 60^\circ) \quad (\text{mV/m}), \quad (7.82a)$$

$$\mathbf{H}(z, t) = \hat{\mathbf{y}} 100 e^{-0.126 z} \cos(2 \pi \times 10^3 t - 0.126 z + 15^\circ) \quad (\text{mA/m}). \quad (7.82b)$$

► 由于有损介质中的 $\mathbf{E}(z, t)$ 和 $\mathbf{H}(z, t)$ 不再具有相同的恒定相位角，因此它们不再作为 z 和 t 的函数彼此同步振荡。◀

(b) $\mathbf{E}$ 的振幅在 $z = 0$ 处减小至其初始值的 1% 时的深度由下式获得

$$0.01 = e^{-0.126 z}$$

或者

$$z = \frac{\ln(0.01)}{-0.126} = 36.55 \text{ m} \approx 37 \text{ m}.$$

练习7-7：铜的本构参数为 $\mu = \mu_0 = 4 \pi \times 10^{-7}$ (H/m)、 $\epsilon = \epsilon_0 \approx (1/36 \pi) \times 10^{-9}$ (F/m)和 $\sigma = 5.8 \times 10^7$ (S/m)。假设这些参数与频率无关，那么在电磁频谱的什么频率范围（见图 1-16）中铜是良导体？

答案：f < 1.04 × 10¹⁶ Hz，其中包括电磁频谱的无线电、红外、可见光和部分紫外区域。（参见^{EM}。）

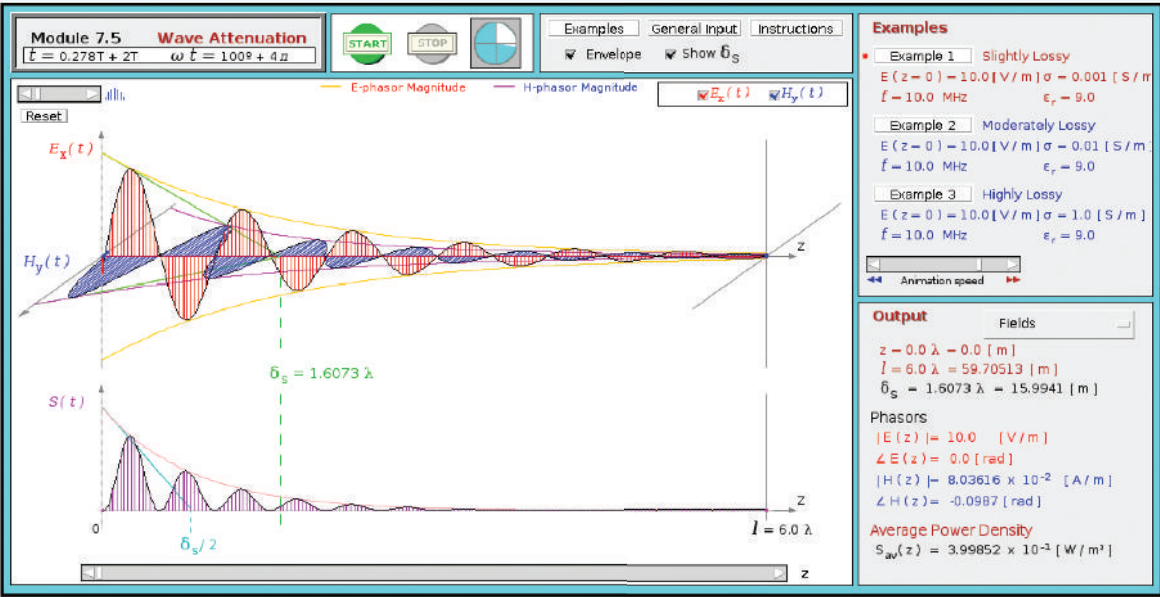
练习 7-8：在什么频率范围内，具有 $\epsilon_r = 3$ 、 $\mu_r = 1$ 和 $\sigma = 10^{-4}$ (S/m) 的干燥土壤可以被视为低损耗电介质？

Answer: $f > 60$ MHz. (See **EM**.)

练习7-9：对于在趋肤深度为 δ_s 的介质中传播的波，与初始值相比，距离 $3\delta_s$ 处 $\mathbf{E}$ 的振幅是多少？

Answer: $e^{-3} \approx 0.05$ or 5%. (See **EM**.)

Module 7.5 Wave Attenuation Observe the profile of a plane wave propagating in a lossy medium. Determine the skin depth, the propagation parameters, and the intrinsic impedance of the medium.



7-5 Current Flow in a Good Conductor

When a dc voltage source is connected across the ends of a conducting wire, the current flowing through the wire is uniformly distributed over its cross section. That is, the current density $\mathbf{J}$ is the same along the axis of the wire and along its outer perimeter (Fig. 7-14(a)). This is not true in the ac case. As we see shortly, a time-varying current density is maximum along the perimeter of the wire and decreases exponentially as a function of distance toward the axis of the wire (Fig. 7-14(b)). In fact, at very high frequencies, most of the current flows in a thin layer near the wire surface, and if the wire material is a perfect conductor, the current flows entirely on the surface of the wire.

Before analyzing a wire with circular cross section, let us consider the simpler geometry of a semi-infinite conducting solid, as shown in Fig. 7-15(a). The solid's planar interface with a perfect dielectric is the x - y plane. If at $z = 0^-$ (just above the surface) an x -polarized electric field with $\tilde{\mathbf{E}} = \hat{\mathbf{x}}E_0$ exists in the dielectric, a similarly polarized field is induced in the conducting medium and propagates as a plane wave along the $+z$ direction. As a consequence of the boundary condition mandating continuity of the tangential component of $\mathbf{E}$ across the boundary between any two contiguous media, the electric field at $z = 0^+$ (just below the boundary) is $\tilde{\mathbf{E}}(0) = \hat{\mathbf{x}}E_0$ also.

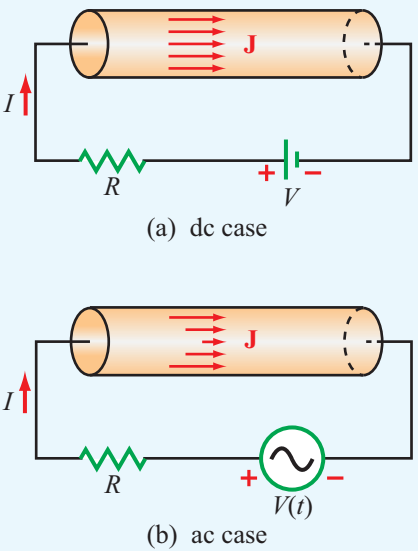
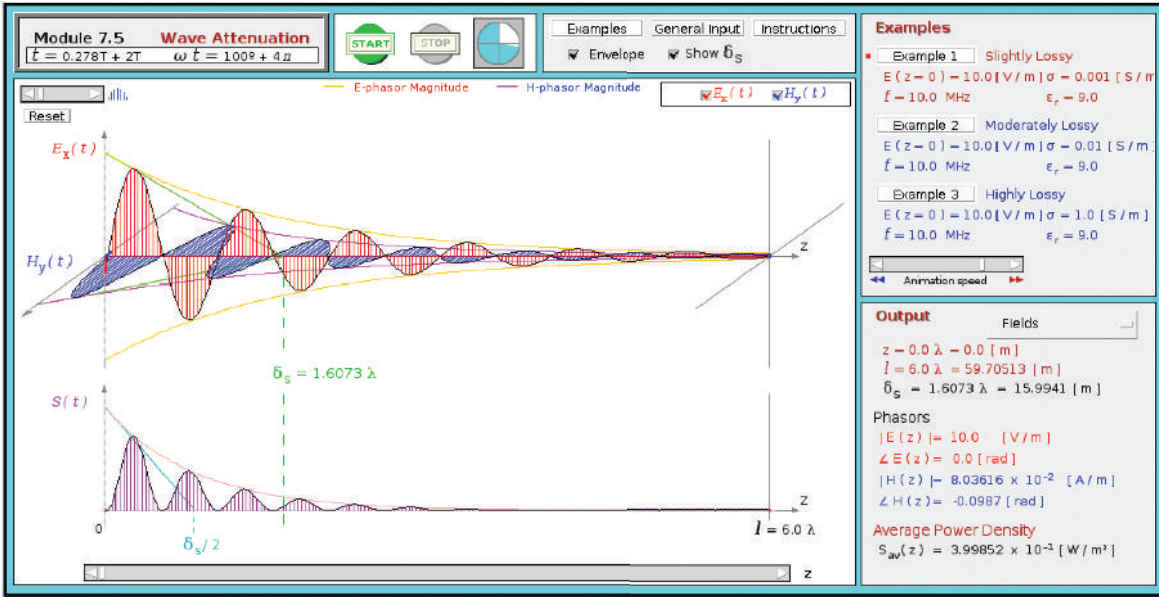


Figure 7-14 Current density $\mathbf{J}$ in a conducting wire is (a) uniform across its cross section in the dc case, but (b) in the ac case, $\mathbf{J}$ is highest along the wire's perimeter.

模块 7.5 波衰减 观察有损介质中传播的平面波的轮廓。确定介质的趋肤深度、传播参数和固有阻抗。



7-5 良导体中的电流

当直流电压源连接在导线的两端时，流过导线的电流均匀分布在其横截面上。也就是说，沿着导线的轴线和沿着其外周的电流密度 $\mathbf{J}$ 是相同的（图 7-14 (a)）。但在交流案例中情况并非如此。正如我们很快看到的，随时间变化的电流密度沿着导线的周长最大，并且随着朝向导线轴线的距离而呈指数下降（图 7-14 (b)）。事实上，在非常高的频率下，大部分电流在靠近导线表面的薄层中流动，如果导线材料是理想的导体，则电流完全在导线的表面上流动。

在分析圆形横截面的导线之前，让我们考虑一下半无限导电固体的更简单的几何形状，如图 7-15(a) 所示。固体与理想电介质的平面界面是 x - y 平面。如果在表面) 上方的 $z = 0^-$ (处，电介质中存在具有 $\tilde{\mathbf{E}} = \hat{\mathbf{x}}E_0$ 的 x 极化电场，则在导电介质中会感应出类似的极化场，并沿 $+z$ 方向作为平面波传播。由于边界条件要求 $\mathbf{E}$ 的切向分量在任何两个相邻介质之间的边界上连续，因此边界) 正下方 $z = 0^+$ (处的电场也是 $\tilde{\mathbf{E}}(0) = \hat{\mathbf{x}}E_0$ 。

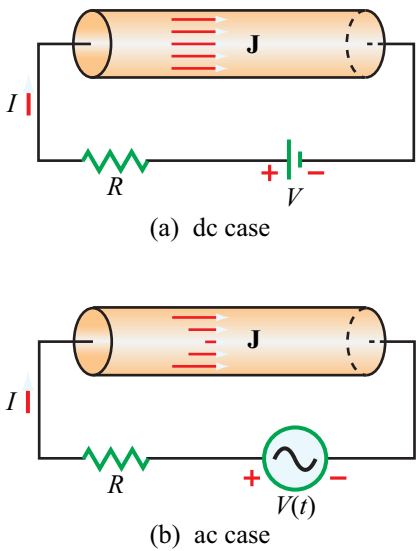


图 7-14 导线中的电流密度 $\mathbf{J}$ (a) 在直流情况下在其横截面上是均匀的，但 (b) 在交流情况下， $\mathbf{J}$ 沿导线周边最高。

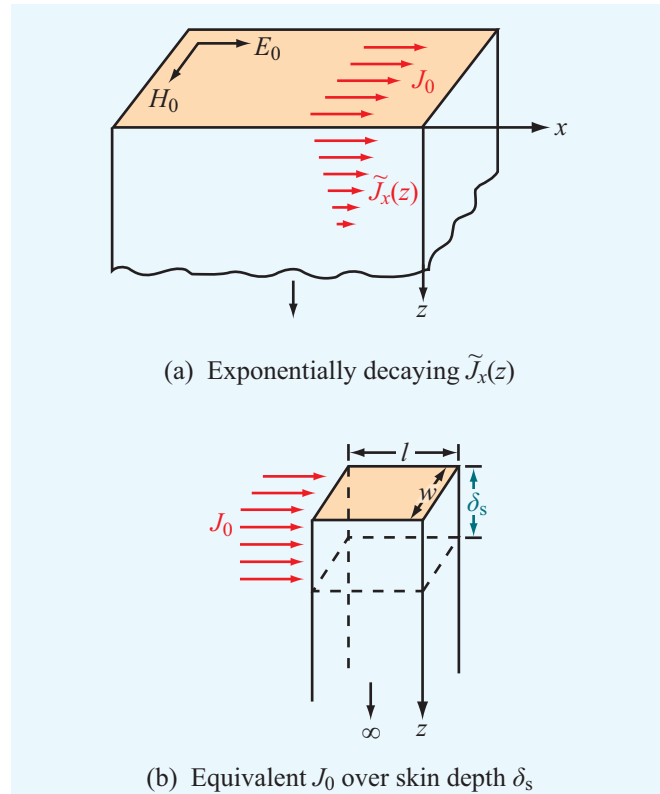


Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s .

The EM fields at any depth z in the conductor are given by

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} E_0 e^{-\alpha z} e^{-j\beta z}, \quad (7.83a)$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z}. \quad (7.83b)$$

From $\mathbf{J} = \sigma \mathbf{E}$, the current flows in the x direction, and its density is

$$\tilde{\mathbf{J}}(z) = \hat{\mathbf{x}} \tilde{J}_x(z), \quad (7.84)$$

with

$$\tilde{J}_x(z) = \sigma E_0 e^{-\alpha z} e^{-j\beta z} = J_0 e^{-\alpha z} e^{-j\beta z}, \quad (7.85)$$

where $J_0 = \sigma E_0$ is the amplitude of the current density at the surface. In terms of the skin depth $\delta_s = 1/\alpha$ defined by Eq. (7.72) and using the fact that in a good conductor $\alpha = \beta$

as expressed by Eq. (7.77b), Eq. (7.85) can be written as

$$\tilde{J}_x(z) = J_0 e^{-(1+j)z/\delta_s} \quad (\text{A/m}^2). \quad (7.86)$$

The current flowing through a rectangular strip of width w along the y direction and extending between zero and ∞ in the z direction is

$$\tilde{I} = w \int_0^\infty \tilde{J}_x(z) dz = w \int_0^\infty J_0 e^{-(1+j)z/\delta_s} dz = \frac{J_0 w \delta_s}{(1+j)} \quad (\text{A}). \quad (7.87)$$

The numerator of Eq. (7.87) is reminiscent of a uniform current density J_0 flowing through a thin surface of width w and depth δ_s . Because $\tilde{J}_x(z)$ decreases exponentially with depth z , a conductor of finite thickness d can be considered electrically equivalent to one of infinite depth as long as d exceeds a few skin depths. Indeed, if $d = 3\delta_s$ [instead of ∞ in the integral of Eq. (7.87)], the error incurred in using the result on the right-hand side of Eq. (7.87) is less than 5%; if $d = 5\delta_s$, the error is less than 1%.

The voltage across a length l at the surface (Fig. 7-15(b)) is given by

$$\tilde{V} = E_0 l = \frac{J_0}{\sigma} l. \quad (7.88)$$

Hence, the impedance of a slab of width w , length l , and depth $d = \infty$ (or, in practice, $d > 5\delta_s$) is

$$Z = \frac{\tilde{V}}{\tilde{I}} = \frac{1+j}{\sigma \delta_s} \frac{l}{w} \quad (\Omega). \quad (7.89)$$

It is customary to represent Z as

$$Z = Z_s \frac{l}{w}, \quad (7.90)$$

where Z_s is called the **internal** or **surface impedance** of the conductor and is defined as the impedance Z for a length $l = 1$ m and a width $w = 1$ m. Thus,

$$Z_s = \frac{1+j}{\sigma \delta_s} \quad (\Omega). \quad (7.91)$$

Since the reactive part of Z_s is positive, Z_s can be defined as

$$Z_s = R_s + j\omega L_s$$

with

$$R_s = \frac{1}{\sigma \delta_s} = \sqrt{\frac{\pi f \mu}{\sigma}} \quad (\Omega), \quad (7.92a)$$

$$L_s = \frac{1}{\omega \sigma \delta_s} = \frac{1}{2} \sqrt{\frac{\mu}{\pi f \sigma}} \quad (\text{H}), \quad (7.92b)$$

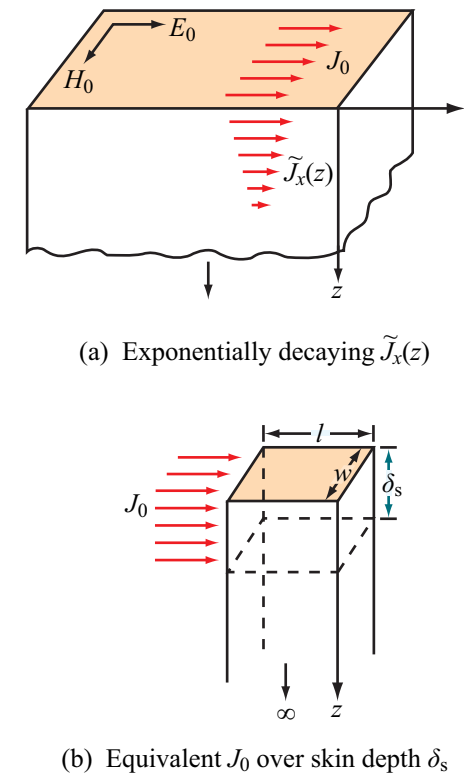


图 7-15 实心导体中电流密度 $\tilde{J}_x(z)$ 随 z 的指数衰减。流过 (a) 在 $z = 0$ 和 $z = \infty$ 之间延伸的宽度为 w 的截面的总电流相当于 (b) 流过深度为 δ_s 的截面的恒定电流密度 J_0 。

导体中任意深度 z 处的电磁场由下式给出

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} E_0 e^{-\alpha z} e^{-j\beta z}, \quad (7.83a)$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z}. \quad (7.83b)$$

从 $\mathbf{J} = \sigma \mathbf{E}$ 开始，电流沿 x 方向流动，其密度为

$$\tilde{\mathbf{J}}(z) = \hat{\mathbf{x}} \tilde{J}_x(z), \quad (7.84)$$

和

$$\tilde{J}_x(z) = \sigma E_0 e^{-\alpha z} e^{-j\beta z} = J_0 e^{-\alpha z} e^{-j\beta z}, \quad (7.85)$$

其中 $J_0 = \sigma E_0$ 是表面电流密度的幅度。就趋肤深度 $\delta_s = 1/\alpha$ 而言，由方程定义：(7.72) 并利用良导体 $\alpha = \beta$

如方程所示 (7.77b)，等式。(7.85) 可以写成

$$\tilde{J}_x(z) = J_0 e^{-(1+j)z/\delta_s} \quad (\text{A/m}^2). \quad (7.86)$$

沿 y 方向流经宽度为 w 的矩形带并沿 z 方向在 0 到 ∞ 之间延伸的电流为

$$\tilde{I} = w \int_0^\infty \tilde{J}_x(z) dz = w \int_0^\infty J_0 e^{-(1+j)z/\delta_s} dz = \frac{J_0 w \delta_s}{(1+j)} \quad (\text{A}). \quad (7.87)$$

方程的分子。(7.87) 让人想起流过宽度为 w 、深度为 δ_s 的薄表面的均匀电流密度 J_0 。因为 $\tilde{J}_x(z)$ 随深度 z 呈指数减小，只要 d 超过几个趋肤深度，则可以将有限厚度 d 的导体视为电等效于无限深度的导体。事实上，如果 $d = 3\delta_s$ [而不是等式积分中的 ∞ (7.87)]，使用等式右侧结果所产生的误差。(7.87) 小于 5%；如果 $d = 5\delta_s$ ，则误差小于 1%。

表面长度 l 上的电压 (图 7-15(b)) 由下式给出

$$\tilde{V} = E_0 l = \frac{J_0}{\sigma} l. \quad (7.88)$$

因此，宽度为 w 、长度为 l 、深度为 $d = \infty$ (或实际上为 $d > 5\delta_s$) 的板的阻抗为

$$Z = \frac{\tilde{V}}{\tilde{I}} = \frac{1+j}{\sigma \delta_s} \frac{l}{w} \quad (\Omega). \quad (7.89)$$

通常将 Z 表示为

$$Z = Z_s \frac{l}{w}, \quad (7.90)$$

其中 Z_s 称为导体的内部或表面阻抗，定义为长度 $l = 1$ m 和宽度 $w = 1$ m 的阻抗 Z 。因此，

$$Z_s = \frac{1+j}{\sigma \delta_s} \quad (\Omega). \quad (7.91)$$

由于 Z_s 的电抗部分为正，因此 Z_s 可以定义为

$$Z_s = R_s + j\omega L_s$$

和

$$R_s = \frac{1}{\sigma \delta_s} = \sqrt{\frac{\pi f \mu}{\sigma}} \quad (\Omega), \quad (7.92a)$$

$$L_s = \frac{1}{\omega \sigma \delta_s} = \frac{1}{2} \sqrt{\frac{\mu}{\pi f \sigma}} \quad (\text{H}), \quad (7.92b)$$

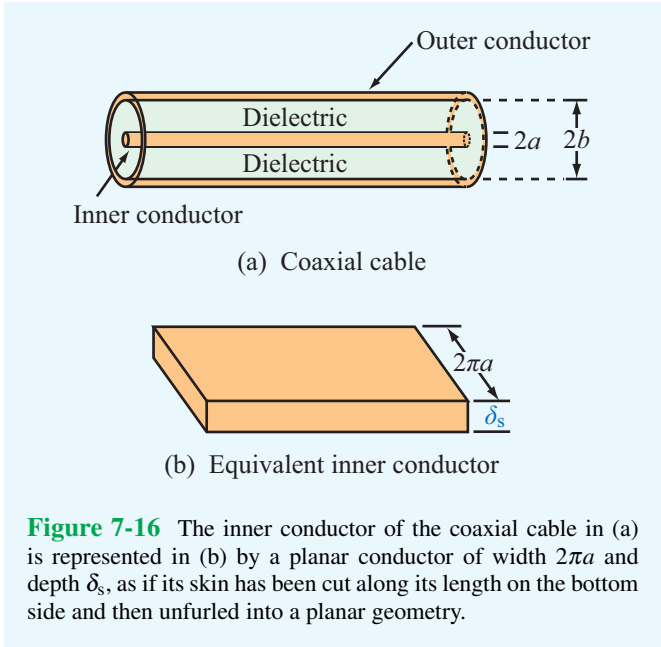


Figure 7-16 The inner conductor of the coaxial cable in (a) is represented in (b) by a planar conductor of width $2\pi a$ and depth δ_s , as if its skin has been cut along its length on the bottom side and then unfurled into a planar geometry.

where we used the relation $\delta_s = 1/\alpha \approx 1/\sqrt{\pi f \mu \sigma}$ given by Eq. (7.77a). In terms of the **surface resistance** R_s , the **ac resistance** of a slab of width w and length l is

$$R = R_s \frac{l}{w} = \frac{l}{\sigma \delta_s w} \quad (\Omega). \quad (7.93)$$

The expression for the ac resistance R is equivalent to the dc resistance of a plane conductor of length l and cross section $A = \delta_s w$.

The results obtained for the planar conductor can be extended to the coaxial cable shown in **Fig. 7-16(a)**. If the conductors are made of copper with $\sigma = 5.8 \times 10^7$ S/m, the skin depth at 1 MHz is $\delta_s = 1/\sqrt{\pi f \mu \sigma} = 0.066$ mm, and since δ_s varies as $1/\sqrt{f}$, it becomes smaller at higher frequencies. As long as the inner conductor's radius a is greater than $5\delta_s$, or 0.33 mm at 1 MHz, its "depth" may be regarded as infinite. A similar criterion applies to the thickness of the outer conductor. To compute the resistance of the inner conductor, note that the current is concentrated near its outer surface and approximately equivalent to a uniform current flowing through a thin layer of depth δ_s and circumference $2\pi a$. In other words, the inner conductor's resistance is nearly the same as that of a planar conductor of depth δ_s and width $w = 2\pi a$, as shown in **Fig. 7-16(b)**. The corresponding resistance per unit length is obtained by setting $w = 2\pi a$ and dividing by l in Eq. (7.93):

$$R'_1 = \frac{R}{l} = \frac{R_s}{2\pi a} \quad (\Omega/\text{m}). \quad (7.94)$$

Similarly, for the outer conductor, the current is concentrated within a thin layer of depth δ_s on the inside surface of the conductor adjacent to the insulating medium between the two conductors, which is where the EM fields exist. The resistance per unit length for the outer conductor with radius b is

$$R'_2 = \frac{R_s}{2\pi b} \quad (\Omega/\text{m}), \quad (7.95)$$

and the coaxial cable's total ac resistance per unit length is

$$R' = R'_1 + R'_2 = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) \quad (\Omega/\text{m}). \quad (7.96)$$

This expression was used in Chapter 2 for characterizing the resistance per unit length of a coaxial transmission line.

Concept Question 7-6: How does β of a low-loss dielectric medium compare to that of a lossless medium?

Concept Question 7-7: In a good conductor, does the phase of $\mathbf{H}$ lead or lag that of $\mathbf{E}$ and by how much?

Concept Question 7-8: Attenuation means that a wave loses energy as it propagates in a lossy medium. What happens to the lost energy?

Concept Question 7-9: Is a conducting medium dispersive or dispersionless? Explain.

Concept Question 7-10: Compare the flow of current through a wire in the dc and ac cases. Compare the corresponding dc and ac resistances of the wire.

7-6 Electromagnetic Power Density

This section deals with the flow of power carried by an electromagnetic wave. For any wave with an electric field $\mathbf{E}$ and magnetic field $\mathbf{H}$, the **Poynting vector** $\mathbf{S}$ is defined as

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (\text{W}/\text{m}^2). \quad (7.97)$$

The unit of $\mathbf{S}$ is $(\text{V}/\text{m}) \times (\text{A}/\text{m}) = (\text{W}/\text{m}^2)$, and the direction of $\mathbf{S}$ is along the wave's direction of propagation. Thus, $\mathbf{S}$ represents the power per unit area (or **power density**) carried by the wave. If the wave is incident upon an aperture of area A with outward surface unit vector $\hat{\mathbf{n}}$ as shown in **Fig. 7-17**,

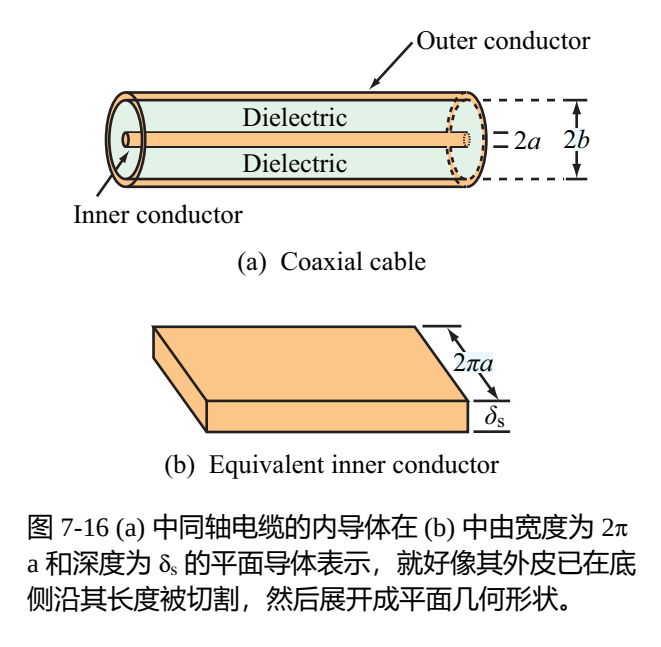


图 7-16 (a) 中同轴电缆的内导体在 (b) 中由宽度为 $2\pi a$ 和深度为 δ_s 的平面导体表示, 就好像其外皮已在底侧沿其长度被切割, 然后展开成平面几何形状。

其中我们使用等式给出的关系 $\delta_s = 1/\alpha \approx 1/\sqrt{\pi f \mu \sigma}$ (7.77a)。就表面电阻 R_s 而言, 宽度为 w 、长度为 l 的平板的交流电阻为

$$R = R_s \frac{l}{w} = \frac{l}{\sigma \delta_s w} \quad (\Omega). \quad (7.93)$$

交流电阻 R 的表达式相当于长度为 l 、横截面为 $A = \delta_s w$ 的平面导体的直流电阻。

针对平面导体获得的结果可以扩展到图7-16(a)所示的同轴电缆。如果导体由 $\sigma = 5.8 \times 10^7$ S/m 的铜制成, 则 1 MHz 下的集肤深度为 $\delta_s = 1/\sqrt{\pi f \mu \sigma} = 0.066$ mm, 并且由于 δ_s 变化为 $1/\sqrt{f}$, 因此在较高频率下趋肤深度会变小。只要内导体的半径 a 大于 $5\delta_s$, 即 1 MHz 时的 0.33 mm, 其 "深度" 就可以被视为无限。类似的标准适用于外导体的厚度。要计算内导体的电阻, 请注意电流集中在其外表面附近, 大约相当于流过深度为 δ_s 且周长为 $2\pi a$ 的薄层的均匀电流。换句话说, 内导体的电阻与深度为 δ_s 、宽度为 $w = 2\pi a$ 的平面导体的电阻几乎相同, 如图7-16(b)所示。式(1)中设置 $w = 2\pi a$ 除以 l 即可得到相应的单位长度电阻。 (7.93) :

$$R'_1 = \frac{R}{l} = \frac{R_s}{2\pi a} \quad (\Omega/\text{m}). \quad (7.94)$$

类似地, 对于外导体, 电流集中在与两个导体之间的绝缘介质相邻的导体内表面上深度为 δ_s 的薄层内, 这就是电磁场存在的地方。半径为 b 的外导体单位长度的电阻为

$$R'_2 = \frac{R_s}{2\pi b} \quad (\Omega/\text{m}), \quad (7.95)$$

同轴电缆单位长度的总交流电阻为

$$R' = R'_1 + R'_2 = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) \quad (\Omega/\text{m}). \quad (7.96)$$

该表达式在第 2 章中用于表征同轴传输线单位长度的电阻。

Concept Question 7-6: How does β of a low-loss dielectric medium compare to that of a lossless medium?

Concept Question 7-7: In a good conductor, does the phase of $\mathbf{H}$ lead or lag that of $\mathbf{E}$ and by how much?

Concept Question 7-8: Attenuation means that a wave loses energy as it propagates in a lossy medium. What happens to the lost energy?

Concept Question 7-9: Is a conducting medium dispersive or dispersionless? Explain.

Concept Question 7-10: Compare the flow of current through a wire in the dc and ac cases. Compare the corresponding dc and ac resistances of the wire.

7-6 电磁功率密度

本节讨论电磁波携带的功率流。对于任何具有电场 $\mathbf{E}$ 和磁场 $\mathbf{H}$ 的波, 坡印廷矢量 $\mathbf{S}$ 定义为

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (\text{W}/\text{m}^2). \quad (7.97)$$

$\mathbf{S}$ 的单位是 $(\text{V}/\text{m}) \times (\text{A}/\text{m}) = (\text{W}/\text{m}^2)$, $\mathbf{S}$ 的方向沿着波的传播方向。因此, $\mathbf{S}$ 代表波所携带的每单位面积的功率 (或功率密度)。如果波以向外表面单位矢量 $\hat{\mathbf{n}}$ 入射到区域 A 的孔径上, 如图7-17所示,

Module 7.6 Current in a Conductor

Module displays exponential decay of current density in a conductor.

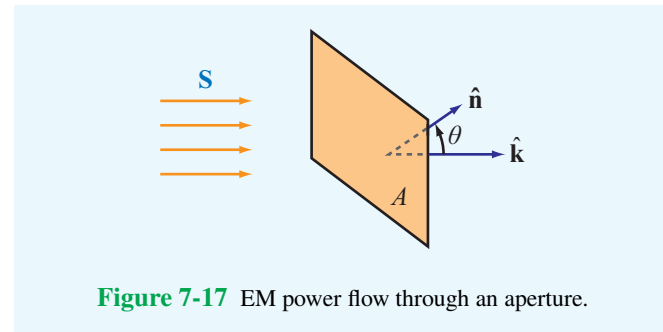
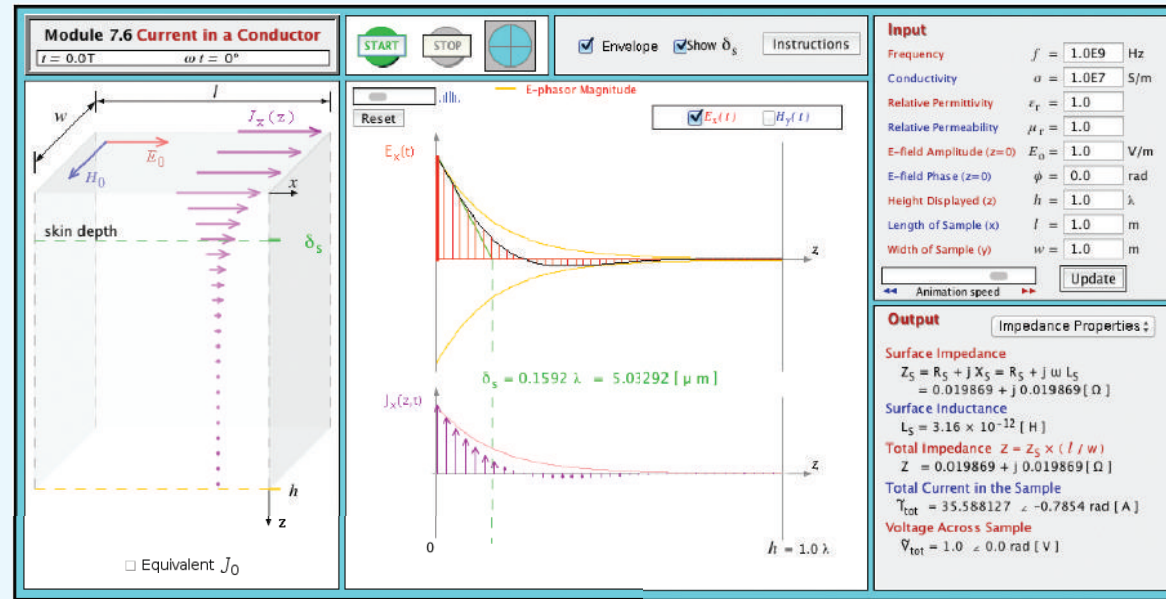


Figure 7-17 EM power flow through an aperture.

Since both $\mathbf{E}$ and $\mathbf{H}$ are functions of time, so is the Poynting vector $\mathbf{S}$. In practice, however, the quantity of greater interest is the **average power density** of the wave, $\mathbf{S}_{av}$, which is the time-average value of $\mathbf{S}$:

$$\mathbf{S}_{av} = \frac{1}{2} \Re \left[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right] \quad (\text{W/m}^2). \quad (7.100)$$

This expression may be regarded as the electromagnetic equivalent of Eq. (2.107) for the time-average power carried by a transmission line, namely

$$P_{av}(z) = \frac{1}{2} \Re \left[\tilde{V}(z) \tilde{I}^*(z) \right], \quad (7.101)$$

where $\tilde{V}(z)$ and $\tilde{I}(z)$ are the phasors corresponding to $v(z,t)$ and $i(z,t)$, respectively.

7-6.1 Plane Wave in a Lossless Medium

Recall that the general expression for the electric field of a uniform plane wave with arbitrary polarization traveling in the $+z$ direction is

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x(z) + \hat{\mathbf{y}} \tilde{E}_y(z) = (\hat{\mathbf{x}} E_{x0} + \hat{\mathbf{y}} E_{y0}) e^{-jkz}, \quad (7.102)$$

then the total power that flows through or is intercepted by the aperture is

$$P = \int_A \mathbf{S} \cdot \hat{\mathbf{n}} dA \quad (\text{W}). \quad (7.98)$$

For a uniform plane wave propagating in a direction $\hat{\mathbf{k}}$ that makes an angle θ with $\hat{\mathbf{n}}$, $P = SA \cos \theta$, where $S = |\mathbf{S}|$.

Except for the fact that the units of $\mathbf{S}$ are per unit area, Eq. (7.97) is the vector analogue of the scalar expression for the instantaneous power $P(z,t)$ flowing through a transmission line,

$$P(z,t) = v(z,t) i(z,t), \quad (7.99)$$

where $v(z,t)$ and $i(z,t)$ are the instantaneous voltage and current on the line.

模块 7.6 导体中的电流

模块显示导体中电流密度的指数衰减。

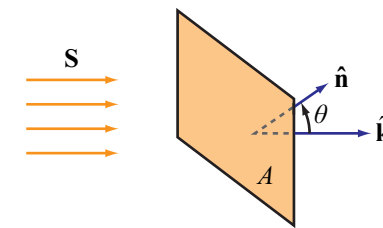
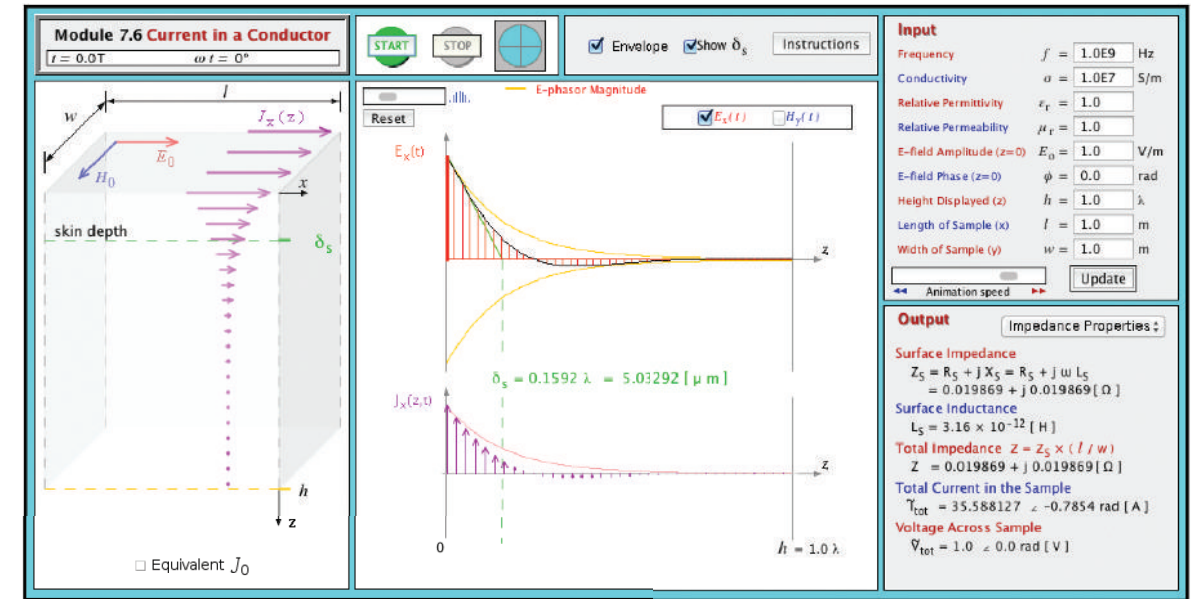


图 7-17 通过孔径的电磁功率流。

由于 $\mathbf{E}$ 和 $\mathbf{H}$ 都是时间的函数，因此坡印廷向量 $\mathbf{S}$ 也是时间的函数。然而，在实践中，更令人感兴趣的量是波的平均功率密度 $\mathbf{S}_{av}$ ，它是 $\mathbf{S}$ 的时间平均值：

$$\mathbf{S}_{av} = \frac{1}{2} \Re \left[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right] \quad (\text{W/m}^2). \quad (7.100)$$

这个表达式可以被认为是方程的电磁等价。(2.107) 为传输线承载的时间平均功率，即

$$P_{av}(z) = \frac{1}{2} \Re \left[\tilde{V}(z) \tilde{I}^*(z) \right], \quad (7.101)$$

其中 $\tilde{V}(z)$ 和 $\tilde{I}(z)$ 分别是对应于 $v(z,t)$ 和 $i(z,t)$ 的相量。

那么流经孔径或被孔径拦截的总功率为

$$P = \int_A \mathbf{S} \cdot \hat{\mathbf{n}} dA \quad (\text{W}). \quad (7.98)$$

对于沿方向 $\hat{\mathbf{k}}$ 传播且与 $\hat{\mathbf{n}}$ 形成角度 θ 的均匀平面波， $P = SA \cos \theta$ ，其中 $S = |\mathbf{S}|$ 。

除了 $\mathbf{S}$ 的单位是每单位面积这一事实外，方程：(7.97) 是流经传输线的瞬时功率 $P(z,t)$ 标量表达式的矢量模拟，

$$P(z,t) = v(z,t) i(z,t), \quad (7.99)$$

其中 $v(z,t)$ 和 $i(z,t)$ 是线路上的瞬时电压和电流。

7-6.1 无损介质中的平面波

回想一下，沿 $+z$ 方向传播的具有任意偏振的均匀平面波的电场的一般表达式为

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x(z) + \hat{\mathbf{y}} \tilde{E}_y(z) = (\hat{\mathbf{x}} E_{x0} + \hat{\mathbf{y}} E_{y0}) e^{-jkz}, \quad (7.102)$$

7-6 ELECTROMAGNETIC POWER DENSITY

329

where, in the general case, E_{x0} and E_{y0} may be complex quantities. The magnitude of $\tilde{\mathbf{E}}$ is

$$|\tilde{\mathbf{E}}| = (\tilde{\mathbf{E}} \cdot \tilde{\mathbf{E}}^*)^{1/2} = [|E_{x0}|^2 + |E_{y0}|^2]^{1/2}. \quad (7.103)$$

The phasor magnetic field associated with $\tilde{\mathbf{E}}$ is obtained by applying Eq. (7.39a):

$$\begin{aligned} \tilde{\mathbf{H}}(z) &= (\hat{\mathbf{x}}\tilde{H}_x + \hat{\mathbf{y}}\tilde{H}_y)e^{-jkz} \\ &= \frac{1}{\eta} \hat{\mathbf{z}} \times \tilde{\mathbf{E}} = \frac{1}{\eta} (-\hat{\mathbf{x}}E_{y0} + \hat{\mathbf{y}}E_{x0})e^{-jkz}. \end{aligned} \quad (7.104)$$

The wave can be considered as the sum of two waves: one comprising fields $(\tilde{E}_x, \tilde{H}_y)$ and another comprising fields $(\tilde{E}_y, \tilde{H}_x)$. Use of Eqs. (7.102) and (7.104) in Eq. (7.100) leads to

$$\mathbf{S}_{av} = \hat{\mathbf{z}} \frac{1}{2\eta} (|E_{x0}|^2 + |E_{y0}|^2) = \hat{\mathbf{z}} \frac{|\tilde{\mathbf{E}}|^2}{2\eta} \quad (\text{W/m}^2), \quad (7.105)$$

(lossless medium)

which states that power flows in the z direction with average power density equal to the sum of the average power densities of the $(\tilde{E}_x, \tilde{H}_y)$ and $(\tilde{E}_y, \tilde{H}_x)$ waves. Note that, because $\mathbf{S}_{av}$ depends only on η and $|\tilde{\mathbf{E}}|$, waves characterized by different polarizations carry the same amount of average power as long as their electric fields have the same magnitudes.

Example 7-5: Solar Power

If solar illumination is characterized by a power density of 1 kW/m^2 on the Earth's surface, find (a) the total power radiated by the sun, (b) the total power intercepted by the Earth, and (c) the electric field of the power density incident upon the Earth's surface, assuming that all the solar illumination is at a single frequency. The radius of the Earth's orbit around the sun, R_s , is approximately $1.5 \times 10^8 \text{ km}$, and the Earth's mean radius R_e is $6,380 \text{ km}$.

Solution: (a) Assuming that the sun radiates isotropically (equally in all directions), the total power it radiates is $S_{av} A_{sph}$, where A_{sph} is the area of a spherical shell of radius R_s (Fig. 7-18(a)). Thus,

$$\begin{aligned} P_{\text{sun}} &= S_{av}(4\pi R_s^2) = 1 \times 10^3 \times 4\pi \times (1.5 \times 10^{11})^2 \\ &= 2.8 \times 10^{26} \text{ W}. \end{aligned}$$

(b) With reference to Fig. 7-18(b), the power intercepted by the Earth's cross section $A_e = \pi R_e^2$ is

$$P_{\text{int}} = S_{av}(\pi R_e^2) = 1 \times 10^3 \times \pi \times (6.38 \times 10^6)^2 = 1.28 \times 10^{17} \text{ W}.$$

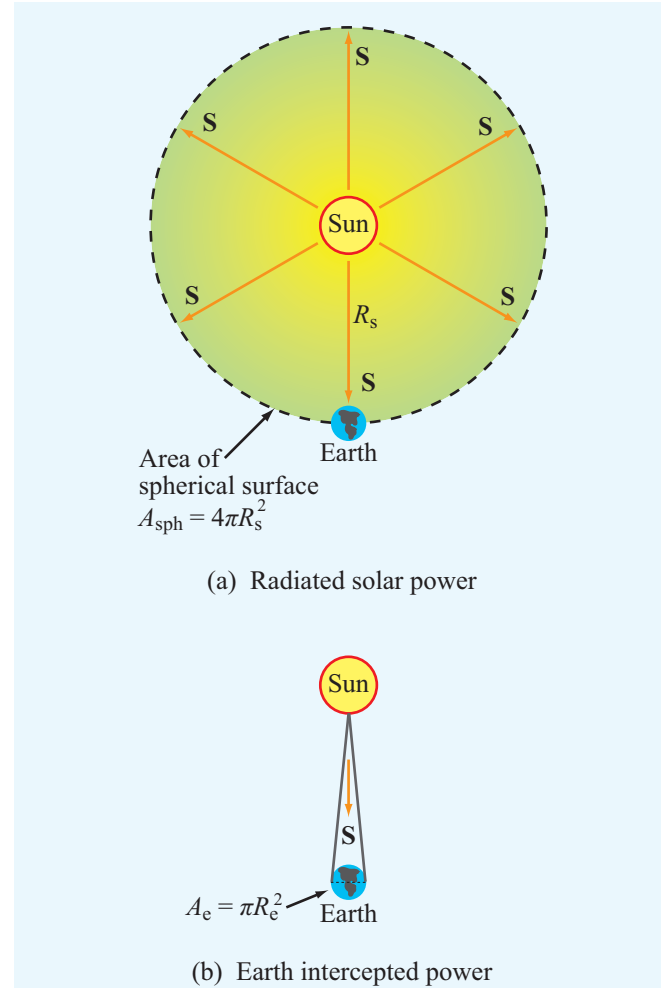


Figure 7-18 Solar radiation intercepted by (a) a spherical surface of radius R_s and (b) the Earth's surface (Example 7-5).

(c) The power density S_{av} is related to the magnitude of the electric field $|\tilde{\mathbf{E}}| = E_0$ by

$$S_{av} = \frac{E_0^2}{2\eta_0},$$

where $\eta_0 = 377 \text{ } (\Omega)$ for air. Hence,

$$E_0 = \sqrt{2\eta_0 S_{av}} = \sqrt{2 \times 377 \times 10^3} = 870 \quad (\text{V/m}).$$

7-6.2 Plane Wave in a Lossy Medium

The expressions given by Eqs. (7.68) and (7.69) characterize the electric and magnetic fields of an x -polarized plane wave

7-6 ELECTROMAGNETIC POWER DENSITY

329

其中, 在一般情况下, E_{x0} 和 E_{y0} 可以是复数。 $\tilde{\mathbf{E}}$ 的大小为

$$|\tilde{\mathbf{E}}| = (\tilde{\mathbf{E}} \cdot \tilde{\mathbf{E}}^*)^{1/2} = [|E_{x0}|^2 + |E_{y0}|^2]^{1/2}. \quad (7.103)$$

与 $\tilde{\mathbf{E}}$ 相关的相量磁场可通过应用等式获得。 (7.39a) :

$$\begin{aligned} \tilde{\mathbf{H}}(z) &= (\hat{\mathbf{x}}\tilde{H}_x + \hat{\mathbf{y}}\tilde{H}_y)e^{-jkz} \\ &= \frac{1}{\eta} \hat{\mathbf{z}} \times \tilde{\mathbf{E}} = \frac{1}{\eta} (-\hat{\mathbf{x}}E_{y0} + \hat{\mathbf{y}}E_{x0})e^{-jkz}. \end{aligned} \quad (7.104)$$

该波可以被视为两个波的总和: 一个波包含场 $(\tilde{E}_x, \tilde{H}_y)$, 另一个波包含场 $(\tilde{E}_y, \tilde{H}_x)$ 。使用等式 (7.102) 和 (7.104) (7.100) 导致

$$\mathbf{S}_{av} = \hat{\mathbf{z}} \frac{1}{2\eta} (|E_{x0}|^2 + |E_{y0}|^2) = \hat{\mathbf{z}} \frac{|\tilde{\mathbf{E}}|^2}{2\eta} \quad (\text{W/m}^2), \quad (7.105)$$

(lossless medium)

其中指出功率沿 z 方向流动, 平均功率密度等于 $(\tilde{E}_x, \tilde{H}_y)$ 和 $(\tilde{E}_y, \tilde{H}_x)$ 波的平均功率密度之和。请注意, 由于 $\mathbf{S}_{av}$ 仅取决于 η 和 $|\tilde{\mathbf{E}}|$, 因此只要电场强度相同, 具有不同偏振特征的波就会携带相同的平均功率。

例7-5: 太阳能

如果太阳光照的特征是地球表面的功率密度为 1 kW/m^2 , 则假设所有太阳光照都处于单一频率, 则求 (a) 太阳辐射的总功率, (b) 地球截获的总功率, 以及 (c) 入射到地球表面的功率密度的电场。地球绕太阳轨道的半径 R_s 约为 $1.5 \times 10^8 \text{ 公里}$, 地球平均半径 R_e 为 $6,380 \text{ 公里}$ 。

解: (a) 假设太阳各向同性 (向各个方向均匀) 辐射, 则其辐射的总功率为 $S_{av} A_{sph}$, 其中 A_{sph} 是半径为 R_s 的球表面积。7-18 (a) 。因此,

$$\begin{aligned} P_{\text{sun}} &= S_{av}(4\pi R_s^2) = 1 \times 10^3 \times 4\pi \times (1.5 \times 10^{11})^2 \\ &= 2.8 \times 10^{26} \text{ W}. \end{aligned}$$

(b) 参考图7-18(b), 地球横截面截获的功率 $A_e = \pi R_e^2$ 为

$$P_{\text{int}} = S_{av}(\pi R_e^2) = 1 \times 10^3 \times \pi \times (6.38 \times 10^6)^2 = 1.28 \times 10^{17} \text{ W}.$$

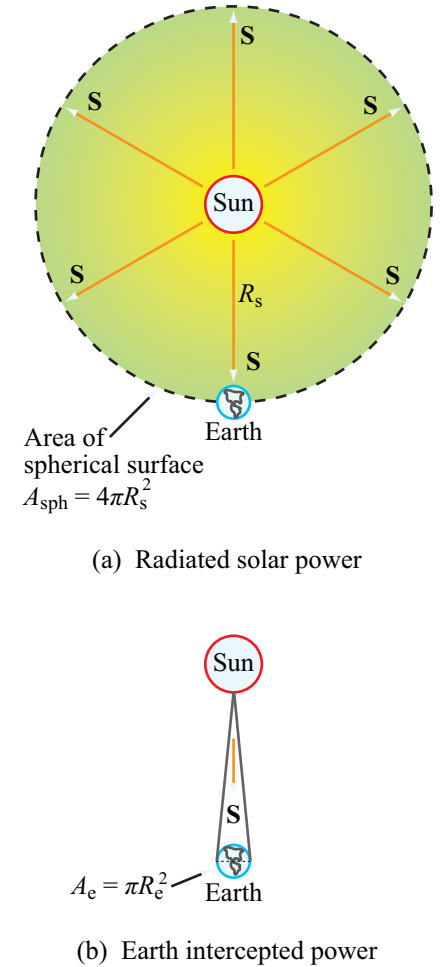


图 7-18 (a) 半径为 R_s 的球面和 (b) 地球表面截获的太阳辐射 (例 7-5)。

(c) 功率密度 S_{av} 与电场 $|\tilde{\mathbf{E}}| = E_0$ 的大小相关:

$$S_{av} = \frac{E_0^2}{2\eta_0},$$

其中 $\eta_0 = 377 \text{ } (\Omega)$ 代表空气。因此,

$$E_0 = \sqrt{2\eta_0 S_{av}} = \sqrt{2 \times 377 \times 10^3} = 870 \quad (\text{V/m}).$$

7-6.2 有损介质中的平面波

等式给出的表达式。 (7.68) 和 (7.69) 描述了 x 偏振平面波的电场和磁场

propagating along the z direction in a lossy medium with propagation constant $\gamma = \alpha + j\beta$. By extending these expressions to the more general case of a wave with components along both x and y , we have

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \widetilde{E}_x(z) + \hat{\mathbf{y}} \widetilde{E}_y(z) = (\hat{\mathbf{x}} E_{x0} + \hat{\mathbf{y}} E_{y0}) e^{-\alpha z} e^{-j\beta z},$$

(7.106a)

$$\widetilde{\mathbf{H}}(z) = \frac{1}{\eta_c} (-\hat{\mathbf{x}} E_{y0} + \hat{\mathbf{y}} E_{x0}) e^{-\alpha z} e^{-j\beta z},$$

(7.106b)

where η_c is the intrinsic impedance of the lossy medium. Application of Eq. (7.100) gives

$$\begin{aligned} \mathbf{S}_{\text{av}}(z) &= \frac{1}{2} \Re \left[\widetilde{\mathbf{E}} \times \widetilde{\mathbf{H}}^* \right] \\ &= \frac{\hat{\mathbf{z}} (|E_{x0}|^2 + |E_{y0}|^2)}{2} e^{-2\alpha z} \Re \left(\frac{1}{\eta_c^*} \right). \end{aligned}$$

(7.107)

By expressing η_c in polar form as

$$\eta_c = |\eta_c| e^{j\theta_\eta},$$

(7.108)

Eq. (7.107) can be rewritten as

$$\mathbf{S}_{\text{av}}(z) = \hat{\mathbf{z}} \frac{|\widetilde{E}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \quad (\text{W/m}^2),$$

(7.109)

(lossy medium)

where $|\widetilde{E}(0)|^2 = [|E_{x0}|^2 + |E_{y0}|^2]^{1/2}$ is the magnitude of $\widetilde{\mathbf{E}}(z)$ at $z = 0$.

► Whereas the fields $\widetilde{\mathbf{E}}(z)$ and $\widetilde{\mathbf{H}}(z)$ decay with z as $e^{-\alpha z}$, the power density $\mathbf{S}_{\text{av}}$ decreases as $e^{-2\alpha z}$. ◀

When a wave propagates through a distance $z = \delta_s = 1/\alpha$, the magnitudes of its electric and magnetic fields decrease to $e^{-1} \approx 37\%$ of their initial values, and its average power density decreases to $e^{-2} \approx 14\%$ of its initial value.

7-6.3 Decibel Scale for Power Ratios

The unit for power P is watts (W). In many engineering problems, the quantity of interest is the ratio of two power levels, P_1 and P_2 , such as the incident and reflected powers on a transmission line, and often the ratio P_1/P_2 may vary over several orders of magnitude. The decibel (dB) scale is logarithmic, thereby providing a convenient representation of

Table 7-2 Power ratios in natural numbers and in decibels.

G	G [dB]
10^x	10x dB
4	6 dB
2	3 dB
1	0 dB
0.5	−3 dB
0.25	−6 dB
0.1	−10 dB
10^{-3}	−30 dB

the power ratio, particularly when numerical values of P_1/P_2 are plotted against some variable of interest. If

$$G = \frac{P_1}{P_2},$$

(7.110)

then

$$\begin{aligned} G \text{ [dB]} &= 10 \log G \\ &= 10 \log \left(\frac{P_1}{P_2} \right) \quad (\text{dB}). \end{aligned}$$

(7.111)

Table 7-2 provides a comparison between values of G and the corresponding values of G [dB]. Even though decibels are defined for power ratios, they can sometimes be used to represent other quantities. For example, if $P_1 = V_1^2/R$ is the power dissipated in a resistor R with voltage V_1 across it at time t_1 , and $P_2 = V_2^2/R$ is the power dissipated in the same resistor at time t_2 , then

$$\begin{aligned} G \text{ [dB]} &= 10 \log \left(\frac{P_1}{P_2} \right) \\ &= 10 \log \left(\frac{V_1^2/R}{V_2^2/R} \right) \\ &= 20 \log \left(\frac{V_1}{V_2} \right) \\ &= 20 \log(g) \\ &= g \text{ [dB]}, \end{aligned}$$

(7.112)

where $g = V_1/V_2$ is the voltage ratio. *Note that for voltage (or current) ratios the scale factor is 20 rather than 10*, which results in G [dB] = g [dB].

The **attenuation rate**, representing the rate of decrease of the magnitude of $\mathbf{S}_{\text{av}}(z)$ as a function of propagation distance,

在有损介质中沿 z 方向传播，传播常数为 $\gamma = \alpha + j\beta$ 。通过将这些表达式扩展到具有沿 x 和 y 分量的波的更一般情况，我们有

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \widetilde{E}_x(z) + \hat{\mathbf{y}} \widetilde{E}_y(z) = (\hat{\mathbf{x}} E_{x0} + \hat{\mathbf{y}} E_{y0}) e^{-\alpha z} e^{-j\beta z},$$

(7.106a)

$$\widetilde{\mathbf{H}}(z) = \frac{1}{\eta_c} (-\hat{\mathbf{x}} E_{y0} + \hat{\mathbf{y}} E_{x0}) e^{-\alpha z} e^{-j\beta z},$$

(7.106b)

其中 η_c 是有损介质的固有阻抗。方程的应用(7.100) 给出

$$\begin{aligned} \mathbf{S}_{\text{av}}(z) &= \frac{1}{2} \Re \left[\widetilde{\mathbf{E}} \times \widetilde{\mathbf{H}}^* \right] \\ &= \frac{\hat{\mathbf{z}} (|E_{x0}|^2 + |E_{y0}|^2)}{2} e^{-2\alpha z} \Re \left(\frac{1}{\eta_c^*} \right). \end{aligned}$$

(7.107)

将 η_c 以极坐标形式表示为

$$\eta_c = |\eta_c| e^{j\theta_\eta},$$

(7.108)

等式。(7.107) 可以重写为

$$\mathbf{S}_{\text{av}}(z) = \hat{\mathbf{z}} \frac{|\widetilde{E}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \quad (\text{W/m}^2),$$

(7.109)

(lossy medium)

其中 $|\widetilde{E}(0)|^2 = [|E_{x0}|^2 + |E_{y0}|^2]^{1/2}$ 是 $\widetilde{\mathbf{E}}(z)$ 在 $z = 0$ 处的幅度。

► 当 z 为 $e^{-\alpha z}$ 时，场 $\widetilde{\mathbf{E}}(z)$ 和 $\widetilde{\mathbf{H}}(z)$ 衰减，而功率密度 $\mathbf{S}_{\text{av}}$ 则随着 $e^{-2\alpha z}$ 降低。 ◀

当波传播距离 $z = \delta_s = 1/\alpha$ 时，其电场和磁场的强度减小至初始值的 $e^{-1} \approx 37\%$ ，平均功率密度减小至初始值的 $e^{-2} \approx 14\%$ 。

7-6.3 功率比的分贝标度

功率P的单位是瓦（W）。在许多工程问题中，感兴趣的量是两个功率电平 P_1 和 P_2 的比率，例如传输线上的入射功率和反射功率，并且通常比率 P_1/P_2 可能会变化几个数量级。分贝 (dB) 刻度是对数的，从而可以方便地表示

表 7-2 以自然数和分贝为单位的功率比。

G	G [dB]
10^x	10x dB
4	6 dB
2	3 dB
1	0 dB
0.5	−3 dB
0.25	−6 dB
0.1	−10 dB
10^{-3}	−30 dB

功率比，特别是当 P_1/P_2 的数值针对某些感兴趣的变量绘制时。如果

$$G = \frac{P_1}{P_2},$$

(7.110)

然后

$$\begin{aligned} G \text{ [dB]} &= 10 \log G \\ &= 10 \log \left(\frac{P_1}{P_2} \right) \quad (\text{dB}). \end{aligned}$$

(7.111)

表 7-2 提供了 G 值和相应的 G [dB] 值之间的比较。尽管分贝是针对功率比定义的，但有时它们也可以用来表示其他量。例如，如果 $P_1 = V_1^2/R$ 是在时间 t_1 时电阻器 R 上的电压为 V_1 时消耗的功率，而 $P_2 = V_2^2/R$ 是在时间 t_2 时同一电阻器中消耗的功率，则

$$\begin{aligned} G \text{ [dB]} &= 10 \log \left(\frac{P_1}{P_2} \right) \\ &= 10 \log \left(\frac{V_1^2/R}{V_2^2/R} \right) \\ &= 20 \log \left(\frac{V_1}{V_2} \right) \\ &= 20 \log(g) \\ &= g \text{ [dB]}, \end{aligned}$$

(7.112)

其中 $g = V_1/V_2$ 是电压比。请注意，对于电压（或电流）比，比例因子为 20 而不是 10，结果为 G [dB] = g [dB]。
衰减率，表示 $\mathbf{S}_{\text{av}}(z)$ 幅度随传播距离变化的减小率，

is defined as

$$\begin{aligned} A &= 10 \log \left[\frac{S_{\text{av}}(z)}{S_{\text{av}}(0)} \right] \\ &= 10 \log(e^{-2\alpha z}) \\ &= -20\alpha z \log e = -8.68\alpha z = -\alpha \text{ [dB/m]} z \quad (\text{dB}), \end{aligned} \tag{7.113}$$

where

$$\alpha \text{ [dB/m]} = 8.68\alpha \quad (\text{Np/m}). \tag{7.114}$$

Since $S_{\text{av}}(z)$ is directly proportional to $|\mathbf{E}(z)|^2$,

$$A = 10 \log \left[\frac{|E(z)|^2}{|E(0)|^2} \right] = 20 \log \left[\frac{|E(z)|}{|E(0)|} \right] \quad (\text{dB}). \tag{7.115}$$

Example 7-6: Power Received by a Submarine Antenna

A submarine at a depth of 200 m below the sea surface uses a wire antenna to receive signal transmissions at 1 kHz. Determine the power density incident upon the submarine antenna due to the EM wave of Example 7-4.

Solution: From Example 7-4, $|\widetilde{E}(0)| = |E_{x0}| = 4.44 \text{ (mV/m)}$, $\alpha = 0.126 \text{ (Np/m)}$, and $\eta_c = 0.044/\underline{45^\circ} \text{ } (\Omega)$. Application of Eq. (7.109) gives

$$\begin{aligned} \mathbf{S}_{\text{av}}(z) &= \hat{\mathbf{z}} \frac{|E_0|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \\ &= \hat{\mathbf{z}} \frac{(4.44 \times 10^{-3})^2}{2 \times 0.044} e^{-0.252z} \cos 45^\circ \\ &= \hat{\mathbf{z}} 0.16 e^{-0.252z} \quad (\text{mW/m}^2). \end{aligned}$$

At $z = 200 \text{ m}$, the incident power density is

$$\begin{aligned} \mathbf{S}_{\text{av}} &= \hat{\mathbf{z}} (0.16 \times 10^{-3} e^{-0.252 \times 200}) \\ &= 2.1 \times 10^{-26} \quad (\text{W/m}^2). \end{aligned}$$

Exercise 7-10: Convert the following values of the power ratio G to decibels: (a) 2.3, (b) 4×10^3 , and (c) 3×10^{-2} .

Answer: (a) 3.6 dB, (b) 36 dB, (c) -15.2 dB . (See )

Exercise 7-11: Find the voltage ratio g corresponding to the following decibel values of the power ratio G : (a) 23 dB, (b) -14 dB , and (c) -3.6 dB .

Answer: (a) 14.13, (b) 0.2, (c) 0.66. (See )

Chapter 7 Summary

Concepts

- A spherical wave radiated by a source becomes approximately a uniform plane wave at large distances from the source.
- The electric and magnetic fields of a transverse electromagnetic (TEM) wave are orthogonal to each other, and both are perpendicular to the direction of wave travel.
- The magnitudes of the electric and magnetic fields of a TEM wave are related by the intrinsic impedance of the medium.
- Wave polarization describes the shape of the locus of the tip of the $\mathbf{E}$ vector at a given point in space as a function of time. The polarization state, which may be linear, circular, or elliptical, is governed by the ratio of

the magnitudes of and the difference in phase between the two orthogonal components of the electric field vector.

- Media are classified as lossless, low-loss, quasi-conducting, or good conducting on the basis of the ratio $\epsilon''/\epsilon' = \sigma/\omega\epsilon$.
- Unlike the dc case, where the current flowing through a wire is distributed uniformly across its cross section, the ac case has most of the current is concentrated along the outer perimeter of the wire.
- Power density carried by a plane EM wave traveling in an unbounded medium is akin to the power carried by the voltage/current wave on a transmission line.

定义为

$$\begin{aligned} A &= 10 \log \left[\frac{S_{\text{av}}(z)}{S_{\text{av}}(0)} \right] \\ &= 10 \log(e^{-2\alpha z}) \\ &= -20\alpha z \log e = -8.68\alpha z = -\alpha \text{ [dB/m]} z \quad (\text{dB}), \end{aligned} \tag{7.113}$$

在哪里

$$\alpha \text{ [dB/m]} = 8.68\alpha \quad (\text{Np/m}). \tag{7.114}$$

由于 $S_{\text{av}}(z)$ 与 $|E(z)|^2$ 成正比,

$$A = 10 \log \left[\frac{|E(z)|^2}{|E(0)|^2} \right] = 20 \log \left[\frac{|E(z)|}{|E(0)|} \right] \quad (\text{dB}). \tag{7.115}$$

例 7-6: 海底天线接收到的功率

位于海面以下 200 m 深度的潜艇使用线天线接收 1 kHz 的信号传输。确定例 7-4 中的电磁波作用在海底天线上的功率密度。

解: 根据例 7-4, $|\widetilde{E}(0)| = |E_{x0}| = 4.44 \text{ (mV/m)}$ 、 $\alpha = 0.126 \text{ (Np/m)}$ 和 $\eta_c = 0.044 \text{ } 45^\circ \text{ } (\Omega)$ 。方程的应用(7.109) 给出

$$\begin{aligned} \mathbf{S}_{\text{av}}(z) &= \hat{\mathbf{z}} \frac{|E_0|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \\ &= \hat{\mathbf{z}} \frac{(4.44 \times 10^{-3})^2}{2 \times 0.044} e^{-0.252z} \cos 45^\circ \\ &= \hat{\mathbf{z}} 0.16 e^{-0.252z} \quad (\text{mW/m}^2). \end{aligned}$$

在 $z = 200 \text{ m}$ 处, 入射功率密度为

$$\begin{aligned} \mathbf{S}_{\text{av}} &= \hat{\mathbf{z}} (0.16 \times 10^{-3} e^{-0.252 \times 200}) \\ &= 2.1 \times 10^{-26} \quad (\text{W/m}^2). \end{aligned}$$

练习7-10: 将下列功率比G值转换为分贝: (a) 2.3、(b) 4×10^3 和(c) 3×10^{-2} 。

Answer: (a) 3.6 dB, (b) 36 dB, (c) -15.2 dB . (See )

练习7-11: 求与以下功率比G的分贝值对应的电压比g: (a)23dB, (b) -14dB , (c) -3.6dB 。

答案: (a)14.13, (b)0.2, (c)0.66。 (参见^{EM}。)

第7章总结

概念

- 由源辐射的球面波在距源较远的距离处近似变为均匀的平面波。
- 横向电磁 (TEM) 波的电场和磁场彼此正交, 并且都垂直于波传播的方向。
- TEM 波的电场和磁场的大小与介质的固有阻抗相关。
- 波偏振描述了空间中给定点处 $\mathbf{E}$ 矢量尖端轨迹的形状随时间的变化。偏振态可以是线性、圆形或椭圆形, 由以下比率决定:

电场矢量的两个正交分量之间的幅度和相位差。

- 根据比率 $\epsilon''/\epsilon' = \sigma/\omega\epsilon$, 介质被分类为无损、低损耗、准传导或良好传导。
- 与直流情况不同, 流经电线的电流在其横截面上均匀分布, 而交流情况则大部分电流集中在电线的外周上。
- 在无界介质中传播的平面电磁波所承载的功率密度类似于传输线上的电压/电流波所承载的功率。

Mathematical and Physical Models

Complex Permittivity $\varepsilon_c = \varepsilon' - j\varepsilon''$ $\varepsilon' = \varepsilon$ $\varepsilon'' = \frac{\sigma}{\omega}$	Maxwell’s Equations for Time-Harmonic Fields $\nabla \cdot \tilde{\mathbf{E}} = 0$ $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$ $\nabla \cdot \tilde{\mathbf{H}} = 0$ $\nabla \times \tilde{\mathbf{H}} = j\omega\varepsilon_c\tilde{\mathbf{E}}$
Lossless Medium $k = \omega\sqrt{\mu\varepsilon}$ $\eta = \sqrt{\frac{\mu}{\varepsilon}} \quad (\Omega)$ $u_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}} \quad (\text{m/s})$ $\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m})$	Lossy Medium $\alpha = \omega \left\{ \frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'}\right)^2} - 1 \right] \right\}^{1/2} \quad (\text{Np/m})$ $\beta = \omega \left\{ \frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'}\right)^2} + 1 \right] \right\}^{1/2} \quad (\text{rad/m})$ $\eta_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon'}} \left(1 - j\frac{\varepsilon''}{\varepsilon'} \right)^{-1/2} \quad (\Omega)$ $\delta_s = \frac{1}{\alpha} \quad (\text{m})$
Wave Polarization $\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{k}} \times \tilde{\mathbf{E}}$ $\tilde{\mathbf{E}} = -\eta \hat{\mathbf{k}} \times \tilde{\mathbf{H}}$	Power Density $\mathbf{S}_{\text{av}} = \tfrac{1}{2} \Re \left[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right] \quad (\text{W/m}^2)$

Important Terms

Provide definitions or explain the meaning of the following terms:

attenuation constant α	in phase	propagation constant γ
attenuation rate A	inclination angle	quasi-conductor
auxiliary angle ψ_0	internal or surface impedance	rotation angle γ
average power density $\mathbf{S}_{\text{av}}$	intrinsic impedance η	skin depth δ_s
axial ratio	LHC and RHC polarizations	spherical wave
circular polarization	linear polarization	surface resistance R_s
complex permittivity ε_c	lossy medium	TEM wave
dc and ac resistances	low-loss dielectric	unbounded
elliptical polarization	out of phase	unbounded wave
ellipticity angle χ	phase constant β	uniform plane wave
good conductor	phase velocity	wave polarization
guided wave	polarization state	wavefront
homogeneous wave equation	Poynting vector $\mathbf{S}$	wavenumber k

数学与物理模型

复介电率 $\varepsilon_c = \varepsilon' - j\varepsilon''$ $\varepsilon' = \varepsilon$ $\varepsilon'' = \frac{\sigma}{\omega}$	时谐场的麦克斯韦方程组 $\nabla \cdot \tilde{\mathbf{E}} = 0$ $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$ $\nabla \cdot \tilde{\mathbf{H}} = 0$ $\nabla \times \tilde{\mathbf{H}} = j\omega\varepsilon_c\tilde{\mathbf{E}}$
无损介质 $k = \omega\sqrt{\mu\varepsilon}$ $\eta = \sqrt{\frac{\mu}{\varepsilon}} \quad (\Omega)$ $u_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}} \quad (\text{m/s})$ $\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m})$	有损介质 $\alpha = \omega \left\{ \frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'}\right)^2} - 1 \right] \right\}^{1/2} \quad (\text{Np/m})$ $\beta = \omega \left\{ \frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'}\right)^2} + 1 \right] \right\}^{1/2} \quad (\text{rad/m})$ $\eta_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon'}} \left(1 - j\frac{\varepsilon''}{\varepsilon'} \right)^{-1/2} \quad (\Omega)$ $\delta_s = \frac{1}{\alpha} \quad (\text{m})$
波的极化 $\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{k}} \times \tilde{\mathbf{E}}$ $\tilde{\mathbf{E}} = -\eta \hat{\mathbf{k}} \times \tilde{\mathbf{H}}$	平均功率密度 $\mathbf{S}_{\text{av}} = \tfrac{1}{2} \Re \left[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right] \quad (\text{W/m}^2)$

重要术语

请给出下列术语的定义或解释其含义。

衰减常数 α	同相	传播常数 γ
衰减率 A	倾角	准导体
辅助角 ψ_0	内阻抗或表面阻抗	旋转角 γ
平均功率密度 $\mathbf{S}_{\text{av}}$	本征阻抗 η	趋肤深度 δ_s
轴比	左旋圆极化与右旋圆极化	球面波
圆极化	线极化	表面电阻 R_s
复介电率 ε_c	有损介质	TEM波
直流电阻与交流电阻	低损耗电介质	无界
椭圆极化	异相	无界波
椭圆率角 χ	相位常数 β	均匀平面波
良导体	相速度	波的极化
导行波	极化状态	波前
齐次波动方程	坡印廷矢量 $\mathbf{S}$	波数 k

适用范围：上表采用 $\text{e}^{\text{j}\omega t}$ 时间约定；场方程用于无外加源的线性、均匀、各向同性介质，传导电流并入复介电率。电、磁场叉积关系表示平面波的场方向与波阻抗关系；有损介质须使用复波阻抗 η_c 。

PROBLEMS

Section 7-2: Propagation in Lossless Media

7.1 The magnetic field of a wave propagating through a certain nonmagnetic material is given by

$$\mathbf{H} = \hat{\mathbf{z}}30 \cos(10^8 t - 0.5y) \quad (\text{mA/m}).$$

Find:

- (a)** the direction of wave propagation,
- (b)** the phase velocity,
- (c)** the wavelength in the material,
- (d)** the relative permittivity of the material, and
- (e)** the electric field phasor.

7.2 Write general expressions for the electric and magnetic fields of a 1 GHz sinusoidal plane wave traveling in the +y direction in a lossless nonmagnetic medium with relative permittivity $\epsilon_r = 9$. The electric field is polarized along the x direction, its peak value is 6 V/m, and its intensity is 4 V/m at $t = 0$ and $y = 2$ cm.

7.3 The electric field phasor of a uniform plane wave is given by $\tilde{\mathbf{E}} = \hat{\mathbf{y}}10e^{j0.2z}$ (V/m). If the phase velocity of the wave is 1.5×10^8 m/s and the relative permeability of the medium is $\mu_r = 2.4$, find:

- (a)** the wavelength,
- (b)** the frequency f of the wave,
- (c)** the relative permittivity of the medium, and
- (d)** the magnetic field $\mathbf{H}(z, t)$.

7.4 The electric field of a plane wave propagating in a nonmagnetic material is given by

$$\mathbf{E} = [\hat{\mathbf{y}}3 \sin(\pi \times 10^7 t - 0.2\pi x) + \hat{\mathbf{z}}4 \cos(\pi \times 10^7 t - 0.2\pi x)] \quad (\text{V/m})$$

Determine:

- (a)** the wavelength
- (b)** ϵ_r
- (c)** $\mathbf{H}$

***7.5** A wave radiated by a source in air is incident upon a soil surface, where a part of the wave is transmitted into the soil medium. If the wavelength of the wave is 60 cm in air and 20 cm in the soil medium, what is the soil’s relative permittivity? Assume the soil to be a very low-loss medium.

* Answer(s) available in Appendix E.

7.6 The electric field of a plane wave propagating in a lossless, nonmagnetic, dielectric material with $\epsilon_r = 2.56$ is given by

$$\mathbf{E} = \hat{\mathbf{y}}20 \cos(6\pi \times 10^9 t - kz) \quad (\text{V/m}).$$

Determine:

- (a)** f , u_p , λ , k , and η .
- (b)** the magnetic field $\mathbf{H}$.

7.7 The magnetic field of a plane wave propagating in a nonmagnetic material is given by

$$\mathbf{H} = \hat{\mathbf{x}}60 \cos(2\pi \times 10^7 t + 0.1\pi y) + \hat{\mathbf{z}}30 \cos(2\pi \times 10^7 t + 0.1\pi y) \quad (\text{mA/m}).$$

Determine:

- (a)** the wavelength
- (b)** ϵ_r
- (c)** $\mathbf{E}$

7.8 A 60 MHz plane wave traveling in the $-x$ direction in dry soil with relative permittivity $\epsilon_r = 4$ has an electric field polarized along the z direction. Assuming dry soil to be approximately lossless, and given that the magnetic field has a peak value of 10 (mA/m) and that its value was measured to be 7 (mA/m) at $t = 0$ and $x = -0.75$ m, develop complete expressions for the wave’s electric and magnetic fields.

Section 7-3: Wave Polarization

***7.9** An RHC-polarized wave with a modulus of 2 (V/m) is traveling in free space in the negative z direction. Write the expression for the wave’s electric field vector given that the wavelength is 6 cm.

7.10 For a wave characterized by the electric field

$$\mathbf{E}(z, t) = \hat{\mathbf{x}}a_x \cos(\omega t - kz) + \hat{\mathbf{y}}a_y \cos(\omega t - kz + \delta),$$

identify the polarization state, determine the polarization angles (γ, χ) , sketch the locus of $\mathbf{E}(0, t)$, and verify your solution using Module 7.3 for each of the following cases.

- (a)** $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = 0$
- (b)** $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = 180^\circ$
- (c)** $a_x = 3$ V/m, $a_y = 3$ V/m, and $\delta = 45^\circ$
- (d)** $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = -135^\circ$

问题

第 7-2 节：无损媒体中的传播

7.1 通过某种非磁性材料传播的波的磁场由下式给出

$$\mathbf{H} = \hat{\mathbf{z}}30 \cos(10^8 t - 0.5y) \quad (\text{mA/m}).$$

寻找：

- (a)** 波传播的方向，
- (b)** 相速度，
- (c)** 材料中的波长，
- (d)** 材料的相对介电常数，以及 **(e)** 电场相量。

7.2 写出 1 GHz 正弦平面波在相对介电常数为 $\epsilon_r =$ 的无损非磁性介质中沿 +y 方向传播的电场和磁场的一般表达式。
 9. 电场沿 x 方向极化，其峰值为 6 V/m，在 $t = 0$ 和 $y = 2$ cm 处强度为 4 V/m。

7.3 均匀平面波的电场相量由 $\tilde{\mathbf{E}} = \hat{\mathbf{y}}10e^{j0.2z}$ (V/m) 给出。如果波的相速度为 1.5×10^8 m/s，介质的相对磁导率为 $\mu_r = 2.4$ ，则可得：

- (a)** 波长，
- (b)** 波的频率 f ，
- (c)** 介质的相对介电常数，以及
- (d)** 磁场 $\mathbf{H}(z, t)$ 。

7.4 在非磁性材料中传播的平面波的电场由下式给出

$$\mathbf{E} = [\hat{\mathbf{y}}3 \sin(\pi \times 10^7 t - 0.2\pi x) + \hat{\mathbf{z}}4 \cos(\pi \times 10^7 t - 0.2\pi x)] \quad (\text{V/m})$$

决定：

- (a)** 波长 **(b)** ϵ_r **(c)** $\mathbf{H}$

***7.5** 空气中的源辐射的波入射到土壤表面，一部分波传输到土壤介质中。如果波的波长在空气中为 60 厘米，在土壤介质中为 20 厘米，那么土壤的相对介电常数是多少？假设土壤是损耗非常低的介质。

* Answer(s) available in Appendix E.

7.6 在 $\epsilon_r = 2.56$ 的无损、非磁性介电材料中传播的平面波的电场由下式给出

$$\mathbf{E} = \hat{\mathbf{y}}20 \cos(6\pi \times 10^9 t - kz) \quad (\text{V/m}).$$

决定：

- (a)** f 、 u_p 、 λ 、 k 和 η
- (b)** 磁场 $\mathbf{H}$ 。

7.7 在非磁性材料中传播的平面波的磁场由下式给出

$$\mathbf{H} = \hat{\mathbf{x}}60 \cos(2\pi \times 10^7 t + 0.1\pi y) + \hat{\mathbf{z}}30 \cos(2\pi \times 10^7 t + 0.1\pi y) \quad (\text{mA/m}).$$

决定：

- (a)** 波长 **(b)** ϵ_r **(c)** $\mathbf{E}$

7.8 60 MHz 平面波在相对介电常数为 $\epsilon_r = 4$ 的干燥土壤中沿 $-x$ 方向传播，具有沿 z 方向极化的电场。假设干燥土壤大致无损耗，并且磁场的峰值为 10 (mA/m)，并且在 $t = 0$ 和 $x = -0.75$ m 处测得其值为 7 (mA/m)，请建立波电场和磁场的完整表达式。

第 7-3 节：波偏振

***7.9** 模数为 2 (V/m) 的 RHC 偏振波在自由空间中沿负 z 方向传播。假设波长为 6 cm，请写出波的电场矢量表达式。

7.10 对于以电场为特征的波

$$\mathbf{E}(z, t) = \hat{\mathbf{x}}a_x \cos(\omega t - kz) + \hat{\mathbf{y}}a_y \cos(\omega t - kz + \delta),$$

识别偏振态，确定偏振角 (γ, χ) ，绘制 $\mathbf{E}(0, t)$ 轨迹，并使用模块 7.3 针对以下每种情况验证您的解决方案。

- (a)** $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = 0$
- (b)** $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = 180^\circ$
- (c)** $a_x = 3$ V/m, $a_y = 3$ V/m, and $\delta = 45^\circ$
- (d)** $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = -135^\circ$

7.11 The electric field of a uniform plane wave propagating in free space is given by

$$\tilde{\mathbf{E}} = (\hat{\mathbf{x}} + j\hat{\mathbf{y}})30e^{-j\pi z/6} \quad (\text{V/m}).$$

Specify the modulus and direction of the electric field intensity at the $z = 0$ plane at $t = 0, 5$, and 10 ns.

***7.12** The magnetic field of a uniform plane wave propagating in a dielectric medium with $\epsilon_r = 36$ is given by

$$\tilde{\mathbf{H}} = 30(\hat{\mathbf{y}} + j\hat{\mathbf{z}})e^{-j\pi x/6} \quad (\text{mA/m}).$$

Specify the modulus and direction of the electric field intensity at the $x = 0$ plane at $t = 0$ and 5 ns.

7.13 A linearly polarized plane wave of the form $\tilde{\mathbf{E}} = \hat{\mathbf{x}}a_x e^{-jkz}$ can be expressed as the sum of an RHC-polarized wave with magnitude a_R and an LHC-polarized wave with magnitude a_L . Prove this statement by finding expressions for a_R and a_L in terms of a_x .

***7.14** The electric field of an elliptically polarized plane wave is given by

$$\begin{aligned} \mathbf{E}(z,t) = & [-\hat{\mathbf{x}}10\sin(\omega t - kz - 60^\circ) \\ & + \hat{\mathbf{y}}30\cos(\omega t - kz)] \quad (\text{V/m}). \end{aligned}$$

Determine:

- (a) the polarization angles (γ, χ) , and
 (b) the direction of rotation.

7.15 Compare the polarization states of each of the following pairs of plane waves:

- (a) Wave 1: $\mathbf{E}_1 = \hat{\mathbf{x}}2\cos(\omega t - kz) + \hat{\mathbf{y}}2\sin(\omega t - kz)$
 Wave 2: $\mathbf{E}_2 = \hat{\mathbf{x}}2\cos(\omega t + kz) + \hat{\mathbf{y}}2\sin(\omega t + kz)$
 (b) Wave 1: $\mathbf{E}_1 = \hat{\mathbf{x}}2\cos(\omega t - kz) - \hat{\mathbf{y}}2\sin(\omega t - kz)$
 Wave 2: $\mathbf{E}_2 = \hat{\mathbf{x}}2\cos(\omega t + kz) - \hat{\mathbf{y}}2\sin(\omega t + kz)$

7.16 Plot the locus of $\mathbf{E}(0,t)$ for a plane wave with

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\sin(\omega t + kz) + \hat{\mathbf{y}}2\cos(\omega t + kz).$$

Determine the polarization state from your plot.

Sections 7-4: Propagation in a Lossy Medium

7.17 For each of the following combinations of parameters, determine if the material is a low-loss dielectric, a quasi-conductor, or a good conductor. Then calculate α , β , λ , u_p , and η_c .

*** (a)** Glass with $\mu_r = 1$, $\epsilon_r = 5$, and $\sigma = 10^{-12}$ S/m at 10 GHz.

(b) Animal tissue with $\mu_r = 1$, $\epsilon_r = 12$, and $\sigma = 0.3$ S/m at 100 MHz.

(c) Wood with $\mu_r = 1$, $\epsilon_r = 3$, and $\sigma = 10^{-4}$ S/m at 1 kHz.

7.18 Dry soil is characterized by $\epsilon_r = 2.5$, $\mu_r = 1$, and $\sigma = 10^{-4}$ (S/m). At each of the following frequencies, determine if dry soil may be considered a good conductor, a quasi-conductor, or a low-loss dielectric. Then calculate α , β , λ , μ_p , and η_c :

- (a) 60 Hz
 (b) 1 kHz
 (c) 1 MHz
 (d) 1 GHz

***7.19** In a medium characterized by $\epsilon_r = 9$, $\mu_r = 1$, and $\sigma = 0.1$ S/m, determine the phase angle by which the magnetic field leads the electric field at 100 MHz.

7.20 Generate a plot for the skin depth δ_s versus frequency for seawater for the range from 1 kHz to 10 GHz (use log-log scales). The constitutive parameters of seawater are $\mu_r = 1$, $\epsilon_r = 80$, and $\sigma = 4$ S/m.

***7.21** Ignoring reflection at the air–soil boundary, if the amplitude of a 3 GHz incident wave is 10 V/m at the surface of a wet soil medium, at what depth will it be down to 1 mV/m? Wet soil is characterized by $\mu_r = 1$, $\epsilon_r = 9$, and $\sigma = 5 \times 10^{-4}$ S/m.

7.22 Ignoring reflection at the air–water boundary, if the amplitude of a 1 GHz incident wave in air is 20 V/m at the water surface, at what depth will it be down to 1μ V/m? Water has $\mu_r = 1$ and at 1 GHz, $\epsilon_r = 80$, and $\sigma = 1$ S/m.

***7.23** The skin depth of a certain nonmagnetic conducting material is $3 \mu\text{m}$ at 5 GHz. Determine the phase velocity in the material.

7.24 Based on wave attenuation and reflection measurements conducted at 1 MHz, it was determined that the intrinsic impedance of a certain medium is $28.1/\underline{45^\circ} \ (\Omega)$ and the skin depth is 2 m. Determine:

- (a) the conductivity of the material,
 (b) the wavelength in the medium, and
 (c) the phase velocity.

***7.25** The electric field of a plane wave propagating in a nonmagnetic medium is given by

$$\mathbf{E} = \hat{\mathbf{z}}25e^{-30x}\cos(2\pi \times 10^9 t - 40x) \quad (\text{V/m}).$$

Obtain the corresponding expression for $\mathbf{H}$.

7.11 在自由空间中传播的均匀平面波的电场由下式给出

$$\tilde{\mathbf{E}} = (\hat{\mathbf{x}} + j\hat{\mathbf{y}})30e^{-j\pi z/6} \quad (\text{V/m}).$$

指定 $z = 0$ 平面在 $t = 0, 5$ 和 10 ns 处的电场强度的模量和方向。

***7.12** 在 $\epsilon_r = 36$ 的介电介质中传播的均匀平面波的磁场由下式给出

$$\tilde{\mathbf{H}} = 30(\hat{\mathbf{y}} + j\hat{\mathbf{z}})e^{-j\pi x/6} \quad (\text{mA/m}).$$

指定 $x = 0$ 平面在 $t = 0$ 和 5 ns 处的电场强度的模量和方向。

7.13 $\tilde{\mathbf{E}} = \hat{\mathbf{x}}a_x e^{-jkz}$ 形式的线偏振平面波可以表示为幅度为 a_R 的 RHC 偏振波与幅度为 a_L 的 LHC 偏振波之和。通过根据 a_x 找到 a_R 和 a_L 的表达式来证明该命题。

***7.14** 椭圆偏振平面波的电场由下式给出

$$\begin{aligned} \mathbf{E}(z,t) = & [-\hat{\mathbf{x}}10\sin(\omega t - kz - 60^\circ) \\ & + \hat{\mathbf{y}}30\cos(\omega t - kz)] \quad (\text{V/m}). \end{aligned}$$

决定:

- (a) 偏振角 (γ, χ) , 以及
 (b) 旋转方向。

7.15 比较下列各对平面波的偏振态:

- (a) 第 1 波: $\mathbf{E}_1 = \hat{\mathbf{x}}2\cos(\omega t - kz) + \hat{\mathbf{y}}2\sin(\omega t - kz)$
 第 2 波: $\mathbf{E}_2 = \hat{\mathbf{x}}2\cos(\omega t + kz) + \hat{\mathbf{y}}2\sin(\omega t + kz)$
 (b) 第 1 波: $\mathbf{E}_1 = \hat{\mathbf{x}}2\cos(\omega t - kz) - \hat{\mathbf{y}}2\sin(\omega t - kz)$
 第 2 波: $\mathbf{E}_2 = \hat{\mathbf{x}}2\cos(\omega t + kz) - \hat{\mathbf{y}}2\sin(\omega t + kz)$

7.16 绘制平面波 $\mathbf{E}(0,t)$ 的轨迹

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\sin(\omega t + kz) + \hat{\mathbf{y}}2\cos(\omega t + kz).$$

从图中确定偏振态。

第 7-4 节: 有损介质中的传播

7.17 对于以下每个参数组合, 确定该材料是低损耗电介质、准导体还是良导体。然后计算 α 、 β 、 λ 、 u_p 和 η_c 。

*** (a)** 10 GHz 时具有 $\mu_r = 1$ 、 $\epsilon_r = 5$ 和 $\sigma = 10^{-12}$ S/m 的玻璃。

(b) 100 MHz 时 $\mu_r = 1$ 、 $\epsilon_r = 12$ 和 $\sigma = 0.3$ S/m 的动物组织。

(c) 木材, 在 1 kHz 时具有 $\mu_r = 1$ 、 $\epsilon_r = 3$ 和 $\sigma = 10^{-4}$ S/m。

7.18 干燥土壤的特征为 $\epsilon_r = 2.5$ 、 $\mu_r = 1$ 和 $\sigma = 10^{-4}$ (S/m)。在以下每个频率下, 确定干燥土壤是否可以被视为良导体、准导体或低损耗电介质。然后计算 α 、 β 、 λ 、 μ_p 和 η_c :

- (a) 60 Hz (
 b) 1 kHz (c
) 1 MHz (d
) 1 GHz

***7.19** 在具有 $\epsilon_r = 9$ 、 $\mu_r = 1$ 和 $\sigma = 0.1$ S/m 特征的介质中, 确定 100 MHz 时磁场超前于电场的相位角。

7.20 生成 1 kHz 至 10 GHz 范围内海水趋肤深度 δ_s 与频率的关系图 (使用双对数刻度)。海水的本构参数为 $\mu_r = 1$ 、 $\epsilon_r = 80$ 和 $\sigma = 4$ S/m。

***7.21** 忽略空气-土壤边界处的反射, 如果 3 GHz 入射波的幅度在湿土壤介质表面为 10 V/m, 那么在什么深度它会降至 1 mV/m? 湿土的特征为 $\mu_r = 1$ 、 $\epsilon_r = 9$ 和 $\sigma = 5 \times 10^{-4}$ S/m。

7.22 忽略空气-水边界处的反射, 如果空气中 1 GHz 入射波的振幅在水面处为 20 V/m, 那么在什么深度处它会降至 1μ V/m? 水的 $\mu_r = 1$ 在 1 GHz、 $\epsilon_r = 80$ 和 $\sigma = 1$ S/m 。

***7.23** 某种非磁性导电材料的集肤深度在 5 GHz 时为 $3 \mu\text{m}$ 。确定材料中的相速度。

7.24 根据在 1 MHz 下进行的波衰减和反射测量, 确定某种介质的固有阻抗为 $28.1 \times 45^\circ \ (\Omega)$, 集肤深度为 2 m。决定:

- (a) 材料的电导率, (b) 介质中的波长,
 , 以及 (c) 相速度。

***7.25** 在非磁性介质中传播的平面波的电场由下式给出

$$\mathbf{E} = \hat{\mathbf{z}}25e^{-30x}\cos(2\pi \times 10^9 t - 40x) \quad (\text{V/m}).$$

求出H对应的表达式。

7.26 The magnetic field of a plane wave propagating in a nonmagnetic medium is given by

$$\mathbf{H} = \hat{\mathbf{y}} 60e^{-10z} \cos(2\pi \times 10^8 t - 12z) \quad (\text{mA/m}).$$

Obtain the corresponding expression for $\mathbf{E}$.

7.27 At 2 GHz, the conductivity of meat is on the order of 1 (S/m). When a material is placed inside a microwave oven and the field is activated, the presence of the electromagnetic fields in the conducting material causes energy dissipation in the material in the form of heat.

- (a) Develop an expression for the time-average power per mm^3 dissipated in a material of conductivity σ if the peak electric field in the material is E_0 .
- (b) Evaluate the result for an electric field $E_0 = 4 \times 10^4$ (V/m).

Section 7-5: Current Flow in Conductors

7.28 In a nonmagnetic, lossy, dielectric medium, a 300 MHz plane wave is characterized by the magnetic field phasor

$$\tilde{\mathbf{H}} = (\hat{\mathbf{x}} - j4\hat{\mathbf{z}})e^{-2y}e^{-j9y} \quad (\text{A/m}).$$

Obtain time-domain expressions for the electric and magnetic field vectors.

***7.29** A rectangular copper block is 30 cm in height (along z). In response to a wave incident upon the block from above, a current is induced in the block in the positive x direction. Determine the ratio of the ac resistance of the block to its dc resistance at 1 kHz. The relevant properties of copper are given in Appendix B.

7.30 Repeat Problem 7.29 at 10 MHz.

7.31 The inner and outer conductors of a coaxial cable have radii of 0.5 cm and 1 cm, respectively. The conductors are made of copper with $\epsilon_r = 1$, $\mu_r = 1$, and $\sigma = 5.8 \times 10^7$ S/m. The outer conductor is 0.5 mm thick. At 10 MHz:

- (a) Are the conductors thick enough to be considered infinitely thick as far as the flow of current through them is concerned?
- (b) Determine the surface resistance R_s .
- (c) Determine the ac resistance per unit length of the cable.

7.32 Repeat Problem 7.31 at 1 GHz.

Section 7-6: EM Power Density

***7.33** The magnetic field of a plane wave traveling in air is given by $\mathbf{H} = \hat{\mathbf{x}} 50 \sin(2\pi \times 10^7 t - ky)$ (mA/m). Determine the average power density carried by the wave.

7.34 A wave traveling in a nonmagnetic medium with $\epsilon_r = 9$ is characterized by an electric field given by

$$\mathbf{E} = [\hat{\mathbf{y}} 3 \cos(\pi \times 10^7 t + kx) - \hat{\mathbf{z}} 2 \cos(\pi \times 10^7 t + kx)] \quad (\text{V/m}).$$

Determine the direction of wave travel and average power density carried by the wave.

7.35 The electric-field phasor of a uniform plane wave traveling downward in water is given by

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}} 5e^{-0.2z}e^{-j0.2z} \quad (\text{V/m}),$$

where $\hat{\mathbf{z}}$ is the downward direction and $z = 0$ is the water surface. If $\sigma = 4$ S/m,

- (a) Obtain an expression for the average power density.
- (b) Determine the attenuation rate.
- * (c) Determine the depth at which the power density has been reduced by 40 dB.

7.36 The amplitudes of an elliptically polarized plane wave traveling in a lossless, nonmagnetic medium with $\epsilon_r = 4$ are $H_{y0} = 3$ (mA/m) and $H_{z0} = 4$ (mA/m). Determine the average power flowing through an aperture in the y - z plane if its area is 20 m^2 .

***7.37** A wave traveling in a lossless, nonmagnetic medium has an electric field amplitude of 24.56 V/m and an average power density of 2.4 W/m^2 . Determine the phase velocity of the wave.

7.38 At microwave frequencies, the power density considered safe for human exposure is $1 \text{ (mW/cm}^2\text{)}$. A radar radiates a wave with an electric field amplitude E that decays with distance as $E(R) = (3,000/R)$ (V/m), where R is the distance in meters. What is the radius of the unsafe region?

7.39 Consider the imaginary rectangular box shown in Fig. P7.39.

- (a) Determine the net power flux $P(t)$ entering the box due to a plane wave in air given by

$$\mathbf{E} = \hat{\mathbf{x}} E_0 \cos(\omega t - ky) \quad (\text{V/m}).$$

- * (b) Determine the net time-average power entering the box.

7.40 Repeat Problem 7.39 for a wave traveling in a lossy medium in which

$$\mathbf{E} = \hat{\mathbf{x}} 100e^{-20y} \cos(2\pi \times 10^9 t - 40y) \quad (\text{V/m}),$$

$$\mathbf{H} = -\hat{\mathbf{z}} 0.64e^{-20y} \cos(2\pi \times 10^9 t - 40y - 36.85^\circ) \quad (\text{A/m}).$$

The box has dimensions $a = 1$ cm, $b = 2$ cm, and $c = 0.5$ cm.

7.26 在非磁性介质中传播的平面波的磁场由下式给出

$$\mathbf{H} = \hat{\mathbf{y}} 60e^{-10z} \cos(2\pi \times 10^8 t - 12z) \quad (\text{mA/m}).$$

求出E对应的表达式。

7.27 在 2 GHz 时，肉的电导率约为 1 (S/m)。当将材料放入微波炉内并激活磁场时，导电材料中存在的电磁场会导致材料以热量的形式耗散能量。

(a) 如果材料中的峰值电场为 E_0 ，则建立电导率 σ 材料中每毫米³ 耗散的时间平均功率的表达式。(b) 评估电场 $E_0 = 4 \times 10^4$ (V/m) 的结果。

第 7-5 节：导体中的电流

7.28 在非磁性、有损耗的介电介质中，300 MHz 平面波的特征在于磁场相量

$$\tilde{\mathbf{H}} = (\hat{\mathbf{x}} - j4\hat{\mathbf{z}})e^{-2y}e^{-j9y} \quad (\text{A/m}).$$

获得电场和磁场矢量的时域表达式。

*7.29 矩形铜块高 30 厘米（沿 z 方向）。响应于从上方入射到块上的波，在块中沿正 x 方向感应出电流。确定 1 kHz 时模块的交流电阻与其直流电阻之比。铜的相关特性在附录B中给出。

7.30 在 10 MHz 下重复问题 7.29。

7.31 同轴电缆的内导体和外导体的半径分别为 0.5 厘米和 1 厘米。导体由 $\epsilon_r = 1$ 、 $\mu_r = 1$ 和 $\sigma = 5.8 \times 10^7$ S/m 的铜制成。外导体厚度为 0.5 毫米。10 MHz 时：

(a) 就流经导体的电流而言，导体的厚度是否足以被视为无限厚？(b) 确定表面电阻 R_s 。(c) 确定电缆单位长度的交流电阻。7.32 在 1 GHz 下重复问题 7.31。

第 7-6 节：电磁功率密度

*7.33 在空气中传播的平面波的磁场由 $\mathbf{H} = \hat{\mathbf{x}} 50 \sin(2\pi \times 10^7 t - ky)$ (mA/m) 给出。确定波所携带的平均功率密度。

7.34 在具有 $\epsilon_r = 9$ 的非磁性介质中传播的波的特征在于电场为：

$$\mathbf{E} = [\hat{\mathbf{y}} 3 \cos(\pi \times 10^7 t + kx) - \hat{\mathbf{z}} 2 \cos(\pi \times 10^7 t + kx)] \quad (\text{V/m}).$$

确定波传播的方向和波所携带的平均功率密度。

7.35 在水中向下传播的均匀平面波的电场相量由下式给出

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}} 5e^{-0.2z}e^{-j0.2z} \quad (\text{V/m}),$$

其中 $\hat{\mathbf{z}}$ 是向下方向， $z = 0$ 是水面。如果 $\sigma = 4$ S/m，

- (a) 获得平均功率密度的表达式。
- (b) 确定衰减率。* (c) 确定功率密度降低 40 dB 的深度。

7.36 在具有 $\epsilon_r = 4$ 的无损非磁性介质中传播的椭圆偏振平面波的振幅为 $H_{y0} = 3$ (mA/m) 和 $H_{z0} = 4$ (mA/m)。如果 y - z 平面上的孔径面积为 20 m^2 ，则确定流经该孔径的平均功率。

*7.37 在无损、非磁性介质中传播的波的电场幅度为 24.56 V/m，平均功率密度为 2.4 W/m^2 。确定波的相速度。

7.38 在微波频率下，人体暴露安全的功率密度为 $1 \text{ (mW/c m}^2\text{)}$ 。雷达辐射电场幅度为 E 的波，该波随距离衰减为 $E(R) = (3,000/R)$ (V/m)，其中 R 是以米为单位的距离。不安全区域的半径是多少？

7.39 考虑图 P7.39 所示的假想矩形框。

- (a) 确定由于空气中的平面波而进入盒子的净功率通量 $P(t)$ ：

$$\mathbf{E} = \hat{\mathbf{x}} E_0 \cos(\omega t - ky) \quad (\text{V/m}).$$

- * (b) 确定进入框中的净时间平均功率。

7.40 对于在无损介质中传播的波，重复问题 7.39，其中

$$\mathbf{E} = \hat{\mathbf{x}} 100e^{-20y} \cos(2\pi \times 10^9 t - 40y) \quad (\text{V/m}),$$

$$\mathbf{H} = -\hat{\mathbf{z}} 0.64e^{-20y} \cos(2\pi \times 10^9 t - 40y - 36.85^\circ) \quad (\text{A/m}).$$

盒子的尺寸为 $a = 1$ 厘米、 $b = 2$ 厘米和 $c = 0.5$ 厘米。

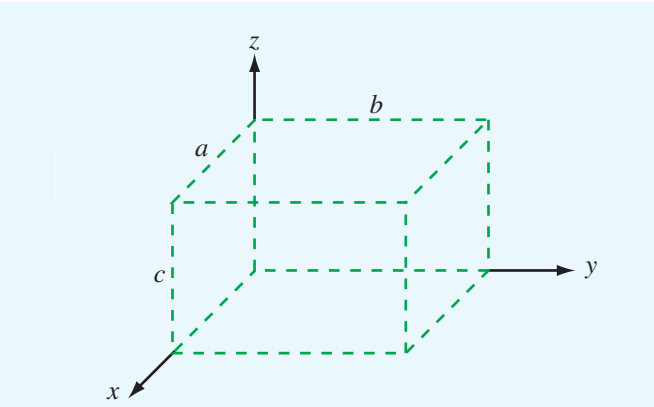


Figure P7.39 Imaginary rectangular box of Problems 7.39 and 7.40.

7.41 Given a wave with $\mathbf{E} = \hat{\mathbf{x}} E_0 \cos(\omega t - kz)$:

*** (a)** Calculate the time-average electric energy density

$$(w_e)_{av} = \frac{1}{T} \int_0^T w_e dt = \frac{1}{2T} \int_0^T \epsilon E^2 dt.$$

(b) Calculate the time-average magnetic energy density

$$(w_m)_{av} = \frac{1}{T} \int_0^T w_m dt = \frac{1}{2T} \int_0^T \mu H^2 dt.$$

(c) Show that $(w_e)_{av} = (w_m)_{av}$.

7.42 A team of scientists is designing a radar as a probe for measuring the depth of the ice layer over the antarctic land mass. In order to measure a detectable echo due to the reflection by the ice-rock boundary, the thickness of the ice sheet should not exceed three skin depths. If $\epsilon'_r = 3$ and $\epsilon''_r = 10^{-2}$ for ice and if the maximum anticipated ice thickness in the area under exploration is 1.2 km, what frequency range is useable with the radar?

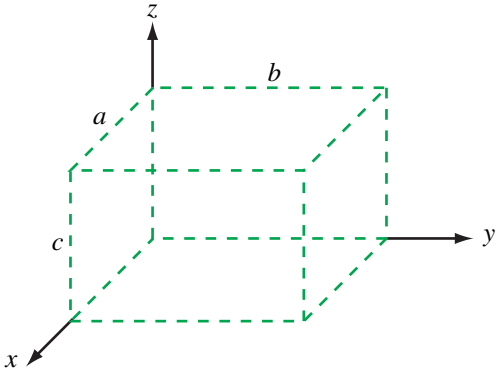


图 P7.39 问题 7.39 和 7.40 的假想矩形框。

7.41 给定一个具有 $\mathbf{E} = \hat{\mathbf{x}} E_0 \cos(\omega t - kz)$ 的波：

*** (a)** 计算时均电能密度

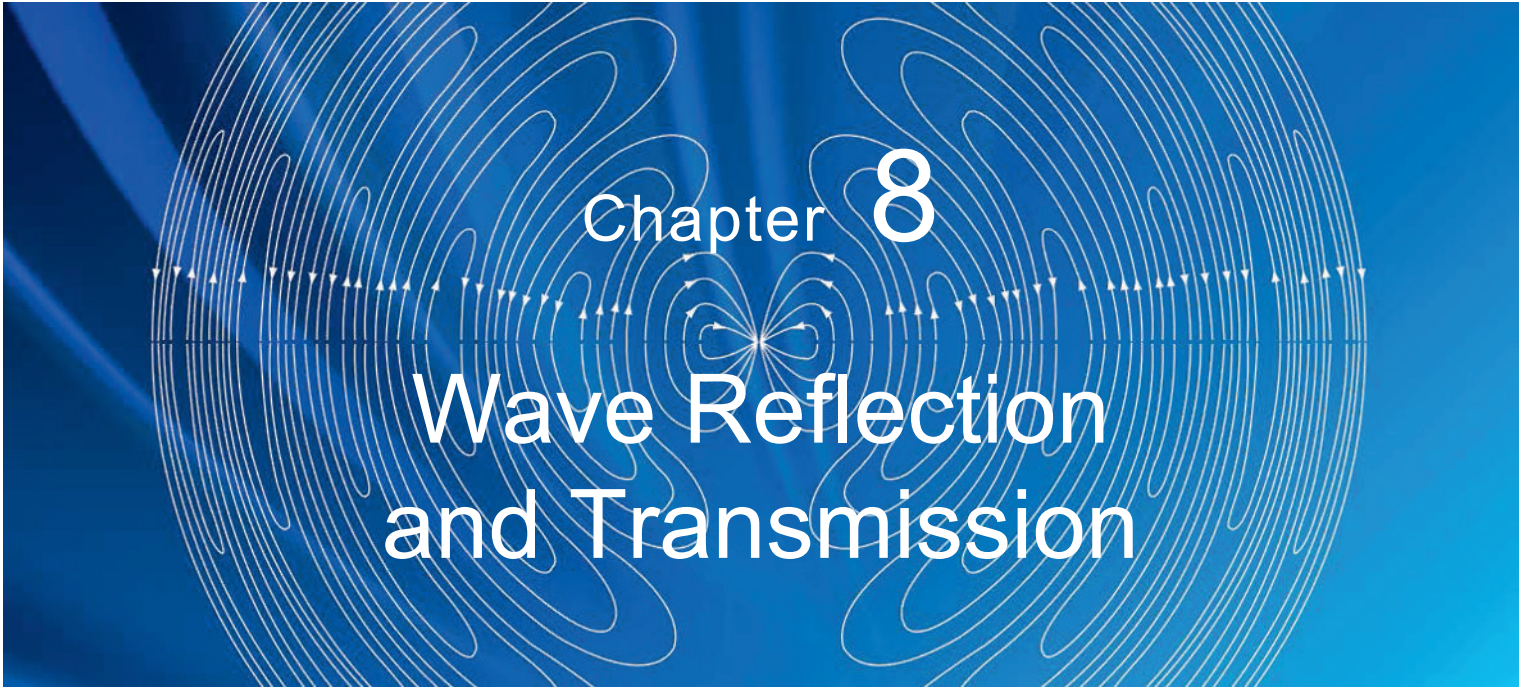
$$(w_e)_{av} = \frac{1}{T} \int_0^T w_e dt = \frac{1}{2T} \int_0^T \epsilon E^2 dt.$$

(b) 计算时均磁能密度

$$(w_m)_{av} = \frac{1}{T} \int_0^T w_m dt = \frac{1}{2T} \int_0^T \mu H^2 dt.$$

(c) 显示 $(w_e)_{av} = (w_m)_{av}$ 。

7.42 一个科学家小组正在设计一种雷达作为探测器，用于测量南极陆地上冰层的深度。为了测量由于冰岩边界反射而产生的可检测回波，冰盖的厚度不应超过三倍趋肤深度。如果冰的情况为 $\epsilon'_r = 3$ 和 $\epsilon''_r = 10^{-2}$ ，并且如果勘探区域的最大预期冰厚度为 1.2 公里，则雷达可使用什么频率范围？



Chapter Contents

	EM Waves at Boundaries, 338
8-1	Wave Reflection and Transmission at Normal Incidence, 338
TB15	Lasers, 347
8-2	Snell’s Laws, 349
8-3	Fiber Optics, 351
8-4	Wave Reflection and Transmission at Oblique Incidence, 353
TB16	Bar-Code Readers, 358
8-5	Reflectivity and Transmissivity, 361
8-6	Waveguides, 364
8-7	General Relations for E and H , 366
8-8	TM Modes in Rectangular Waveguide, 367
8-9	TE Modes in Rectangular Waveguide, 370
8-10	Propagation Velocities, 371
8-11	Cavity Resonators, 374
	Chapter 8 Summary, 377
	Problems, 378

Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Characterize the reflection and transmission behavior of plane waves incident upon plane boundaries for both normal and oblique incidence.
2. Calculate the transmission properties of optical fibers.
3. Characterize wave propagation in a rectangular waveguide.
4. Determine the behavior of resonant modes inside a rectangular cavity.



章节内容

边界处的电磁波, 338 8-1 法向入射时的波反射和传输, 338 TB15 激光器, 347 8-2 斯涅尔定律, 349 8-3 光纤, 351 8-4 倾斜入射时的波反射和传输, 353 TB16 条码阅读器, 358 8-5 反射率和反射率透射率、361 8-6 波导、364 8-7 **E** 和 **H** 的一般关系、366 8-8 矩形波导中的 **TM** 模式、367 8-9 矩形波导中的 **TE** 模式、370 8-10 传播速度、371 8-11 空腔谐振器、374 第 8 章总结、377 问题、第378 章

目标

学习本章介绍的材料后, 您应该能够:

1. 表征垂直和倾斜入射的平面波在平面边界上的反射和传输行为。
2. 计算光纤的传输特性。
3. 表征波在矩形波导中的传播。
4. 确定矩形腔内谐振模式的行为。

EM Waves at Boundaries

Figure 8-1 depicts the propagation path traveled by a signal transmitted by a shipboard antenna and received by an antenna on a submerged submarine. Starting from the transmitter (denoted Tx in **Fig. 8-1**), the signal travels along a transmission line to the transmitting antenna. The relationship between the transmitter (generator) output power, P_t , and the power supplied to the antenna is governed by the transmission-line equations of Chapter 2. If the transmission line is approximately lossless and properly matched to the transmitting antenna, then all of P_t is delivered to the antenna. If the antenna itself is lossless too, it will convert all of the power P_t in the guided wave provided by the transmission line into a spherical wave radiated outward into space. The radiation process is the subject of Chapter 9. From point 1, denoting the location of the shipboard antenna, to point 2, denoting the point of incidence of the wave onto the water's surface, the signal's behavior is governed by the equations characterizing wave propagation in lossless media, which was covered in Chapter 7. As the wave impinges upon the air–water boundary, part of it is reflected by the surface while another part gets transmitted across the boundary into the water. The transmitted wave is refracted, wherein its propagation direction moves closer toward the vertical, compared with that of the direction of the incident wave. Reflection and transmission processes are treated in this chapter. Wave travel from point 3, representing a point just below the water surface, to point 4, which denotes the location of the submarine antenna, is subject to the laws of wave propagation in lossy media, which also was treated in

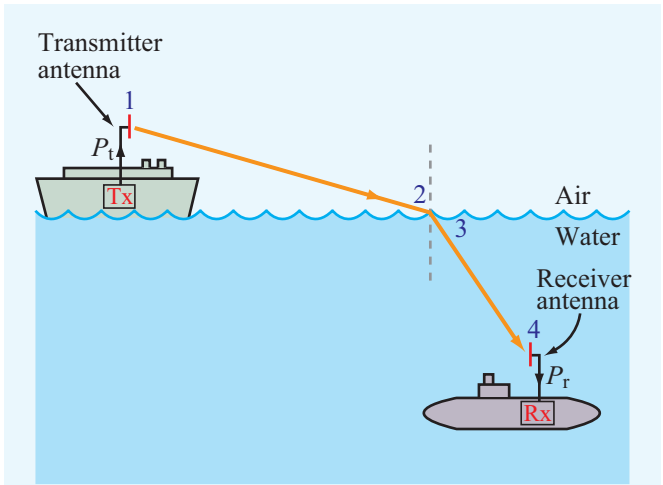


Figure 8-1 Signal path between a shipboard transmitter (Tx) and a submarine receiver (Rx).

Chapter 7. Finally, some of the power carried by the wave traveling in water towards the submarine is intercepted by the receiving antenna. The received power, P_r , is then delivered to the receiver via a transmission line. The receiving properties of antennas are covered in Chapter 9. In summary, each wave-related aspect of the transmission process depicted in **Fig. 8-1** is treated in this book, starting with the transmitter and ending with the receiver.

This chapter begins by examining the reflection and transmission properties of plane waves incident upon planar boundaries and concludes with sections on waveguides and cavity resonators. Applications discussed along the way include fiber and laser optics.

8-1 Wave Reflection and Transmission at Normal Incidence

We know from Chapter 2 that, when a guided wave encounters a junction between two transmission lines with different characteristic impedances, the incident wave is partly reflected back toward the source and partly transmitted across the junction onto the other line. The same happens to a uniform plane wave when it encounters a boundary between two material half-spaces with different characteristic impedances. In fact, the situation depicted in **Fig. 8-2(b)** has an exact analogue in the transmission-line configuration of **Fig. 8-2(a)**. The boundary conditions governing the relationships between the electric and magnetic fields in **Fig. 8-2(b)** map one-to-one onto those we developed in Chapter 2 for the voltages and currents on the transmission line.

For convenience, we divide our treatment of wave reflection by and transmission through planar boundaries into two parts. In this section, we confine our discussion to the normal-incidence case depicted in **Fig. 8-3(a)**, and in Sections 8-2 to 8-4, we examine the more general oblique-incidence case depicted in **Fig. 8-3(b)**. We will show the basis for the analogy between the transmission-line and plane-wave configurations so that we may use transmission-line equivalent models, tools (e.g., Smith chart), and techniques (e.g., quarter-wavelength matching) to expeditiously solve plane-wave problems.

Before proceeding, however, we should explain the notion of rays and wavefronts and the relationship between them, as both are used throughout this chapter to represent electromagnetic waves. A **ray** is a line representing the direction of flow of electromagnetic energy carried by a wave; therefore, it is parallel to the propagation unit vector $\hat{\mathbf{k}}$. A **wavefront** is a surface across which the phase of a wave is constant; it is perpendicular to the wavevector $\hat{\mathbf{k}}$. Hence, rays are perpendicular to wavefronts. The ray representation of wave incidence, reflection, and transmission shown in **Fig. 8-3(b)** is equivalent

边界处的电磁波

图8-1描述了由舰载天线发射并由水下潜艇上的天线接收的信号所经过的传播路径。从发射机（图 8-1 中表示为 Tx）开始，信号沿着传输线传输到发射天线。发射器（发生器）输出功率 P_t 与提供给天线的功率之间的关系由第 2 章的传输线方程控制。如果传输线近似无损并且与发射天线正确匹配，则所有 P_t 都会传递到天线。如果天线本身也是无损的，它会将传输线提供的导波中的所有功率 P_t 转换为向外辐射到空间的球面波。辐射过程是第 9 章的主题。从点 1（表示舰载天线的位置）到点 2（表示波在水面上的入射点），信号的行为受表征无损介质中的波传播的方程控制，这在第 7 章中进行了介绍。当波撞击空气-水边界时，一部分被表面反射，而另一部分则穿过边界传输到水中。发射波被折射，其中与入射波的方向相比，其传播方向更接近垂直方向。本章讨论反射和传输过程。波从代表水面下方的点 3 到代表潜艇天线位置的点 4 的传播受到有损介质中波传播定律的影响，该定律也已在

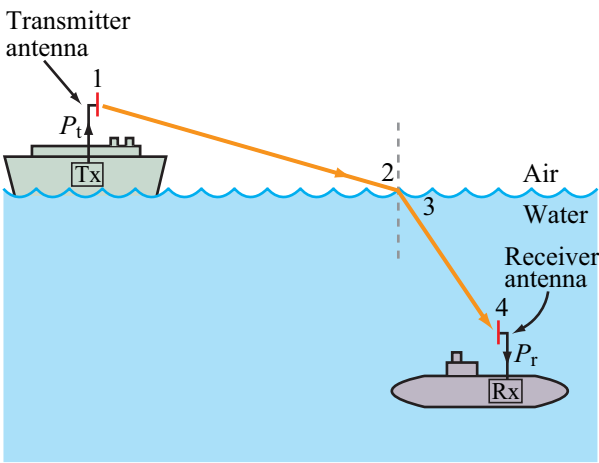


图 8-1 舰载发射器 (Tx) 和海底接收器 (Rx) 之间的信号路径。

第 7 章 最后，在水中传播到潜艇的波所携带的一些功率被接收天线拦截。然后，接收到的功率 P_r 通过传输线传送到接收器。第 9 章介绍了天线的接收特性。总之，本书讨论了图 8-1 中描述的传输过程中与波相关的每个方面，从发射机开始，到接收机结束。

本章首先研究入射到平面边界上的平面波的反射和传输特性，最后介绍波导和空腔谐振器。一路上讨论的应用包括光纤和激光光学。

8-1 法向入射时的波反射和透射

从第二章我们知道，当导波遇到具有不同特性阻抗的两条传输线之间的连接点时，入射波部分被反射回源，部分穿过连接点传输到另一条线路上。当均匀平面波遇到具有不同特性阻抗的两个材料半空间之间的边界时，也会发生同样的情况。事实上，图 8-2(b) 中描述的情况与图 8-2(a) 的传输线配置完全相同。图 8-2(b) 中控制电场和磁场之间关系的边界条件——映射到我们在第 2 章中针对传输线上的电压和电流开发的边界条件。

为了方便起见，我们将通过平面边界对波反射和传输的处理分为两部分。在本节中，我们将讨论限制在图 8-3(a) 所示的正入射情况下，在第 8-2 节到 8-4 节中，我们将研究图 8-3(b) 中所示的更一般的斜入射情况。我们将展示传输线和平面波配置之间的类比基础，以便我们可以使用传输线等效模型、工具（例如史密斯圆图）和技术（例如四分之一波长匹配）来快速解决平面波问题。

然而，在继续之前，我们应该解释射线和波前的概念以及它们之间的关系，因为本章通篇都使用两者来表示电磁波。射线是代表波所携带的电磁能流动方向的线；因此，它平行于传播单位向量 $\hat{\mathbf{k}}$ 。波前是波的相位恒定的表面；它垂直于波矢 $\hat{\mathbf{k}}$ 。因此，光线垂直于波前。图8-3(b)所示的波入射、反射和透射的射线表示是等效的

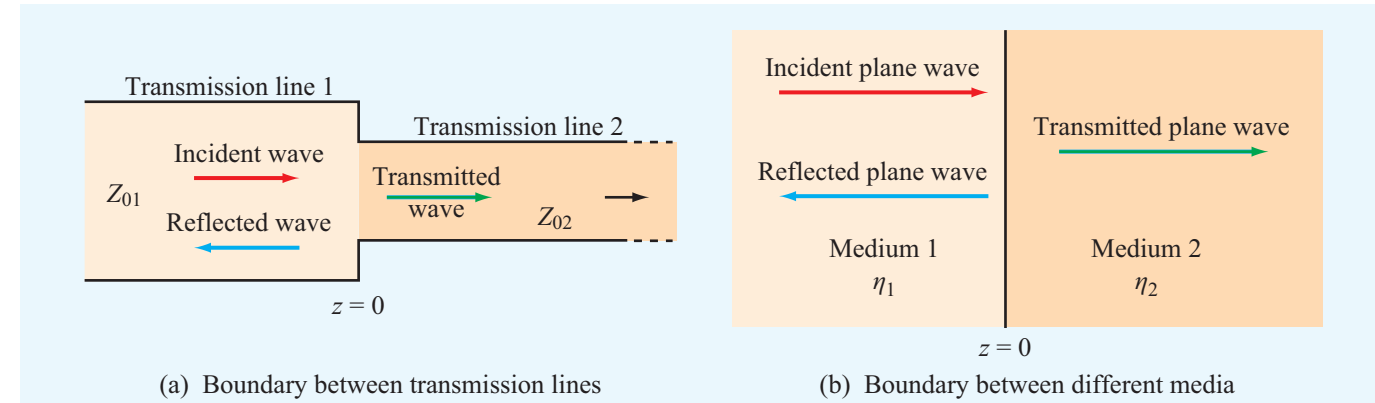


Figure 8-2 Discontinuity between two different transmission lines is analogous to that between two dissimilar media.

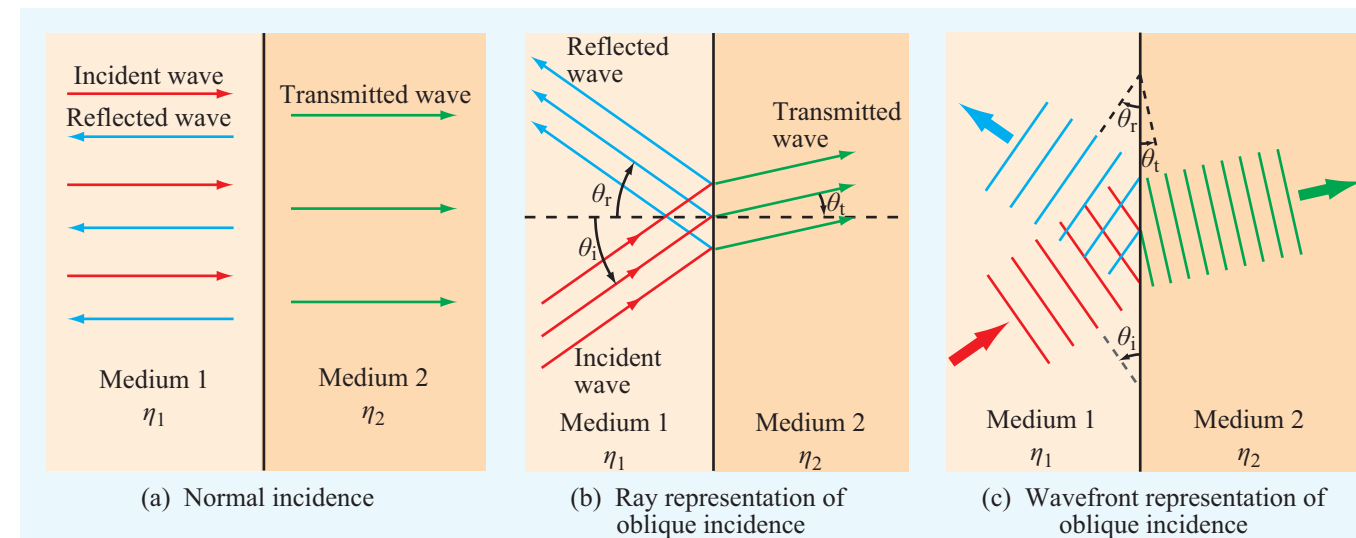


Figure 8-3 Ray representation of wave reflection and transmission at (a) normal incidence and (b) oblique incidence. (c) Wavefront representation of oblique incidence.

to the wavefront representation depicted in **Fig. 8-3(c)**. The two representations are complementary; the ray representation is easier to use in graphical illustrations, whereas the wavefront representation provides greater physical insight into what happens to a wave when it encounters a discontinuous boundary.

8-1.1 Boundary between Lossless Media

A planar boundary located at $z = 0$ (**Fig. 8-4(a)**) separates two lossless, homogeneous, dielectric media. Medium 1 has permittivity ϵ_1 and permeability μ_1 and fills the half-space

$z \leq 0$. Medium 2 has permittivity ϵ_2 and permeability μ_2 and fills the half-space $z \geq 0$. An x -polarized plane wave, with electric and magnetic fields ($\mathbf{E}^i, \mathbf{H}^i$) propagates in medium 1 along direction $\hat{\mathbf{k}}_i = \hat{\mathbf{z}}$ toward medium 2. Reflection and transmission at the boundary at $z = 0$ result in a reflected wave, with electric and magnetic fields ($\mathbf{E}^r, \mathbf{H}^r$), traveling along direction $\hat{\mathbf{k}}_r = -\hat{\mathbf{z}}$ in medium 1, and a transmitted wave, with electric and magnetic fields ($\mathbf{E}^t, \mathbf{H}^t$), traveling along direction $\hat{\mathbf{k}}_t = \hat{\mathbf{z}}$ in medium 2. On the basis of the formulations developed in Sections 7-2 and 7-3 for plane waves, the three waves are described in phasor form by:

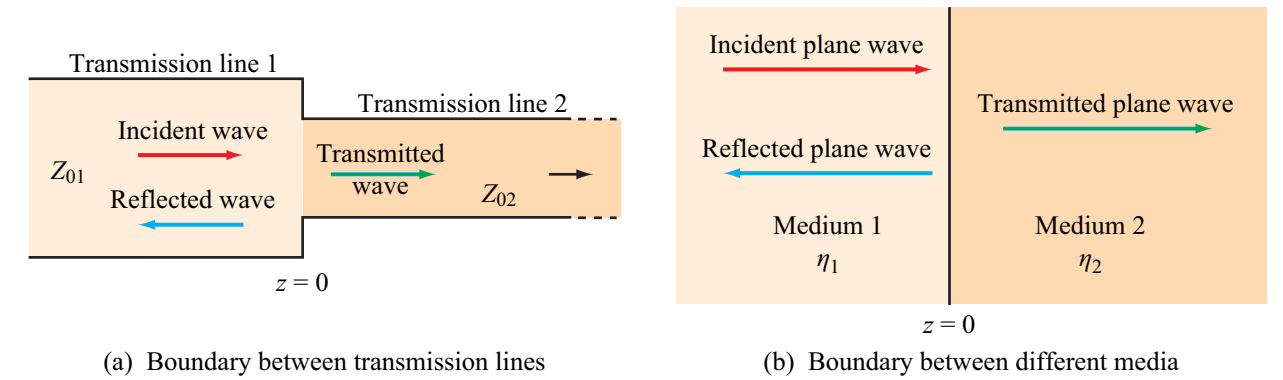
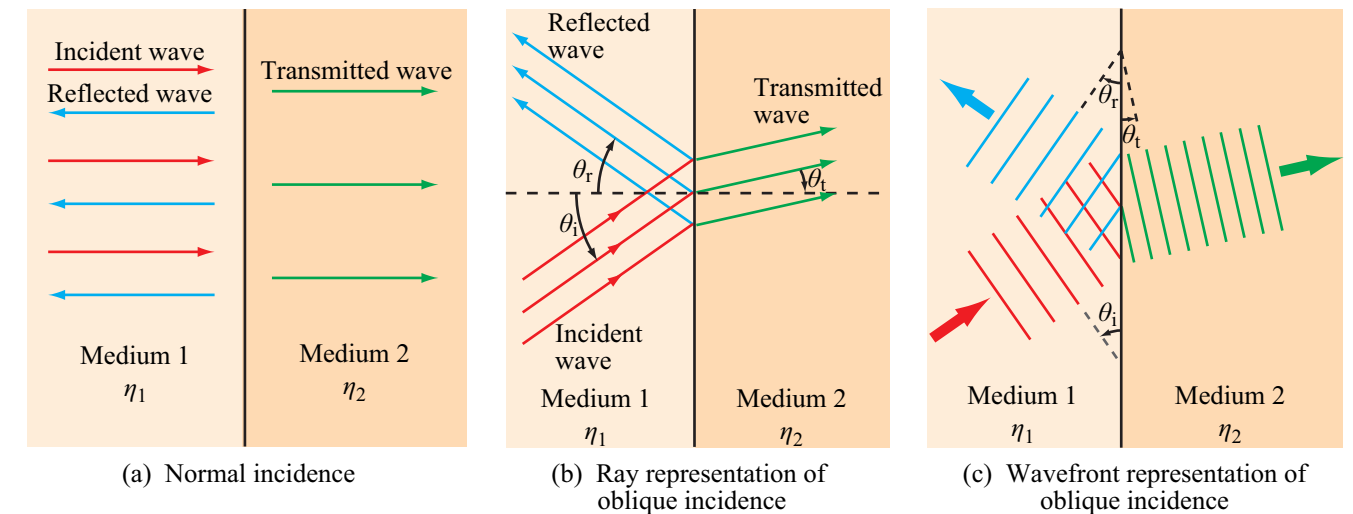


图 8-2 两条不同传输线之间的不连续性类似于两种不同介质之间的不连续性。



菲图 8-3 (a) 垂直入射和 (b) 倾斜入射时波反射和透射的射线表示。(c) 波前代表入射的怨恨。

到图8-3(c)中描绘的波前表示。这两种表示是互补的；射线表示更容易在图形插图使用，而波前表示可以更深入地了解波遇到不连续边界时会发生什么。

8-1.1 无损媒体之间的界限

位于 $z = 0$ 处的平面边界 (图 8-4(a)) 将两种无损、均匀的介电介质分开。介质 1 具有介电常数 ϵ_1 和磁导率 μ_1 并填充半空间

$z \leq 0$ 。介质 2 具有介电常数 ϵ_2 和磁导率 μ_2 ，并填充半空间 $z \geq 0$ 。具有电场和磁场 ($\mathbf{E}^i, \mathbf{H}^i$) 的 x 偏振平面波在介质 1 中沿方向 $\hat{\mathbf{k}}_i = \hat{\mathbf{z}}$ 向介质 2 传播。在 $z = 0$ 边界处的反射和透射产生具有电场和磁场 ($\mathbf{E}^r, \mathbf{H}^r$) 的反射波，在介质 1 中沿 $\hat{\mathbf{k}}_r = -\hat{\mathbf{z}}$ 方向传播，以及具有电场和磁场 ($\mathbf{E}^t, \mathbf{H}^t$) 的透射波，在介质 2 中沿 $\hat{\mathbf{k}}_t = \hat{\mathbf{z}}$ 方向传播。7-2和7-3对于平面波，这三个波以相量形式描述为：

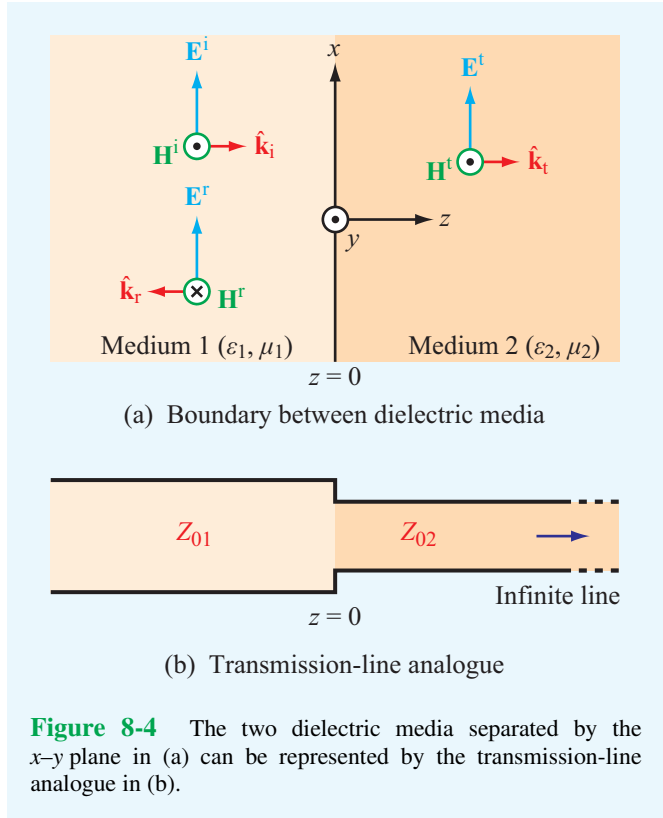


Figure 8-4 The two dielectric media separated by the x - y plane in (a) can be represented by the transmission-line analogue in (b).

Incident Wave

$$\tilde{\mathbf{E}}^i(z) = \hat{\mathbf{x}} E_0^i e^{-jk_1 z}, \quad (8.1a)$$

$$\tilde{\mathbf{H}}^i(z) = \hat{\mathbf{z}} \times \frac{\tilde{\mathbf{E}}^i(z)}{\eta_1} = \hat{\mathbf{y}} \frac{E_0^i}{\eta_1} e^{-jk_1 z}. \quad (8.1b)$$

Reflected Wave

$$\tilde{\mathbf{E}}^r(z) = \hat{\mathbf{x}} E_0^r e^{jk_1 z}, \quad (8.2a)$$

$$\tilde{\mathbf{H}}^r(z) = (-\hat{\mathbf{z}}) \times \frac{\tilde{\mathbf{E}}^r(z)}{\eta_1} = -\hat{\mathbf{y}} \frac{E_0^r}{\eta_1} e^{jk_1 z}. \quad (8.2b)$$

Transmitted Wave

$$\tilde{\mathbf{E}}^t(z) = \hat{\mathbf{x}} E_0^t e^{-jk_2 z}, \quad (8.3a)$$

$$\tilde{\mathbf{H}}^t(z) = \hat{\mathbf{z}} \times \frac{\tilde{\mathbf{E}}^t(z)}{\eta_2} = \hat{\mathbf{y}} \frac{E_0^t}{\eta_2} e^{-jk_2 z}. \quad (8.3b)$$

The quantities E_0^i , E_0^r , and E_0^t are, respectively, the amplitudes of the incident, reflected, and transmitted electric fields at $z=0$ (the boundary between the two media). The wavenumber and intrinsic impedance of medium 1 are $k_1 = \omega\sqrt{\mu_1\epsilon_1}$ and

$\eta_1 = \sqrt{\mu_1/\epsilon_1}$, and those for medium 2 are $k_2 = \omega\sqrt{\mu_2\epsilon_2}$ and $\eta_2 = \sqrt{\mu_2/\epsilon_2}$.

The amplitude E_0^i is imposed by the source responsible for generating the incident wave; therefore, it is assumed known. Our goal is to relate E_0^r and E_0^t to E_0^i . We do so by applying boundary conditions for the total electric and magnetic fields at $z=0$. According to **Table 6-2**, the tangential component of the total electric field is always continuous across a boundary between two contiguous media, and in the absence of current sources at the boundary, the same is true for the total magnetic field. In the present case, the electric and magnetic fields of the incident, reflected, and transmitted waves are all tangential to the boundary.

The total electric field $\tilde{\mathbf{E}}_1(z)$ in medium 1 is the sum of the electric fields of the incident and reflected waves, and a similar statement applies to the magnetic field $\tilde{\mathbf{H}}_1(z)$. Hence,

Medium 1

$$\tilde{\mathbf{E}}_1(z) = \tilde{\mathbf{E}}^i(z) + \tilde{\mathbf{E}}^r(z) = \hat{\mathbf{x}} (E_0^i e^{-jk_1 z} + E_0^r e^{jk_1 z}), \quad (8.4a)$$

$$\tilde{\mathbf{H}}_1(z) = \tilde{\mathbf{H}}^i(z) + \tilde{\mathbf{H}}^r(z) = \hat{\mathbf{y}} \frac{1}{\eta_1} (E_0^i e^{-jk_1 z} - E_0^r e^{jk_1 z}). \quad (8.4b)$$

With only the transmitted wave present in medium 2, the total fields are

Medium 2

$$\tilde{\mathbf{E}}_2(z) = \tilde{\mathbf{E}}^t(z) = \hat{\mathbf{x}} E_0^t e^{-jk_2 z}, \quad (8.5a)$$

$$\tilde{\mathbf{H}}_2(z) = \tilde{\mathbf{H}}^t(z) = \hat{\mathbf{y}} \frac{E_0^t}{\eta_2} e^{-jk_2 z}. \quad (8.5b)$$

At the boundary ($z=0$), the tangential components of the electric and magnetic fields are continuous. Hence,

$$\tilde{\mathbf{E}}_1(0) = \tilde{\mathbf{E}}_2(0) \quad \text{or} \quad E_0^i + E_0^r = E_0^t, \quad (8.6a)$$

$$\tilde{\mathbf{H}}_1(0) = \tilde{\mathbf{H}}_2(0) \quad \text{or} \quad \frac{E_0^i}{\eta_1} - \frac{E_0^r}{\eta_1} = \frac{E_0^t}{\eta_2}. \quad (8.6b)$$

Solving these equations for E_0^r and E_0^t in terms of E_0^i gives

$$E_0^r = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right) E_0^i = \Gamma E_0^i, \quad (8.7a)$$

$$E_0^t = \left(\frac{2\eta_2}{\eta_2 + \eta_1} \right) E_0^i = \tau E_0^i, \quad (8.7b)$$

where

$$\Gamma = \frac{E_0^r}{E_0^i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad \text{(normal incidence)}, \quad (8.8a)$$

$$\tau = \frac{E_0^t}{E_0^i} = \frac{2\eta_2}{\eta_2 + \eta_1} \quad \text{(normal incidence)}. \quad (8.8b)$$

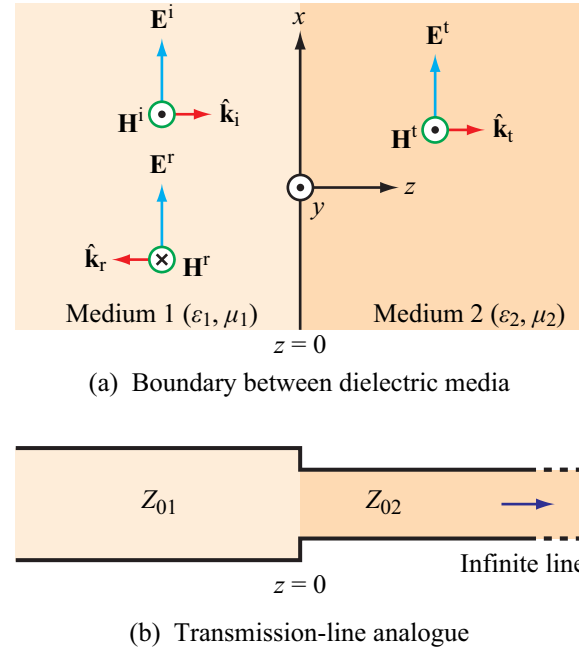


图 8-4 (a) 中由 x - y 平面分隔开的两种介电介质可以用 (b) 中的传输线模拟来表示。

事件波

$$\tilde{\mathbf{E}}^i(z) = \hat{\mathbf{x}} E_0^i e^{-jk_1 z}, \quad (8.1a)$$

$$\tilde{\mathbf{H}}^i(z) = \hat{\mathbf{z}} \times \frac{\tilde{\mathbf{E}}^i(z)}{\eta_1} = \hat{\mathbf{y}} \frac{E_0^i}{\eta_1} e^{-jk_1 z}. \quad (8.1b)$$

反射波

$$\tilde{\mathbf{E}}^r(z) = \hat{\mathbf{x}} E_0^r e^{jk_1 z}, \quad (8.2a)$$

$$\tilde{\mathbf{H}}^r(z) = (-\hat{\mathbf{z}}) \times \frac{\tilde{\mathbf{E}}^r(z)}{\eta_1} = -\hat{\mathbf{y}} \frac{E_0^r}{\eta_1} e^{jk_1 z}. \quad (8.2b)$$

传输波

$$\tilde{\mathbf{E}}^t(z) = \hat{\mathbf{x}} E_0^t e^{-jk_2 z}, \quad (8.3a)$$

$$\tilde{\mathbf{H}}^t(z) = \hat{\mathbf{z}} \times \frac{\tilde{\mathbf{E}}^t(z)}{\eta_2} = \hat{\mathbf{y}} \frac{E_0^t}{\eta_2} e^{-jk_2 z}. \quad (8.3b)$$

量 E_0^i 、 E_0^r 和 E_0^t 分别是 $z=0$ (两种介质之间的边界) 处的入射电场、反射电场和透射电场的振幅。介质 1 的波数和本征阻抗为 $k_1 = \omega\sqrt{\mu_1\epsilon_1}$ 且

$\eta_1 = \sqrt{\mu_1/\epsilon_1}$, 介质 2 的为 $k_2 = \omega\sqrt{\mu_2\epsilon_2}$ 和

$\eta_2 = \sqrt{\mu_2/\epsilon_2}$ 。振幅 E_0^i 由负责生成入射波的源施加; 因此, 假设已知。我们的目标是将 E_0^r 和 E_0^t 与 E_0^i 联系起来。为此, 我们对 $z=0$ 处的总电场和磁场应用边界条件。根据表 6-2, 总电场的切向分量在两个相邻介质之间的边界上始终是连续的, 并且在边界处没有电流源的情况下, 总磁场也是如此。在本例中, 入射波、反射波和透射波的电场和磁场均与边界相切。

介质 1 中的总电场 $\tilde{\mathbf{E}}_1(z)$ 是入射波和反射波的电场之和, 类似的说法也适用于磁场 $\tilde{\mathbf{H}}_1(z)$ 。因此,

中1

$$\tilde{\mathbf{E}}_1(z) = \tilde{\mathbf{E}}^i(z) + \tilde{\mathbf{E}}^r(z) = \hat{\mathbf{x}} (E_0^i e^{-jk_1 z} + E_0^r e^{jk_1 z}), \quad (8.4a)$$

$$\tilde{\mathbf{H}}_1(z) = \tilde{\mathbf{H}}^i(z) + \tilde{\mathbf{H}}^r(z) = \hat{\mathbf{y}} \frac{1}{\eta_1} (E_0^i e^{-jk_1 z} - E_0^r e^{jk_1 z}). \quad (8.4b)$$

当介质 2 中仅存在发射波时, 总场为

中2

$$\tilde{\mathbf{E}}_2(z) = \tilde{\mathbf{E}}^t(z) = \hat{\mathbf{x}} E_0^t e^{-jk_2 z}, \quad (8.5a)$$

$$\tilde{\mathbf{H}}_2(z) = \tilde{\mathbf{H}}^t(z) = \hat{\mathbf{y}} \frac{E_0^t}{\eta_2} e^{-jk_2 z}. \quad (8.5b)$$

在边界 ($z=0$) 处, 电场和磁场的切向分量是连续的。因此,

$$\tilde{\mathbf{E}}_1(0) = \tilde{\mathbf{E}}_2(0) \quad \text{or} \quad E_0^i + E_0^r = E_0^t, \quad (8.6a)$$

$$\tilde{\mathbf{H}}_1(0) = \tilde{\mathbf{H}}_2(0) \quad \text{or} \quad \frac{E_0^i}{\eta_1} - \frac{E_0^r}{\eta_1} = \frac{E_0^t}{\eta_2}. \quad (8.6b)$$

根据 E_0^i 求解这些方程, 得到 E_0^r 和 E_0^t

$$E_0^r = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right) E_0^i = \Gamma E_0^i, \quad (8.7a)$$

$$E_0^t = \left(\frac{2\eta_2}{\eta_2 + \eta_1} \right) E_0^i = \tau E_0^i, \quad (8.7b)$$

在哪里

$$\Gamma = \frac{E_0^r}{E_0^i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad \text{(normal incidence)}, \quad (8.8a)$$

$$\tau = \frac{E_0^t}{E_0^i} = \frac{2\eta_2}{\eta_2 + \eta_1} \quad \text{(normal incidence)}. \quad (8.8b)$$

The quantities Γ and τ are called the **reflection** and **transmission coefficients**. For lossless dielectric media, η_1 and η_2 are real; consequently, both Γ and τ are real also. As we see in Section 8-1.4, the expressions given by Eqs. (8.8a) and (8.8b) are equally applicable when the media are conductive, even though in that case η_1 and η_2 may be complex. Hence, Γ and τ may be complex as well. From Eqs. (8.8a) and (8.8b), it is easily shown that Γ and τ are interrelated as

$$\tau = 1 + \Gamma \quad \textbf{(normal incidence).} \quad (8.9)$$

For nonmagnetic media,

$$\eta_1 = \frac{\eta_0}{\sqrt{\epsilon_{r1}}}, \quad \text{and} \quad \eta_2 = \frac{\eta_0}{\sqrt{\epsilon_{r2}}},$$

where η_0 is the intrinsic impedance of free space. In this case, Eq. (8.8a) may be expressed as

$$\Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} \quad \textbf{(nonmagnetic media).} \quad (8.10)$$

8-1.2 Transmission-Line Analogue

The transmission-line configuration shown in **Fig. 8-4(b)** consists of a lossless transmission line with characteristic impedance Z_{01} connected at $z = 0$ to an infinitely long lossless transmission line with characteristic impedance Z_{02} . The input impedance of an infinitely long line is equal to its characteristic impedance. Hence, at $z = 0$, the voltage reflection coefficient (looking toward the boundary from the vantage point of the first line) is

$$\Gamma = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}},$$

which is identical in form to Eq. (8.8a). The analogy between plane waves and waves on transmission lines does not end here. To demonstrate the analogy further, equations pertinent to the two cases are summarized in **Table 8-1**. Comparison of the two columns shows that there is a one-to-one correspondence between the transmission-line quantities ($\tilde{V}, \tilde{I}, \beta, Z_0$) and the plane-wave quantities ($\tilde{E}, \tilde{H}, k, \eta$).

► This correspondence allows us to use the techniques developed in Chapter 2, including the Smith-chart method for calculating impedance transformations, to solve plane-wave propagation problems. ◀

The simultaneous presence of incident *and* reflected waves in medium 1 (**Fig. 8-4(a)**) gives rise to a standing-wave pattern.

By analogy with the transmission-line case, the **standing-wave ratio** in medium 1 is defined as

$$S = \frac{|\tilde{E}_1|_{\max}}{|\tilde{E}_1|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \quad (8.15)$$

► If the two media have equal impedances ($\eta_1 = \eta_2$), then $\Gamma = 0$ and $S = 1$, and if medium 2 is a perfect conductor with $\eta_2 = 0$ (which is equivalent to a short-circuited transmission line), then $\Gamma = -1$ and $S = \infty$. ◀

The distance from the boundary to where the magnitude of the electric field intensity in medium 1 is a maximum (denoted $l_{\max}$) is described by the same expression as that given by Eq. (2.70) for the voltage maxima on a transmission line, namely

$$-z = l_{\max} = \frac{\theta_r + 2n\pi}{2k_1} = \frac{\theta_r \lambda_1}{4\pi} + \frac{n\lambda_1}{2}, \quad (8.16)$$

$$\begin{cases} n = 1, 2, \dots & \text{if } \theta_r < 0, \\ n = 0, 1, 2, \dots & \text{if } \theta_r \geq 0, \end{cases}$$

where $\lambda_1 = 2\pi/k_1$ and θ_r is the phase angle of Γ (i.e., $\Gamma = |\Gamma|e^{j\theta_r}$, and θ_r is bounded in the range $-\pi < \theta_r \leq \pi$). The expression for $l_{\max}$ is valid not only when the two media are lossless dielectrics, but also when medium 1 is a low-loss dielectric. Moreover, medium 2 may be either a dielectric or a conductor. When both media are lossless dielectrics, $\theta_r = 0$ if $\eta_2 > \eta_1$, and $\theta_r = \pi$ if $\eta_2 < \eta_1$.

The spacing between adjacent maxima is $\lambda_1/2$, and the spacing between a maximum and the nearest minimum is $\lambda_1/4$. The electric-field minima occur at

$$l_{\min} = \begin{cases} l_{\max} + \lambda_1/4, & \text{if } l_{\max} < \lambda_1/4, \\ l_{\max} - \lambda_1/4, & \text{if } l_{\max} \geq \lambda_1/4. \end{cases} \quad (8.17)$$

8-1.3 Power Flow in Lossless Media

Medium 1 in **Fig. 8-4(a)** is host to the incident and reflected waves, which together comprise the total electric and magnetic fields $\tilde{\mathbf{E}}_1(z)$ and $\tilde{\mathbf{H}}_1(z)$ given by Eqs. (8.11a) and (8.12a) of **Table 8-1**. Using Eq. (7.100), the net average power density

量 Γ 和 τ 称为反射系数和透射系数。对于无损电介质， η_1 和 η_2 是实数；因此， Γ 和 τ 也是实数。正如我们在第 8-1.4 节中看到的，方程给出的表达式。当介质导电时，(8.8a) 和 (8.8b) 同样适用，即使在这种情况下 η_1 和 η_2 可能很复杂。因此， Γ 和 τ 也可能很复杂。从方程式。(8.8a) 和 (8.8b) ，很容易证明 Γ 和 τ 是相互关联的

$$\tau = 1 + \Gamma \quad \textbf{(normal incidence).} \quad (8.9)$$

对于非磁性介质，

$$\eta_1 = \frac{\eta_0}{\sqrt{\epsilon_{r1}}}, \quad \text{and} \quad \eta_2 = \frac{\eta_0}{\sqrt{\epsilon_{r2}}},$$

其中 η_0 是自由空间的固有阻抗。在这种情况下，等式。(8.8a) 可表示为

$$\Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} \quad \textbf{(nonmagnetic media).} \quad (8.10)$$

8-1.2 传输线模拟

图 8-4(b) 所示的传输线配置由一条特征阻抗为 Z_{01} 的无损传输线在 $z = 0$ 处连接到一条特征阻抗为 Z_{02} 的无限长无损传输线组成。无限长线路的输入阻抗等于其特性阻抗。因此，在 $z = 0$ 处，电压反射系数（从第一条线的有利位置看向边界）为

$$\Gamma = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}},$$

其形式与等式相同。(8.8a)。平面波和传输线上的波之间的类比还没有结束。为了进一步证明这种类比，表 8-1 总结了与这两种情况相关的方程。两列的比较表明，传输线量 ($\tilde{V}$ 、 $\tilde{I}$ 、 β 、 Z_0) 和平面波量 ($\tilde{E}$ 、 $\tilde{H}$ 、 k 、 η) 之间存在——对应关系。

► 这种对应关系使我们能够使用第 2 章中开发的技术（包括用于计算阻抗变换的史密斯圆图方法）来解决平面波传播问题。 ◀

介质 1 中同时存在入射波和反射波（图 8-4(a)），从而产生驻波图案。

与传输线情况类比，介质 1 中的驻波比定义为

$$S = \frac{|\tilde{E}_1|_{\max}}{|\tilde{E}_1|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \quad (8.15)$$

► 如果两种介质具有相等的阻抗 ($\eta_1 = \eta_2$)，则 $\Gamma = 0$ 且 $S = 1$ ，并且如果介质2是理想导体且 $\eta_2 = 0$ （相当于短路传输线），则 $\Gamma = -1$ 和 $S = \infty$ 。 ◀

从边界到介质 1 中电场强度的最大值（表示为 $l_{\max}$ ）的距离可以用与等式 1 给出的相同的表达式来描述。(2.70) 为传输线上的电压最大值，即

$$-z = l_{\max} = \frac{\theta_r + 2n\pi}{2k_1} = \frac{\theta_r \lambda_1}{4\pi} + \frac{n\lambda_1}{2}, \quad (8.16)$$

$$\begin{cases} n = 1, 2, \dots & \text{if } \theta_r < 0, \\ n = 0, 1, 2, \dots & \text{if } \theta_r \geq 0, \end{cases}$$

其中， $\lambda_1 = 2\pi/k_1$ 和 θ_r 是 Γ （的相位角，即 $\Gamma = |\Gamma|e^{j\theta_r}$ ，而 θ_r 的范围为 $-\pi < \theta_r \leq \pi$ ）。 $l_{\max}$ 的表达式不仅在两种介质是无损电介质时有效，而且在介质 1 是低损耗电介质时也有效。此外，介质2可以是电介质或导体。当两种介质都是无损电介质时，如果 $\eta_2 > \eta_1$ ，则 $\theta_r = 0$ ；如果 $\eta_2 < \eta_1$ ，则 $\theta_r = \pi$ 。

相邻最大值之间的间距为 $\lambda_1/2$ ，最大值与最近的最小值之间的间距为 $\lambda_1/4$ 。电场最小值出现在

$$l_{\min} = \begin{cases} l_{\max} + \lambda_1/4, & \text{if } l_{\max} < \lambda_1/4, \\ l_{\max} - \lambda_1/4, & \text{if } l_{\max} \geq \lambda_1/4. \end{cases} \quad (8.17)$$

8-1.3 无损媒体中的功率流

图 8-4(a) 中的介质 1 是入射波和反射波的宿主，它们共同构成了方程 1 给出的总电场和磁场 $\mathbf{E}_1(z)$ 和 $\mathbf{H}_1(z)$ 。表 8-1的(8.11a)和(8.12a)。使用方程式(7.100)，净平均功率密度

Table 8-1 Analogy between plane-wave equations for normal incidence and transmission-line equations with both under lossless conditions.

Plane Wave (Fig. 8-4(a))	Transmission Line (Fig. 8-4(b))
$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}}E_0^i(e^{-jk_1z} + \Gamma e^{jk_1z})$ (8.11a)	$\widetilde{V}_1(z) = V_0^+(e^{-j\beta_1z} + \Gamma e^{j\beta_1z})$ (8.11b)
$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}}\frac{E_0^i}{\eta_1}(e^{-jk_1z} - \Gamma e^{jk_1z})$ (8.12a)	$\widetilde{I}_1(z) = \frac{V_0^+}{Z_{01}}(e^{-j\beta_1z} - \Gamma e^{j\beta_1z})$ (8.12b)
$\widetilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}}\tau E_0^i e^{-jk_2z}$ (8.13a)	$\widetilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2z}$ (8.13b)
$\widetilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}}\tau\frac{E_0^i}{\eta_2}e^{-jk_2z}$ (8.14a)	$\widetilde{I}_2(z) = \tau\frac{V_0^+}{Z_{02}}e^{-j\beta_2z}$ (8.14b)
$\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$ $\tau = 1 + \Gamma$	$\Gamma = (Z_{02} - Z_{01})/(Z_{02} + Z_{01})$ $\tau = 1 + \Gamma$
$k_1 = \omega\sqrt{\mu_1\epsilon_1}$, $k_2 = \omega\sqrt{\mu_2\epsilon_2}$	$\beta_1 = \omega\sqrt{\mu_1\epsilon_1}$, $\beta_2 = \omega\sqrt{\mu_2\epsilon_2}$
$\eta_1 = \sqrt{\mu_1/\epsilon_1}$, $\eta_2 = \sqrt{\mu_2/\epsilon_2}$	Z_{01} and Z_{02} depend on transmission-line parameters

flowing in medium 1 is

$$\begin{aligned}\mathbf{S}_{\text{av}_1}(z) &= \tfrac{1}{2}\Re\mathfrak{e}[\widetilde{\mathbf{E}}_1(z) \times \widetilde{\mathbf{H}}_1^*(z)] \\ &= \tfrac{1}{2}\Re\mathfrak{e}\left[\hat{\mathbf{x}}E_0^i(e^{-jk_1z} + \Gamma e^{jk_1z})\hat{\mathbf{y}}\frac{E_0^{i*}}{\eta_1}(e^{jk_1z} - \Gamma^*e^{-jk_1z})\right] \\ &= \hat{\mathbf{z}}\frac{|E_0^i|^2}{2\eta_1}(1 - |\Gamma|^2),\end{aligned}\tag{8.18}$$

which is analogous to Eq. (2.106) for the lossless transmission-line case. The first and second terms inside the bracket in Eq. (8.18) represent the average power density of the incident and reflected waves, respectively. Thus,

$$\mathbf{S}_{\text{av}_1} = \mathbf{S}_{\text{av}}^{\text{i}} + \mathbf{S}_{\text{av}}^{\text{r}}\tag{8.19a}$$

with

$$\mathbf{S}_{\text{av}}^{\text{i}} = \hat{\mathbf{z}}\frac{|E_0^i|^2}{2\eta_1},\tag{8.19b}$$

$$\mathbf{S}_{\text{av}}^{\text{r}} = -\hat{\mathbf{z}}|\Gamma|^2\frac{|E_0^i|^2}{2\eta_1} = -|\Gamma|^2\mathbf{S}_{\text{av}}^{\text{i}}.\tag{8.19c}$$

Even though Γ is purely real when both media are lossless dielectrics, we chose to treat it as complex, thereby providing in Eq. (8.19c) an expression that is also valid when medium 2 is conducting.

The average power density of the transmitted wave in

medium 2 is

$$\begin{aligned}\mathbf{S}_{\text{av}_2}(z) &= \tfrac{1}{2}\Re\mathfrak{e}[\widetilde{\mathbf{E}}_2(z) \times \widetilde{\mathbf{H}}_2^*(z)] \\ &= \tfrac{1}{2}\Re\mathfrak{e}\left[\hat{\mathbf{x}}\tau E_0^i e^{-jk_2z} \times \hat{\mathbf{y}}\tau^*\frac{E_0^{i*}}{\eta_2}e^{jk_2z}\right] = \hat{\mathbf{z}}|\tau|^2\frac{|E_0^i|^2}{2\eta_2}.\end{aligned}\tag{8.20}$$

Through the use of Eqs. (8.8a and b), it can be easily shown that for lossless media

$$\frac{\tau^2}{\eta_2} = \frac{1 - \Gamma^2}{\eta_1}, \quad \textbf{(lossless media)}\tag{8.21}$$

which leads to

$$\mathbf{S}_{\text{av}_1} = \mathbf{S}_{\text{av}_2}.$$

This result is expected from considerations of power conservation.

Example 8-1: Radar Radome Design

A 10 GHz aircraft radar uses a narrow-beam scanning antenna mounted on a gimbal behind a dielectric radome, as shown in **Fig. 8-5**. Even though the radome shape is far from planar, it is approximately planar over the narrow extent of the radar beam. If the radome material is a lossless dielectric with $\epsilon_{\text{r}} = 9$ and

表 8-1 法向入射平面波方程与无损条件下的传输线方程之间的类比。

Plane Wave (Fig. 8-4(a))	Transmission Line (Fig. 8-4(b))
$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}}E_0^i(e^{-jk_1z} + \Gamma e^{jk_1z})$ (8.11a)	$\widetilde{V}_1(z) = V_0^+(e^{-j\beta_1z} + \Gamma e^{j\beta_1z})$ (8.11b)
$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}}\frac{E_0^i}{\eta_1}(e^{-jk_1z} - \Gamma e^{jk_1z})$ (8.12a)	$\widetilde{I}_1(z) = \frac{V_0^+}{Z_{01}}(e^{-j\beta_1z} - \Gamma e^{j\beta_1z})$ (8.12b)
$\widetilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}}\tau E_0^i e^{-jk_2z}$ (8.13a)	$\widetilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2z}$ (8.13b)
$\widetilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}}\tau\frac{E_0^i}{\eta_2}e^{-jk_2z}$ (8.14a)	$\widetilde{I}_2(z) = \tau\frac{V_0^+}{Z_{02}}e^{-j\beta_2z}$ (8.14b)
$\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$ $\tau = 1 + \Gamma$	$\Gamma = (Z_{02} - Z_{01})/(Z_{02} + Z_{01})$ $\tau = 1 + \Gamma$
$k_1 = \omega\sqrt{\mu_1\epsilon_1}$, $k_2 = \omega\sqrt{\mu_2\epsilon_2}$	$\beta_1 = \omega\sqrt{\mu_1\epsilon_1}$, $\beta_2 = \omega\sqrt{\mu_2\epsilon_2}$
$\eta_1 = \sqrt{\mu_1/\epsilon_1}$, $\eta_2 = \sqrt{\mu_2/\epsilon_2}$	Z_{01} and Z_{02} depend on transmission-line parameters

在介质 1 中流动的是

$$\begin{aligned}\mathbf{S}_{\text{av}_1}(z) &= \tfrac{1}{2}\Re\mathfrak{e}[\widetilde{\mathbf{E}}_1(z) \times \widetilde{\mathbf{H}}_1^*(z)] \\ &= \tfrac{1}{2}\Re\mathfrak{e}\left[\hat{\mathbf{x}}E_0^i(e^{-jk_1z} + \Gamma e^{jk_1z})\hat{\mathbf{y}}\frac{E_0^{i*}}{\eta_1}(e^{jk_1z} - \Gamma^*e^{-jk_1z})\right] \\ &= \hat{\mathbf{z}}\frac{|E_0^i|^2}{2\eta_1}(1 - |\Gamma|^2),\end{aligned}\tag{8.18}$$

这类似于等式。 (2.106) 适用于无损传输线情况。方程中括号内的第一项和第二项(8.18) 分别表示入射波和反射波的平均功率密度。因此,

$$\mathbf{S}_{\text{av}_1} = \mathbf{S}_{\text{av}}^{\text{i}} + \mathbf{S}_{\text{av}}^{\text{r}}\tag{8.19a}$$

和

$$\mathbf{S}_{\text{av}}^{\text{i}} = \hat{\mathbf{z}}\frac{|E_0^i|^2}{2\eta_1},\tag{8.19b}$$

$$\mathbf{S}_{\text{av}}^{\text{r}} = -\hat{\mathbf{z}}|\Gamma|^2\frac{|E_0^i|^2}{2\eta_1} = -|\Gamma|^2\mathbf{S}_{\text{av}}^{\text{i}}.\tag{8.19c}$$

尽管当两种介质都是无损电介质时 Γ 是纯实数，但我们选择将其视为复数，从而在等式中提供： (8.19c) 当介质 2 号通时该表达式也有效。

发射波的平均功率密度

中 2 是

$$\begin{aligned}\mathbf{S}_{\text{av}_2}(z) &= \tfrac{1}{2}\Re\mathfrak{e}[\widetilde{\mathbf{E}}_2(z) \times \widetilde{\mathbf{H}}_2^*(z)] \\ &= \tfrac{1}{2}\Re\mathfrak{e}\left[\hat{\mathbf{x}}\tau E_0^i e^{-jk_2z} \times \hat{\mathbf{y}}\tau^*\frac{E_0^{i*}}{\eta_2}e^{jk_2z}\right] = \hat{\mathbf{z}}|\tau|^2\frac{|E_0^i|^2}{2\eta_2}.\end{aligned}\tag{8.20}$$

通过使用方程式。 (8.8a和b) ， 可以很容易地证明对于无损媒体

$$\frac{\tau^2}{\eta_2} = \frac{1 - \Gamma^2}{\eta_1}, \quad \textbf{(lossless media)}\tag{8.21}$$

这导致

$$\mathbf{S}_{\text{av}_1} = \mathbf{S}_{\text{av}_2}.$$

这个结果是出于节能考虑的预期。

例 8-1: 雷达天线罩设计

10GHz飞机雷达采用窄波束扫描天线，安装在介质天线罩后面的万向架上，如图8-5所示。尽管天线罩形状远非平面，但在雷达波束的狭窄范围内它近似平面。如果天线罩材料是无损电介质， $\epsilon_{\text{r}} = 9$ 且

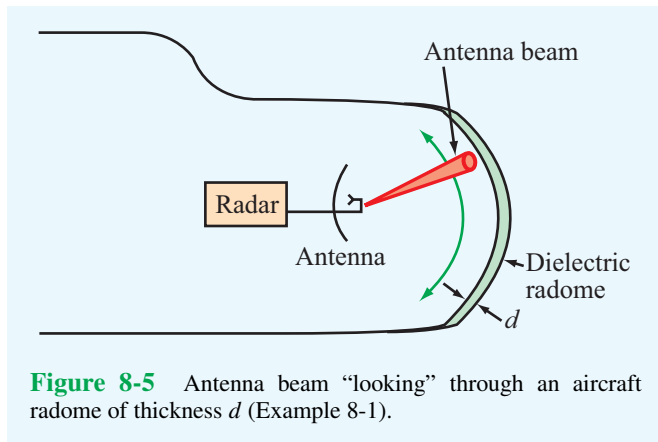


Figure 8-5 Antenna beam “looking” through an aircraft radome of thickness d (Example 8-1).

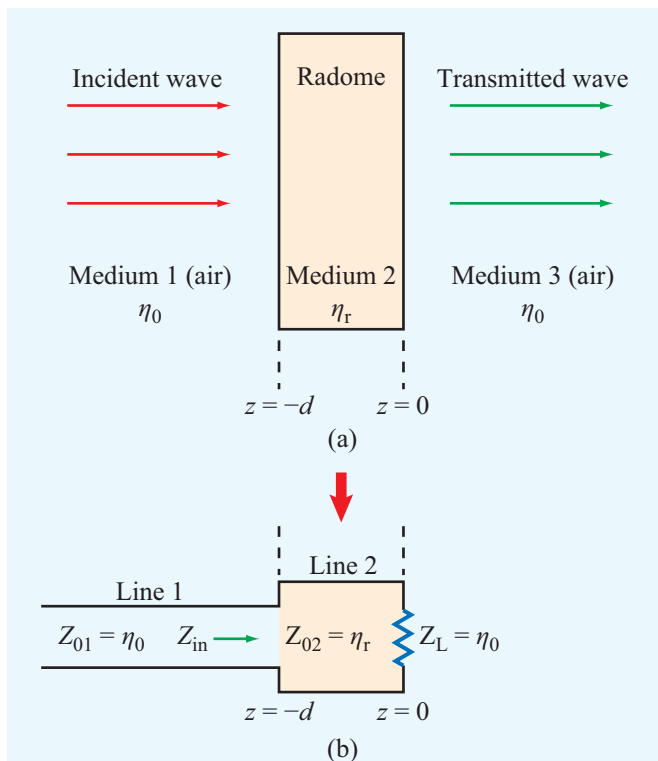


Figure 8-6 (a) Planar section of the radome of **Fig. 8-5** at an expanded scale and (b) its transmission-line equivalent model (Example 8-1).

$\mu_r = 1$, choose its thickness d such that the radome appears transparent to the radar beam. Structural integrity requires d to be greater than 2.3 cm.

Solution: **Figure 8-6(a)** shows a small section of the radome

on an expanded scale. The incident wave can be approximated as a plane wave propagating in medium 1 (air) with intrinsic impedance η_0 . Medium 2 (the radome) is of thickness d and intrinsic impedance η_r , and medium 3 (air) is semi-infinite with intrinsic impedance η_0 . **Figure 8-6(b)** shows an equivalent transmission-line model with $z = 0$ selected to coincide with the outside surface of the radome, and the load impedance $Z_L = \eta_0$ represents the input impedance of the semi-infinite air medium to the right of the radome.

For the radome to “appear” transparent to the incident wave, the reflection coefficient must be zero at $z = -d$, thereby guaranteeing total transmission of the incident power into medium 3. Since $Z_L = \eta_0$ in **Fig. 8-6(b)**, no reflection takes place at $z = -d$ if $Z_{in} = \eta_0$, which can be realized by choosing $d = n\lambda_2/2$ [see Section 2-8.4], where λ_2 is the wavelength in medium 2 and n is a positive integer. At 10 GHz, the wavelength in air is $\lambda_0 = c/f = 3$ cm, while in the radome material it is

$$\lambda_2 = \frac{\lambda_0}{\sqrt{\epsilon_r}} = \frac{3 \text{ cm}}{3} = 1 \text{ cm}.$$

Hence, by choosing $d = 5\lambda_2/2 = 2.5$ cm, the radome is both nonreflecting and structurally stable.

Example 8-2: Yellow Light Incident upon a Glass Surface

A beam of yellow light with wavelength $0.6 \mu\text{m}$ is normally incident in air upon a glass surface. Assume the glass is sufficiently thick as to ignore its back surface. If the surface is situated in the plane $z = 0$ and the relative permittivity of glass is 2.25, determine:

- the locations of the electric field maxima in medium 1 (air),
- the standing-wave ratio, and
- the fraction of the incident power transmitted into the glass medium.

Solution: (a) We begin by determining the values of η_1 , η_2 , and Γ :

$$\begin{aligned} \eta_1 &= \sqrt{\frac{\mu_1}{\epsilon_1}} = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx 120\pi \text{ } (\Omega), \\ \eta_2 &= \sqrt{\frac{\mu_2}{\epsilon_2}} = \sqrt{\frac{\mu_0}{\epsilon_0}} \cdot \frac{1}{\sqrt{\epsilon_r}} \approx \frac{120\pi}{\sqrt{2.25}} = 80\pi \text{ } (\Omega), \\ \Gamma &= \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{80\pi - 120\pi}{80\pi + 120\pi} = -0.2. \end{aligned}$$

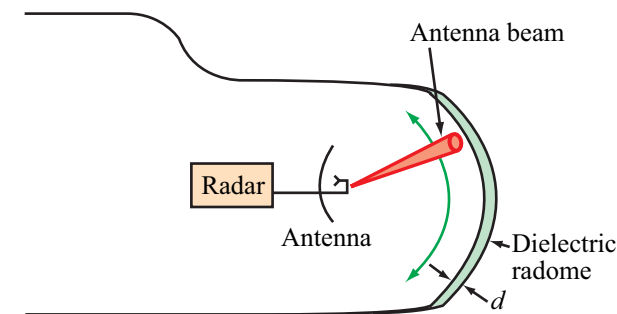


图 8-5 天线波束 “看” 过厚度为 d 的飞机天线罩 (示例 8-1)。

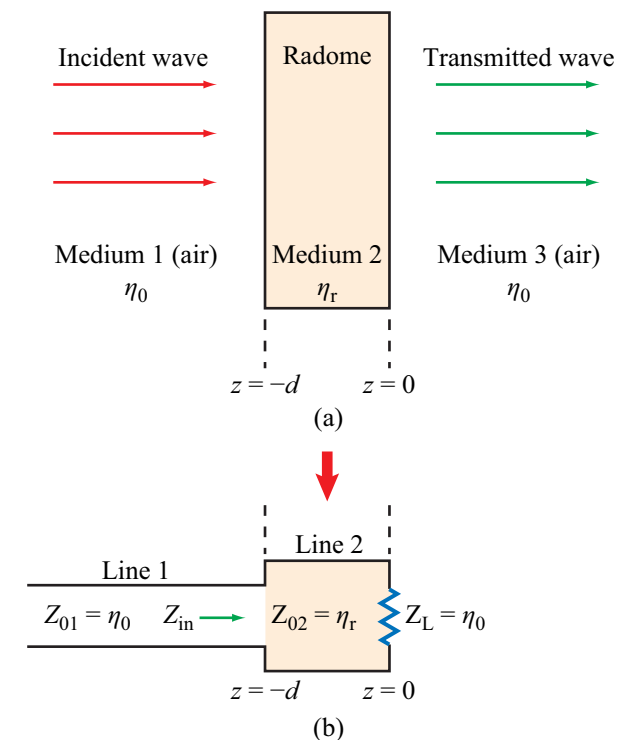


图 8-6 (a) 图 8-5 天线罩放大后的平面剖面, (b) 其传输线等效模型 (例 8-1)。

$\mu_r = 1$, 选择其厚度 d , 使天线罩对雷达波束显得透明。结构完整性要求 d 大于 2.3 厘米。

解: 图8-6(a)显示了天线罩的一小部分

在扩大的规模上。入射波可以近似为在介质1 (空气) 中传播的具有固有阻抗 η_0 的平面波。介质2 (天线罩) 的厚度为 d , 固有阻抗为 η_r , 介质3 (空气) 为半无限, 固有阻抗为 η_0 。图 8-6(b) 显示了一个等效传输线模型, 其中 $z = 0$ 选择与天线罩外表面重合, 负载阻抗 $Z_L = \eta_0$ 表示天线罩右侧半无限空气介质的输入阻抗。

为了让天线罩对入射波 “显得” 透明, 在 $z = -d$ 处的反射系数必须为零, 从而保证入射功率全部传输到介质3中。由于图 8-6 (b) 中的 $Z_L = \eta_0$, 如果 $Z_{in} = \eta_0$, 则在 $z = -d$ 处不会发生反射, 这可以通过选择 $d = n\lambda_2/2$ 来实现2-8.4], 其中 λ_2 是介质2中的波长, n 是正整数。在 10 GHz 时, 空气中的波长为 $\lambda_0 = c/f = 3$ cm, 而在天线罩材料中的波长为

$$\lambda_2 = \frac{\lambda_0}{\sqrt{\epsilon_r}} = \frac{3 \text{ cm}}{3} = 1 \text{ cm}.$$

因此, 通过选择 $d = 5\lambda_2/2 = 2.5$ cm, 天线罩既不反射又结构稳定。

示例 8-2: 黄光入射到玻璃表面

波长为 $0.6 \mu\text{m}$ 的黄光垂直在空气中入射到玻璃表面上。假设玻璃足够厚, 可以忽略其背面。如果表面位于平面 $z = 0$ 且玻璃的相对介电常数为 2.25, 则确定:

- 介质1 (空气) 中电场最大值的位置, (b) 驻波比, 以及 (c) 传输到玻璃介质中的入射功率的比例。

解: (a) 我们首先确定 η_1 、 η_2 和 Γ 的值:

$$\begin{aligned} \eta_1 &= \sqrt{\frac{\mu_1}{\epsilon_1}} = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx 120\pi \text{ } (\Omega), \\ \eta_2 &= \sqrt{\frac{\mu_2}{\epsilon_2}} = \sqrt{\frac{\mu_0}{\epsilon_0}} \cdot \frac{1}{\sqrt{\epsilon_r}} \approx \frac{120\pi}{\sqrt{2.25}} = 80\pi \text{ } (\Omega), \\ \Gamma &= \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{80\pi - 120\pi}{80\pi + 120\pi} = -0.2. \end{aligned}$$

Hence, $|\Gamma| = 0.2$ and $\theta_r = \pi$. From Eq. (8.16), the electric-field magnitude is maximum at

$$l_{\max} = \frac{\theta_r \lambda_1}{4\pi} + n \frac{\lambda_1}{2} = \frac{\lambda_1}{4} + n \frac{\lambda_1}{2} \quad (n = 0, 1, 2, \dots)$$

with $\lambda_1 = 0.6 \mu\text{m}$.

(b)

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.2}{1 - 0.2} = 1.5.$$

(c) The fraction of the incident power transmitted into the glass medium is equal to the ratio of the transmitted power density given by Eq. (8.20) to the incident power density, $S_{\text{av}}^i = |E_0^i|^2 / 2\eta_1$:

$$\frac{S_{\text{av}2}}{S_{\text{av}}^i} = \tau^2 \frac{|E_0^i|^2}{2\eta_2} \bigg/ \left[\frac{|E_0^i|^2}{2\eta_1} \right] = \tau^2 \frac{\eta_1}{\eta_2}.$$

In view of Eq. (8.21),

$$\frac{S_{\text{av}2}}{S_{\text{av}}^i} = 1 - |\Gamma|^2 = 1 - (0.2)^2 = 0.96, \text{ or } 96\%.$$

8-1.4 Boundary between Lossy Media

In Section 8-1.1, we considered a plane wave in a lossless medium incident normally on a planar boundary of another lossless medium. We now generalize our expressions to lossy media. In a medium with constitutive parameters (ϵ, μ, σ) , the propagation constant $\gamma = \alpha + j\beta$ and the intrinsic impedance η_c are both complex. General expressions for α , β , and η_c are given by Eqs. (7.66a), (7.66b), and (7.70), respectively, and approximate expressions are given in **Table 7-2** for the special cases of low-loss media and good conductors. If media 1 and 2 have constitutive parameters $(\epsilon_1, \mu_1, \sigma_1)$ and $(\epsilon_2, \mu_2, \sigma_2)$ (**Fig. 8-7**), then expressions for the electric and magnetic fields in media 1 and 2 can be obtained from Eqs. (8.11) through (8.14) of **Table 8-1** by replacing jk with γ and η with η_c . Thus,

Medium 1

$$\tilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^i (e^{-\gamma_1 z} + \Gamma e^{\gamma_1 z}), \quad (8.22a)$$

$$\tilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \frac{E_0^i}{\eta_{c1}} (e^{-\gamma_1 z} - \Gamma e^{\gamma_1 z}), \quad (8.22b)$$

Medium 2

$$\tilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E_0^i e^{-\gamma_2 z}, \quad (8.23a)$$

$$\tilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \frac{E_0^i}{\eta_{c2}} e^{-\gamma_2 z}. \quad (8.23b)$$

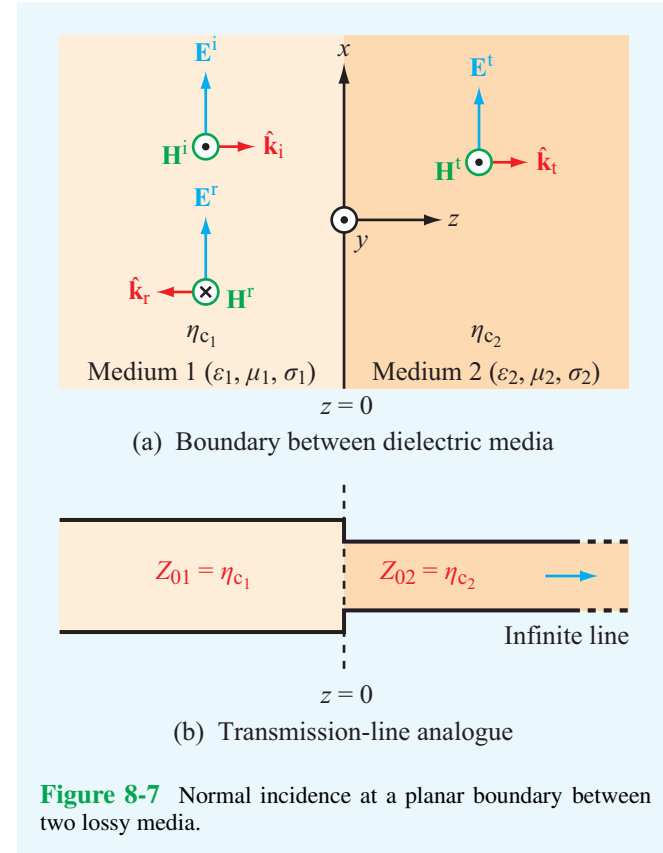


Figure 8-7 Normal incidence at a planar boundary between two lossy media.

Here, $\gamma_1 = \alpha_1 + j\beta_1$, $\gamma_2 = \alpha_2 + j\beta_2$, and

$$\Gamma = \frac{\eta_{c2} - \eta_{c1}}{\eta_{c2} + \eta_{c1}}, \quad (8.24a)$$

$$\tau = 1 + \Gamma = \frac{2\eta_{c2}}{\eta_{c2} + \eta_{c1}}. \quad (8.24b)$$

Because η_{c1} and η_{c2} are, in general, complex, Γ and τ may be complex as well.

Example 8-3: Normal Incidence on a Metal Surface

A 1 GHz x -polarized plane wave traveling in the $+\hat{z}$ direction is incident from air upon a copper surface. The air-to-copper interface is at $z = 0$, and copper has $\epsilon_r = 1$, $\mu_r = 1$, and $\sigma = 5.8 \times 10^7 \text{ S/m}$. If the amplitude of the electric field of the incident wave is 12 (mV/m), obtain expressions for the

因此, $|\Gamma| = 0.2$ 和 $\theta_r = \pi$ 。从方程。 (8.16) 电场强度最大时

$$l_{\max} = \frac{\theta_r \lambda_1}{4\pi} + n \frac{\lambda_1}{2} = \frac{\lambda_1}{4} + n \frac{\lambda_1}{2} \quad (n = 0, 1, 2, \dots)$$

与 $\lambda_1 = 0.6 \mu\text{m}$ 。

(b)

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.2}{1 - 0.2} = 1.5.$$

(c) 传输到玻璃介质中的入射功率的比例等于方程 (1) 给出的传输功率密度之比。 (8.20) 到入射功率密度 $S_{\text{av}}^i = |E_0^i|^2 / 2\eta_1$:

$$\frac{S_{\text{av}2}}{S_{\text{av}}^i} = \tau^2 \frac{|E_0^i|^2}{2\eta_2} \bigg/ \left[\frac{|E_0^i|^2}{2\eta_1} \right] = \tau^2 \frac{\eta_1}{\eta_2}.$$

鉴于方程。 (8.21) ,

$$\frac{S_{\text{av}2}}{S_{\text{av}}^i} = 1 - |\Gamma|^2 = 1 - (0.2)^2 = 0.96, \text{ or } 96\%.$$

8-1.4 有损媒体之间的边界

在第 8-1.1 节中, 我们考虑了无损介质中的平面波垂直入射到另一个无损介质的平面边界上。我们现在将我们的表达推广到有损媒体。在具有本构参数 (ϵ, μ, σ) 的介质中, 传播常数 $\gamma = \alpha + j\beta$ 和固有阻抗 η_c 都是复数。 α 、 β 和 η_c 的一般表达式由等式 1 给出。分别为 (7.66a)、(7.66b) 和 (7.70), 表7-2给出了低损耗介质和良导体特殊情况的近似表达式。如果介质1和介质2具有本构参数 $(\epsilon_1, \mu_1, \sigma_1)$ 和 $(\epsilon_2, \mu_2, \sigma_2)$ (图8-7), 则介质1和介质2中的电场和磁场的表达式可以由式(1)获得。表 8-1 的 (8.11) 到 (8.14), 用 γ 替换 jk , 用 η_c 替换 η 。因此,

中1

$$\tilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^i (e^{-\gamma_1 z} + \Gamma e^{\gamma_1 z}), \quad (8.22a)$$

$$\tilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \frac{E_0^i}{\eta_{c1}} (e^{-\gamma_1 z} - \Gamma e^{\gamma_1 z}), \quad (8.22b)$$

中2

$$\tilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E_0^i e^{-\gamma_2 z}, \quad (8.23a)$$

$$\tilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \frac{E_0^i}{\eta_{c2}} e^{-\gamma_2 z}. \quad (8.23b)$$

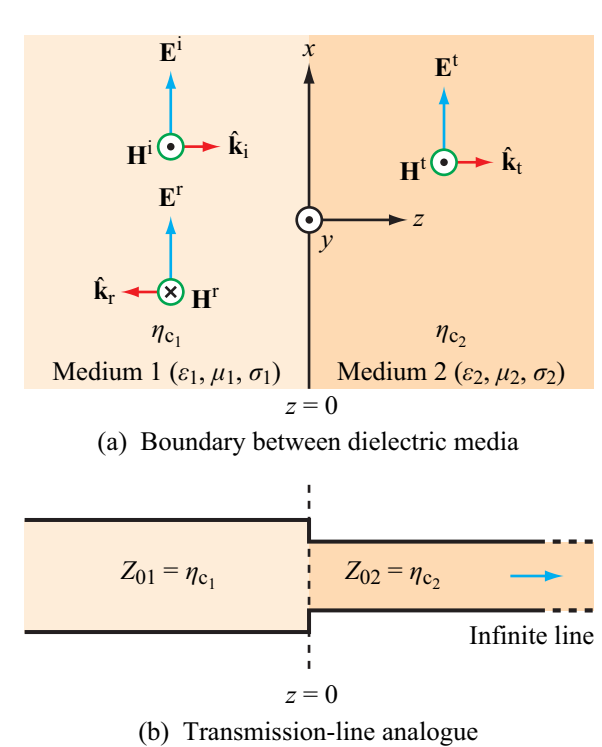


图 8-7 两个有损介质之间平面边界处的法向入射。

这里, $\gamma_1 = \alpha_1 + j\beta_1$ 、 $\gamma_2 = \alpha_2 + j\beta_2$ 和

$$\Gamma = \frac{\eta_{c2} - \eta_{c1}}{\eta_{c2} + \eta_{c1}}, \quad (8.24a)$$

$$\tau = 1 + \Gamma = \frac{2\eta_{c2}}{\eta_{c2} + \eta_{c1}}. \quad (8.24b)$$

由于 η_{c1} 和 η_{c2} 通常是复杂的, 因此 Γ 和 τ 也可能是复杂的。

示例 8-3: 金属表面上的法向入射

沿 $+\hat{z}$ 方向行进的 1 GHz x 偏振平面波从空气入射到铜表面。空气-铜界面位于 $z = 0$, 铜具有 $\epsilon_r = 1$ 、 $\mu_r = 1$ 和 $\sigma = 5.8 \times 10^7 \text{ S/m}$ 。如果入射波的电场幅度为 12 (mV/m), 则得到表达式

instantaneous electric and magnetic fields in the air medium. Assume the metal surface to be several skin depths deep.

Solution: In medium 1 (air), $\alpha = 0$,

$$\beta = k_1 = \frac{\omega}{c} = \frac{2\pi \times 10^9}{3 \times 10^8} = \frac{20\pi}{3} \quad (\text{rad/m}),$$

$$\eta_1 = \eta_0 = 377 \, (\Omega), \quad \lambda = \frac{2\pi}{k_1} = 0.3 \, \text{m}.$$

At $f = 1 \, \text{GHz}$, copper is an excellent conductor because

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega\epsilon_r\epsilon_0} = \frac{5.8 \times 10^7}{2\pi \times 10^9 \times (10^{-9}/36\pi)} = 1 \times 10^9 \gg 1.$$

Use of Eq. (7.77c) gives

$$\eta_{c2} = (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} = (1+j)\left[\frac{\pi \times 10^9 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}\right]^{1/2} = 8.25(1+j) \quad (\text{m}\Omega).$$

Since η_{c2} is so small compared to $\eta_0 = 377 \, (\Omega)$ for air, the copper surface acts, in effect, like a short circuit. Hence,

$$\Gamma = \frac{\eta_{c2} - \eta_0}{\eta_{c2} + \eta_0} \approx -1.$$

Upon setting $\Gamma = -1$ in Eqs. (8.11) and (8.12) of **Table 8-1**, we obtain

$$\tilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}}E_0^i(e^{-jk_1z} - e^{jk_1z}) = -\hat{\mathbf{x}}j2E_0^i \sin k_1z, \quad (8.25a)$$

$$\tilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}}\frac{E_0^i}{\eta_1}(e^{-jk_1z} + e^{jk_1z}) = \hat{\mathbf{y}}2\frac{E_0^i}{\eta_1} \cos k_1z. \quad (8.25b)$$

With $E_0^i = 12 \, (\text{mV/m})$, the instantaneous fields associated with these phasors are

$$\begin{aligned} \mathbf{E}_1(z,t) &= \Re\{\tilde{\mathbf{E}}_1(z)e^{j\omega t}\} \\ &= \hat{\mathbf{x}}2E_0^i \sin k_1z \sin \omega t \\ &= \hat{\mathbf{x}}24 \sin(20\pi z/3) \sin(2\pi \times 10^9 t) \quad (\text{mV/m}), \end{aligned}$$

$$\begin{aligned} \mathbf{H}_1(z,t) &= \Re\{\tilde{\mathbf{H}}_1(z)e^{j\omega t}\} \\ &= \hat{\mathbf{y}}2\frac{E_0^i}{\eta_1} \cos k_1z \cos \omega t \\ &= \hat{\mathbf{y}}64 \cos(20\pi z/3) \cos(2\pi \times 10^9 t) \quad (\mu\text{A/m}). \end{aligned}$$

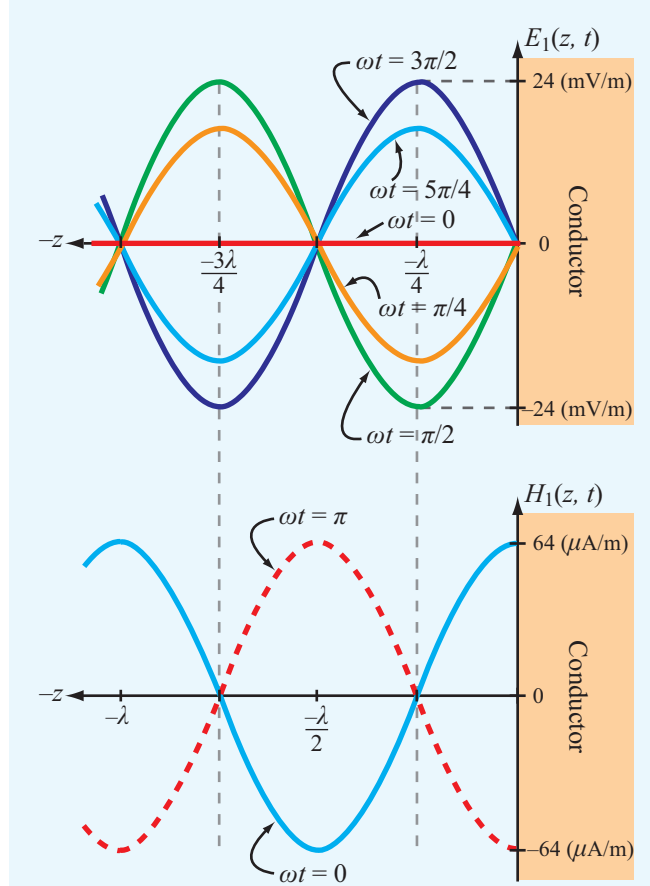


Figure 8-8 Wave patterns for fields $E_1(z,t)$ and $H_1(z,t)$ of Example 8-3.

Plots of the magnitude of $\mathbf{E}_1(z,t)$ and $\mathbf{H}_1(z,t)$ are shown in **Fig. 8-8** as a function of negative z for various values of ωt . The wave patterns exhibit a repetition period of $\lambda/2$, and E and H are in phase quadrature (90° phase shift) in both space and time. This behavior is identical with that for voltage and current waves on a shorted transmission line.

Concept Question 8-1: What boundary conditions were used in the derivations of the expressions for Γ and τ ?

Concept Question 8-2: In the radar radome design of Example 8-1, all the incident energy in medium 1 ends up getting transmitted into medium 3, and vice versa. Does this imply that no reflections take place within medium 2? Explain.

空气介质中的瞬时电场和磁场。假设金属表面有几个趋肤深度。

解：在介质1（空气）中， $\alpha = 0$,

$$\beta = k_1 = \frac{\omega}{c} = \frac{2\pi \times 10^9}{3 \times 10^8} = \frac{20\pi}{3} \quad (\text{rad/m}),$$

$$\eta_1 = \eta_0 = 377 \, (\Omega), \quad \lambda = \frac{2\pi}{k_1} = 0.3 \, \text{m}.$$

在 $f = 1 \, \text{GHz}$ 时，铜是一种优良的导体，因为

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega\epsilon_r\epsilon_0} = \frac{5.8 \times 10^7}{2\pi \times 10^9 \times (10^{-9}/36\pi)} = 1 \times 10^9 \gg 1.$$

使用等式 (7.77c) 给出

$$\eta_{c2} = (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} = (1+j)\left[\frac{\pi \times 10^9 \times 4\pi \times 10^{-7}}{5.8 \times 10^7}\right]^{1/2} = 8.25(1+j) \quad (\text{m}\Omega).$$

由于 η_{c2} 与空气中的 $\eta_0 = 377 \, (\Omega)$ 相比非常小，因此铜表面实际上就像短路一样。因此，

$$\Gamma = \frac{\eta_{c2} - \eta_0}{\eta_{c2} + \eta_0} \approx -1.$$

在等式中设置 $\Gamma = -1$ 后。由表8-1的(8.11)和(8.12)，我们得到

$$\tilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}}E_0^i(e^{-jk_1z} - e^{jk_1z}) = -\hat{\mathbf{x}}j2E_0^i \sin k_1z, \quad (8.25a)$$

$$\tilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}}\frac{E_0^i}{\eta_1}(e^{-jk_1z} + e^{jk_1z}) = \hat{\mathbf{y}}2\frac{E_0^i}{\eta_1} \cos k_1z. \quad (8.25b)$$

对于 $E_0^i = 12 \, (\text{mV/m})$ ，与这些相量相关的瞬时场为

$$\begin{aligned} \mathbf{E}_1(z,t) &= \Re\{\tilde{\mathbf{E}}_1(z)e^{j\omega t}\} \\ &= \hat{\mathbf{x}}2E_0^i \sin k_1z \sin \omega t \\ &= \hat{\mathbf{x}}24 \sin(20\pi z/3) \sin(2\pi \times 10^9 t) \quad (\text{mV/m}), \end{aligned}$$

$$\begin{aligned} \mathbf{H}_1(z,t) &= \Re\{\tilde{\mathbf{H}}_1(z)e^{j\omega t}\} \\ &= \hat{\mathbf{y}}2\frac{E_0^i}{\eta_1} \cos k_1z \cos \omega t \\ &= \hat{\mathbf{y}}64 \cos(20\pi z/3) \cos(2\pi \times 10^9 t) \quad (\mu\text{A/m}). \end{aligned}$$

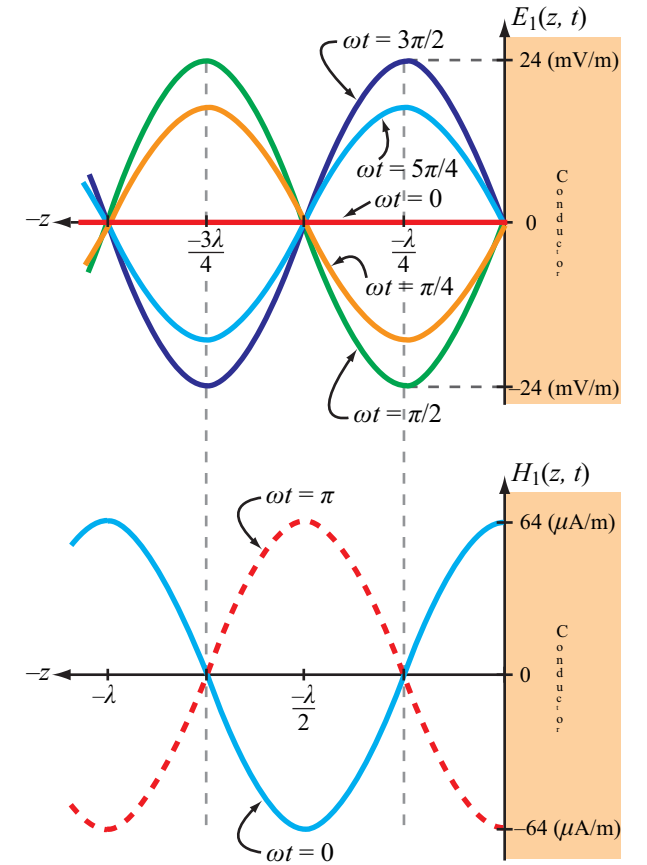


图 8-8 例 8-3 中 $E_1(z,t)$ 和 $H_1(z,t)$ 场的波形。

$E_1(z,t)$ 和 $H_1(z,t)$ 的幅值图如图 8-8 所示，作为不同 ωt 值的负 z 的函数。波形的重复周期为 $\lambda/2$ ， E 和 H 在空间和时间上相位正交 (90° 相移)。这种行为与短路传输线上的电压和电流波的行为相同。

概念问题 8-1：在 Γ 和 τ 表达式的推导中使用了哪些边界条件？

概念问题 8-2：在例 8-1 的雷达天线罩设计中，介质 1 中的所有入射能量最终都会传输到介质 3 中，反之亦然。这是否意味着介质 2 内不会发生反射？解释。